

RF and Microwave Engineering

Complete Exam Notes

EX 716 (BEI, IV/I) · EX 752 (BEX, IV/II)
Institute of Engineering, Tribhuvan University

Built from the syllabus, the Pulchowk lecture decks, three textbooks and **all 24 past papers (2069–2082 BS)** across both course codes. Every topic carries a frequency tier counted from those papers, and `src/check.py` checks each tier chip against its own year list *and* against the sorted PYQ document, so no chapter can cite a paper that asks it nothing.

Solved problems live in the companion volume, **RF and Microwave Engineering: Complete Numerical Problems and Solutions**: **56 problems** over five chapters, each opening with the question exactly as its paper prints it. **Chapters 1, 5 and 8 set no numerical** in 24 papers, and each says so in its own PYQ-mapping section.

Short on time? Go straight to §9, **Last-Minute Revision**.

Ch	Topic	Hours	Papers	Marks	Problems
1	Introduction	3	19/24	184	—
2	RF and M/W Transmission Lines	6	24/24	297	24
3	RF and M/W Network Theory and Analysis	4	18/24	193	12
4	RF/Microwave Components and Devices	8	24/24	317	8
5	Microwave Generators	5	23/24	166	—
6	RF Design Practices	10	24/24	426	10
7	Microwave Antennas and Propagation	3	23/24	162	2
8	RF/Microwave Measurements	6	22/24	181	—
Total		45		1926	56 problems

Read the marks column, not the hours column. Chapter 6 is **22.4 % of every mark ever set** against a 22 % hour share, so it is the one chapter that is exactly as big as it looks, and it is the biggest. Chapter 1 is **3 syllabus hours** and **9.7 % of the marks**, 1.5 times its hour share, because “compare microwave with conventional circuits” alone is asked in **18 of the 24 papers**. Chapter 4 is the opposite: 8 hours for 16.7 %, close to fair. Nothing here is safe to skip, because the least-asked chapter, Chapter 3, still appears in **18 of 24 papers**.

The marks column sums to 1926 against a theoretical $24 \times 80 = 1920$, because a two-part question whose halves belong to different chapters, such as 82 Bh’s [4 + 8] on hazards *and* on a BTS survey instrument, is counted into both. The percentage each chapter prints for itself was computed on an archive total of 1902, taken before the Chapter-7 recount, so every one of them runs a shade high. Treat these as shares, not as an audit.

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1. Introduction

3 hrs · asked in **19 of 24 papers** · 3 syllabus sub-topics · core foundational chapter → qualitative comparisons, equivalent circuits, frequency tables, and system protocols.

READ THIS FIRST

This chapter provides guaranteed theory marks in almost every paper (typically 8–14 marks). The examiner repeatedly asks three core questions:

1. **Conventional low frequency vs RF/microwave circuit behavior** (asked in 18 papers) → contrast lumped vs distributed analysis ($d \ll \lambda$ vs $d \gtrsim 0.01\lambda$), breakdown of Kirchhoff's laws, skin effect, and equivalent circuit parasitics of R , L , C .
 2. **Detailed real-world protocols and specifications of microwave communication systems** (asked in 11 papers) → prepare full parametric tables for GSM 900/1800, Satellite C/Ku-band, Terrestrial Microwave link, and WiFi 802.11.
 3. **IEEE microwave frequency band classification** (asked in 7 papers) → memorise the letter-code frequency boundaries (L, S, C, X, Ku, K, Ka, mmWave) and their primary application.
-

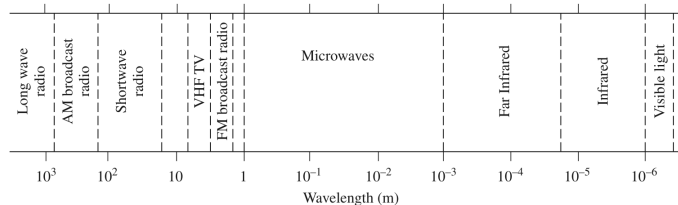
1.1 Standard Frequency Bands and Features

How is microwave frequency band classified by IEEE? Enumerate features, advantages and disadvantages of microwaves 2+4 3+3 4+4 5+5

TOP 7/24 70 Bh, 70 Ma, 76 Bh, 80 Ba, 79 Bh, 82 Ba, 69 Bh · standard frequency classification

Microwave definition and spectrum

- **Frequency range:** 300 MHz to 300 GHz.
- **Wavelength range:** $\lambda = c/f = 1$ m down to 1 mm (free space).
- **Millimeter waves (mmWave):** 30 GHz to 300 GHz ($\lambda = 10$ mm to 1 mm).
- **Sub-millimeter / Terahertz:** 300 GHz to 3 THz ($\lambda < 1$ mm).



Wavelength in metres: microwaves span 1 m to 1 mm, between VHF/FM and far infrared. Pozar Fig 1.1, bar only: its band list uses older letter-band edges, so quote the IEEE table below.

IEEE Radar / Microwave Band Designations

Band	Frequency	Primary Uses
UHF	300–1000 MHz	TV, GSM, GPS, RFID
L	1–2 GHz	Mobile (GSM/LTE), GPS, ATC radar
S	2–4 GHz	Wi-Fi (2.4 GHz), Bluetooth, MW oven
C	4–8 GHz	Satellite TV, weather radar
X	8–12 GHz	Military radar, marine radar, deep space
Ku	12–18 GHz	Direct Broadcast Satellite (DBS TV)
K	18–27 GHz	Police speed radar, water vapor absorption
Ka	27–40 GHz	High-throughput satellite, 5G NR
V	40–75 GHz	60 GHz unlicensed WiGig, inter-sat
W	75–110 GHz	Automotive radar (77 GHz), imaging
mm	110–300 GHz	Radio astronomy, security imaging

Advantages of microwaves

- **High bandwidth:**
 - Bandwidth is 1–10% of carrier frequency.
 - Carrier at 10 GHz offers 1 GHz bandwidth → accommodates thousands of high-speed data channels.
- **High directivity and gain:**
 - Antenna gain $G \propto (D/\lambda)^2$.
 - Small physical antenna diameter D produces sharp, pencil-beam radiation → high EIRP, low interference.
- **Compact physical size:**
 - Component dimensions scale with λ ($\lambda/2$, $\lambda/4$).
 - RF circuits fit on compact MMICs and microstrip boards.
- **Ionospheric transparency:**
 - Waves above 100 MHz penetrate ionosphere without total reflection → enables satellite and space communication.
- **Dielectric (dipolar) heating:**
 - Polar water molecules follow the alternating field, and the friction of that rotation heats the material → cooking, industrial drying, vulcanisation.
 - verified/added Not a “resonance of water”: that is a **Debye relaxation** peaking near 20 GHz. 2.45 GHz is an **ISM** band whose absorption is weak enough to penetrate centimetres — a true resonance would cook only the surface.

Disadvantages of microwaves

- **Atmospheric attenuation:**
 - Rain fade and gaseous absorption (H_2O at 22.2 GHz, O_2 at 60 GHz) degrade signals over long paths.
- **Line of sight (LOS) required:**
 - Cannot bend around obstacles (diffraction is minimal); shadows and terrain block transmission.
- **High manufacturing cost:**
 - Tight mechanical tolerances ($< \mu\text{m}$); requires GaAs, InP, or GaN instead of cheap silicon.
- **Complex circuit analysis:**
 - Lumped analysis fails; distributed transmission lines, scattering parameters, and electromagnetic field equations required.

1.2Behaviour of Circuits: Conventional vs RF/Microwave Bands

Compare the behavior of microwave circuits against conventional low frequency circuits. Differentiate lumped vs distributed analysis 4 5 6 8 10

TOP 18/24 70 Bh, 70 Ma, 71 Ma, 72 Ash, 72 Ma, 73 Ma, 75 Bh, 76 Bh, 77 Ch, 78 Ch, 80 Ba, 80 Bh, 80 Ch, 81 Bh, 81 Ba, 82 Bh, 82 Ba, 69 Bh · top

repeated topic

- The fundamental governing criterion: d/λ
- Let d = maximum physical dimension of the circuit.
 - Let $\lambda = c/f$ = operating wavelength.
 - Low frequency** ($d \ll \lambda$, typically $d < 0.01\lambda$):
 - Propagation time of EM wave across the circuit is negligible compared to wave period $T = 1/f$.
 - Phase change across any conductor:
 $\Delta\theta = \beta d = \frac{2\pi d}{\lambda} \approx 0$.
 - Voltage is instantaneous and uniform along an ideal node.
 - Kirchhoff's Laws (KVL, KCL) hold strictly.**
 - RF / Microwave** ($d \gtrsim 0.01\lambda$ to $d \approx \lambda$):
 - Propagation delay cannot be ignored; wave travel causes finite phase shift along conductors.
 - Voltage and current become functions of **both position z and time t** : $v(z, t), i(z, t)$.
 - Reflections arise at every impedance discontinuity.
 - Wires exhibit distributed inductance and capacitance; open lines radiate energy like antennas.
 - Distributed transmission line theory** and Maxwell's equations replace KVL/KCL.

Comparison: Low Frequency vs RF/Microwave Systems

Parameter	Low Frequency / Conventional	RF / Microwave
Circuit theory	Lumped elements (R, L, C discrete blocks).	Distributed elements (R', L', G', C' per meter).
Laws applied	KVL, KCL, Ohm's law.	Maxwell's equations, Telegrapher's wave equations.
Variables	Absolute voltage V , current I .	Incident / reflected wave amplitudes (a, b), S-parameters, power.
Lines used	Wires, twisted pair, PCB traces.	Coaxial cables, Waveguides, Microstrip, Stripline.
Interconnects	Ideal short circuits (zero phase delay).	Transmission lines with characteristic impedance Z_0 and phase delay βz .
Discontinuities	No reflection; current is continuous.	Causes wave reflection, standing waves ($VSWR > 1$).
Radiation	Negligible; energy guided in wire.	Significant radiation if unshielded ($d \approx \lambda/4$).
Tuning	Variable lumped capacitors/inductors.	Reactive stubs, slides, screws, waveguide plungers.
Active devices	BJTs, Op-Amps, Si MOSFETs.	GaAs MESFETs, HEMTs, Gunn diodes, Klystrons, Magnetrons, TWTs.
Measurement	Voltmeter, ammeter, oscilloscope.	Power meter, Bolometer, VSWR meter, Spectrum/Vector Network Analyzer.

Why are S-parameters used in microwave network analysis instead of Z, Y, or h-parameters? 3 4 5

TOP 10/24 70 Bh, 70 Ma, 71 Ma, 72 Ash, 73 Ma, 78 Ch, 80 Ba, 80 Ch, 79 Bh, 81 Bh · core justification

Limitations of H, Z, Y parameters at microwave bands

- **Requires open and short circuits:**
 - Z-parameters require open circuits ($I = 0$):
 $Z_{11} = V_1/I_1|_{I_2=0}$.
 - Y-parameters require short circuits ($V = 0$):
 $Y_{11} = I_1/V_1|_{V_2=0}$.
 - h-parameters require both short and open circuits.
- **Ideal opens and shorts are physically impossible:**
 - An open-circuited line at microwave frequencies radiates energy like an open-ended antenna → radiation loss causes $I \neq 0$.
 - Stray fringe capacitance exists at open ends.
 - A short circuit has finite lead inductance ($L_{\text{lead}} > 0$)
→ $\omega L \gg 0$, not a true short ($V \neq 0$).
- **Active device instability:**
 - Transistors (MESFETs, BJTs) frequently become conditionally unstable and oscillate violently under pure open/short terminations.
- **Direct voltage/current measurement impossible:**
 - Probes perturb fields; standing waves make V and I position-dependent.

Why S-parameters succeed

- **Terminated in matched characteristic impedance Z_0 :**
 - S-parameter measurement terminates ports in matched loads ($Z_L = Z_0 = 50 \Omega$).
 - Matched load absorbs all incident energy → zero reflection ($a_2 = 0$).
 - Precision 50Ω broadband loads are easy to manufacture.
- **No parasitic radiation or lead reactance:**
 - Complete absorption eliminates open-circuit radiation and short-circuit inductive ringing.
- **Active devices remain stable:**
 - Resistive 50Ω loading prevents self-oscillation of active transistors during testing.
- **Based on traveling power waves:**
 - Relies on incident wave a and reflected wave b :

$$a = \frac{V + Z_0 I}{2\sqrt{Z_0}}, \quad b = \frac{V - Z_0 I}{2\sqrt{Z_0}}$$

- Directly measurable using directional couplers and Vector Network Analyzers (VNA).

1.3 High-Frequency Parasitics in Passive Components (R, L, C)

Explain the characteristic behavior of passive components (resistor, inductor, capacitor) at RF/microwave frequencies 4 6 8

HOT 4/24 70 Bh, 77 Ch, 80 Bh, 82 Bh · equivalent circuits and frequency responses

1. Resistors at RF/Microwave

- **Parasitic effects:**
 - **Lead inductance L :** metal leads introduce serial inductance ($L \approx 1\text{--}5 \text{ nH}$).
 - **Stray capacitance C_a :** capacitive coupling between resistor terminals.
 - **Skin effect:** current concentrates in outer conductor layer ($\delta = 1/\sqrt{\pi f \mu \sigma}$) → cross-sectional area decreases → AC resistance increases: $R_{ac} \propto \sqrt{f}$.
- **Equivalent circuit:**
 - Resistor R in series with lead inductance L , shunted by terminal capacitance C_a .
 - At low freq: $Z \approx R$.
 - At medium freq: L causes inductive impedance rise.
 - At high freq: shunt C_a bypasses R , causing impedance to drop.

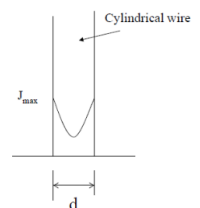
Resistor equivalent model

Resistance

AC → B-fields → Force on charge carriers

Skin depth

$$\delta = \sqrt{\frac{1}{\pi f \mu \sigma}}$$



Equivalent circuit: ideal R with series lead L and parallel parasitic C_a .

2. Inductors at RF/Microwave

– Parasitic effects:

- **Winding resistance** R_s : coil wire resistance, increased by skin and proximity effects ($R_{ac} \propto \sqrt{f}$).
- **Inter-turn capacitance** C_d : electric field lines between adjacent coil turns create distributed parallel capacitance.

– Self-Resonant Frequency (SRF), f_{SR} :

$$f_{SR} = \frac{1}{2\pi\sqrt{LC_d}}$$

– Behavioral regions:

- $f < f_{SR}$: inductive reactance dominates ($\omega L > 1/\omega C_d$); X_L increases with f .
- $f = f_{SR}$: parallel resonance $\rightarrow$ impedance peaks to a very large purely resistive value; quality factor $Q \rightarrow 0$.
- $f > f_{SR}$: **Inductor acts as a capacitor!** Inter-turn capacitance bypasses the coil ($\omega C_d > 1/\omega L$).

3. Capacitors at RF/Microwave

– Parasitic effects:

- **Lead inductance** L_s (**ESL**): connecting leads and plates introduce series inductance ($L_s \approx 1\text{--}2$ nH).
- **Dielectric loss** R_p : leakage resistance across dielectric ($\tan \delta = \epsilon''/\epsilon'$).
- **Equivalent series resistance** R_s (**ESR**): resistance of plates and leads.

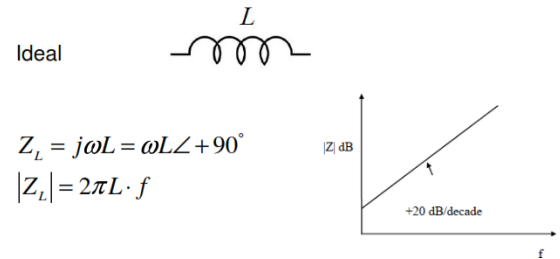
– Series Self-Resonant Frequency, f_{SR} :

$$f_{SR} = \frac{1}{2\pi\sqrt{L_s C}}$$

– Behavioral regions:

- $f < f_{SR}$: capacitive reactance dominates ($1/\omega C > \omega L_s$); impedance drops with f .
- $f = f_{SR}$: series resonance $\rightarrow$ impedance reaches a minimum equal to ESR (R_s); optimal for bypass/decoupling.
- $f > f_{SR}$: **Capacitor acts as an inductor!** ESL dominates ($\omega L_s > 1/\omega C$); impedance rises with f .

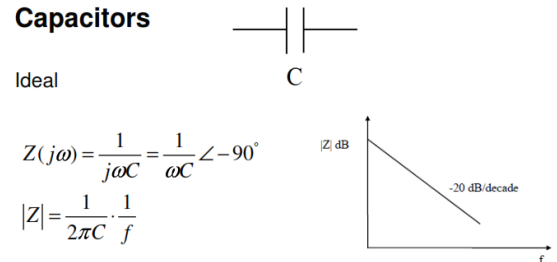
Inductor equivalent model



Inductor equivalent circuit: inductance L with series loss R_s shunted by distributed winding capacitance C_d .

Capacitor equivalent model

Capacitors



Capacitor equivalent circuit: ideal C with series lead inductance L_s , series resistance R_s , and dielectric leakage R_p .

1.4 Comparison with Other Wave Types

Compare and contrast microwave signals with acoustic and seismic waves 45

HOT 3/24 72 Ma, 72 Ash, 75 Bh · physical wave mechanisms

Physical wave nature

- **Microwave signals:**
 - Electromagnetic transverse waves ($\vec{E} \perp \vec{H} \perp \vec{k}$).
 - Propagates in vacuum (no medium required).
 - Propagation speed: $c \approx 3 \times 10^8$ m/s.
 - Frequencies: 300 MHz to 300 GHz; wavelengths: 1 m to 1 mm.
- **Acoustic (sound) waves:**
 - Mechanical longitudinal pressure waves (compression / rarefaction).
 - Requires material medium (gas, liquid, solid); zero propagation in vacuum.
 - Speed: ~ 343 m/s in air, ~ 1500 m/s in water.
 - Frequencies: 20 Hz to 20 kHz (ultrasound > 20 kHz).
- **Seismic waves:**
 - Mechanical waves traveling through Earth's geological strata.
 - Body waves (P -waves longitudinal, S -waves transverse shear) + Surface waves (Rayleigh, Love).
 - Speed: 2–8 km/s depending on rock density.
 - Infrasonic frequencies: 0.01 Hz to 50 Hz.

Comparison Table

Property	Microwaves	Acoustic	Seismic
Wave type	EM (Transverse)	Mechanical (Longitudinal)	Mechanical (Long. & Shear)
Medium	Not required (vacuum)	Mandatory (gas/liq/solid)	Solid earth / rock
Velocity	3×10^8 m/s	~ 343 m/s (air)	2000–8000 m/s
Frequency	10^8 – 10^{11} Hz	20–20000 Hz	0.01–50 Hz
Wavelength	1 mm – 1 m	17 mm – 17 m	100 m – 100 km
Information	Massive data rate (Gbps)	Audio/speech, sonar	Geological fault analysis
Absorption	Rain / moisture fade	Viscous fluid damping	Geometric spreading, rock damping
Application	Radar, satellite, 5G	Audio, ultrasound NDT	Earthquake detection, oil survey

1.5 Microwave Applications and Real-World System Protocols

Prepare the detail real-world protocols and specifications of any 3 (or 5) major microwave communication systems 4+6 2+6 4+4 8 10

TOP 11/24 70 Bh, 70 Ma, 76 Bh, 77 Ch, 78 Ch, 79 Bh, 80 Bh, 81 Ba, 81 Bh, 82 Bh, 82 Ba · top repeated exam requirement

General microwave communication system architecture

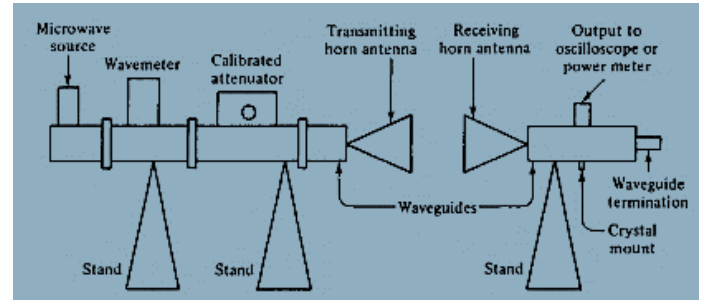
- **Transmitter (Tx):** Baseband source → Modulator (QPSK/QAM) → Up-converter (Mixer + Local Oscillator) → Bandpass Filter → High Power Amplifier (HPA: TWT or GaN SSPA) → Duplexer/Circulator → Directional Antenna.
- **Channel:** Free space line-of-sight (LOS) path with atmospheric attenuation and multipath fading.
- **Receiver (Rx):** Directive Antenna → Duplexer → Low Noise Amplifier (LNA) → Down-converter (Mixer + LO) → IF Filter / Amplifier → Demodulator → Baseband output.

Major application areas

- **Telecommunications:** Cellular BTS, satellite, microwave LOS trunk radio.
- **Radar & Navigation:** Air traffic control (ASR), weather radar, GPS/GLONASS.
- **Industrial & Domestic:** Microwave ovens (2.45 GHz), rubber vulcanization, food drying.
- **Medical:** Hyperthermia cancer treatment, microwave diathermy, tomography.
- **Scientific:** Radio astronomy, remote sensing, atmospheric sounding.

Detailed Specifications and Protocols of Five Major Microwave Communication Systems

Microwave system block diagram



Block diagram of general microwave communication link: Tx upconversion, HPA, antenna, propagation, LNA, downconversion, and detector.

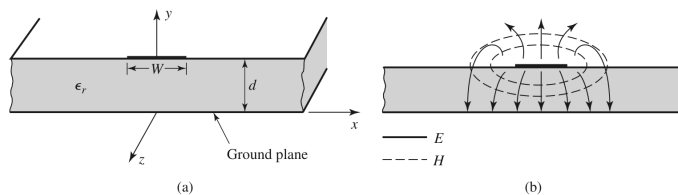
Parameter	1. GSM 900 / 1800	2. Satellite (Ku-band)	3. Terrestrial MW Link	4. Wi-Fi (802.11ac)	5. Airport Radar (ASR)
Operating band	UHF / L-band (900/1800 MHz)	Ku-band (12–18 GHz)	Lower 6 GHz band (L6)	U-NII, 5 GHz only	S-band (2.7–2.9 GHz)
Uplink / Tx Freq	890–915 MHz (GSM 900)	14.0–14.5 GHz	5.925–6.425 GHz (go)	5.15–5.85 GHz	2.7–2.9 GHz pulsed
Downlink / Rx Freq	935–960 MHz (GSM 900)	11.7–12.2 GHz	6.425–6.925 GHz (return)	5.15–5.85 GHz	Reflected echo
Channel bandwidth	200 kHz per carrier	36–54 MHz transponder	28–56 MHz per channel	20, 40, 80, 160 MHz	1–5 MHz chirp
Duplexing mode	FDD (45 MHz duplex split)	FDD (2–2.5 GHz split)	FDD (frequency separation)	TDD (CSMA/CA)	Monostatic simplex (pulsed)
Multiple access	TDMA (8 time slots / carrier)	FDMA / TDMA / DVB-S2	Point-to-Point dedicated	CSMA/CA, OFDMA	Pulse timing / PRF
Modulation scheme	GMSK ($BT = 0.3$)	QPSK, 8PSK, 16-APSK	64-QAM, 256-QAM, 1024-QAM	OFDM w/ 256-QAM	Linear FM Chirp / Barker
Data rate	270.833 kbps gross	30–90 Mbps per transponder	Up to 1 Gbps per link	Up to 1.3 Gbps	Target range & velocity
Tx Output Power	20–40 W (+43–46 dBm)	50–150 W TWTA	1–10 W (+30–40 dBm)	100–200 mW (+20 dBm)	1–25 kW peak
Antenna type	Sectorized panel (120°)	Parabolic reflector (1.2–3.0 m)	Parabolic dish w/ radome	Printed patch array / dipole	Parabolic cosecant-squared
Antenna gain	15–18 dBi	38–45 dBi	35–44 dBi	3–6 dBi	32–35 dBi
Coverage / Range	1–35 km radius	36000 km (GEO orbit)	30–60 km per hop	30–100 m indoor	60–80 NM (100–150 km)

1.6 Microstrip Lines vs Strip-Lines

Differentiate between microstrip lines and strip-lines. Discuss their transmission properties 5

PIN 1/24 71 Ma · planar transmission lines

Microstrip Line



(a) geometry, (b) field lines: part in the substrate, part in air, hence quasi-TEM. Pozar Fig 3.25.

– Geometry:

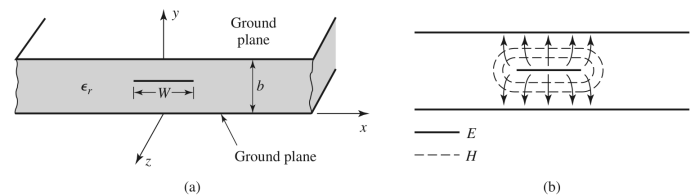
- Conducting strip of width W on top of dielectric substrate (h, ϵ_r).
- Ground plane on the bottom; upper surface exposed to air (ϵ_0).

– Wave mode:

- **Quasi-TEM mode:** fields exist partly in dielectric (ϵ_r) and partly in air ($\epsilon_0 = 1$).
- Phase velocities in the two media differ → pure TEM cannot be supported.
- Characterized by effective dielectric constant ϵ_{eff} :

$$1 < \epsilon_{\text{eff}} < \epsilon_r, \quad v_p = \frac{c}{\sqrt{\epsilon_{\text{eff}}}}$$

Strip-line



(a) geometry, (b) field lines: all inside one dielectric between two ground planes, hence pure TEM. Pozar Fig 3.22.

– Geometry:

- Conducting strip embedded symmetrically inside dielectric medium between **two parallel ground planes**.

– Wave mode:

- **Pure TEM mode:** dielectric is completely homogeneous around the conductor.
- Phase velocity is frequency-independent:

$$v_p = \frac{c}{\sqrt{\epsilon_r}}$$

- **Dispersion:** Moderate dispersion at higher frequencies as fields pull into substrate.
- **Fabrication:** Very easy; standard photolithography; surface accessible for discrete component mounting (SMD, diodes, FETs).
- **Radiation:** Unshielded; radiates at bends and discontinuities.
- **Dispersion:** Non-dispersive; wide broadband operation.
- **Shielding:** Completely enclosed between two ground planes → **zero radiation loss**, no crosstalk with adjacent circuits.
- **Fabrication:** Difficult; multi-layer sandwich lamination; impossible to mount surface components after manufacturing.
- **Applications:** High-performance broadband passive circuits (couplers, filters, power dividers).

1.7 PYQ Mapping

Theory questions – covered in this document

- IEEE microwave band classification; features, (+) and (-) of microwaves – §1.1. 69 Bh 4, 80 Ba 2+4, 79 Bh 3+3, 76 Bh 5+5
- Conventional low-frequency vs RF/microwave circuit behaviour; lumped vs distributed analysis – §1.2. **The most repeated topic in the chapter, 18 papers.**
- Why S-parameter based analysis is a must – §1.2. Full S-parameter theory is Chapter 3; only the justification is asked here. 81 Bh 5+3
- Behaviour of passive components (R , L , C) at microwave bands – §1.3, with equivalent circuits and self-resonance.
- Microwaves vs acoustic and seismic waves – §1.4. 72 Ma 5, 72 Ash 5, 75 Bh 8
- Areas of application; real-world protocols and specifications of major microwave communication systems – §1.5. **Prepare the full parametric table; asked in 11 papers.**
- Microstrip vs strip-lines – §1.6. Kept here because §1.1–1.6 introduce the guiding structures; Chapter 2 (§2.1) cross-references it rather than repeating it. 71 Ma 5

Numerical content

- **This chapter has no numerical component.** Every Chapter 1 question across all 24 papers is qualitative – classification, comparison, enumeration or specification. There is no numerical companion for Chapter 1.
- The only arithmetic worth rehearsing is the one-line conversion $\lambda = c/f$ used to place a band, and the antenna-gain scaling $G \propto (D/\lambda)^2$ quoted in §1.1.

Syllabus coverage check

- Standard frequency bands – §1.1 ✓
- Behaviour of circuits at conventional and RF/microwave bands – §1.2, §1.3, §1.4 ✓
- Microwave applications – §1.5 ✓

2. RF and M/W Transmission Lines

6 hrs · asked in **all 24 papers** · **297 marks**, 12.4 per paper · 4 syllabus sub-topics → the Smith chart chapter: almost every mark here is a graphical construction.

READ THIS FIRST

This is the second-heaviest chapter in the subject and the most predictable one. Of its 297 marks, roughly 230 are stub-matching constructions, and **double-stub tuning alone appears in 17 of the 24 papers** — the single most reliably asked item in RF and Microwave Engineering. Learn the construction, not the answers.

1. **Double-stub matching** (17 papers, usually 8–12 marks) → master one procedure: spacing circle, constant- g arc, rotate, cancel. It works for every variant the examiner prints (stub spacing $\lambda/8$ or $3\lambda/8$, short or open, first stub at the load or offset).
2. **The S-matrix rider** (10 papers) → “prepare the S-matrix of the matched network” is tacked onto most stub questions and is worth 2–6 marks for two lines of work. See §2.8; do not leave it.
3. **Single-stub matching** (6 papers) and **the immittance chart** (5 papers) → both are short, self-contained and frequently the easy half of a long question.

Numerical work is in the companion, *Ch2 – RF and MW Transmission Lines – Numerical Problems & Solutions*, where all 24 PYQ constructions are solved step by step.

2.1 Types of Transmission Lines

Differentiate the behaviour of a transmission line at conventional low frequency and at microwave bands. Compare microwave transmission lines with conventional low-frequency lines 4 5 10

HOT 3/24 69 Bh, 71 Ma, 79 Ch · builds on §1.2; here the focus is the *line* rather than the circuit

A transmission line is a guiding structure

- At low frequency, power is thought of as travelling **in the wire**; the pair of conductors merely completes a circuit.
- At microwave frequency, power travels **in the $\vec{E}$ and $\vec{H}$ fields** between and around the conductors. Any structure that guides that field from place to place is a transmission line.
- The governing criterion is again d/λ (§1.2): once the line length is a noticeable fraction of λ , voltage and current are functions of position.

Classification

- Two-conductor (TEM) lines** — support a pure TEM mode, no low-frequency cut-off:
 - Two-wire open line, twisted pair (UTP / STP), coaxial cable, stripline.
- Quasi-TEM lines** — inhomogeneous dielectric, so no exact TEM:
 - Microstrip, coplanar waveguide (CPW), slot line, fin line.
- Single-conductor (waveguide) lines** — TE/TM modes only, and a **cut-off frequency** below which nothing propagates:
 - Rectangular and circular waveguide, dielectric waveguide, optical fibre.
- Balanced vs unbalanced**: two-wire and twisted pair are balanced (equal and opposite currents, common-mode noise cancels); coaxial is unbalanced (signal referred to the grounded outer). A **balun** converts between them.

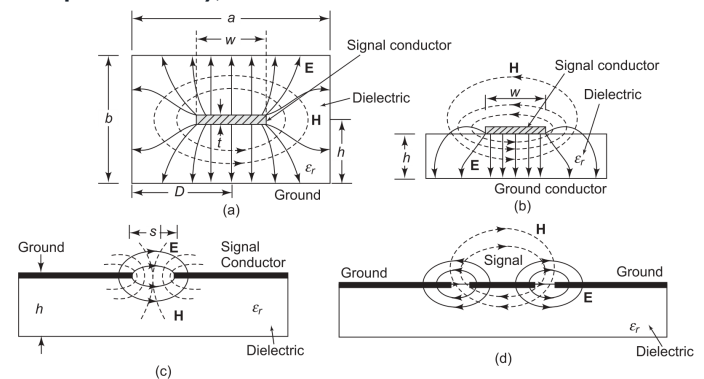
Choosing a line — the four practical trade-offs

- Loss**: waveguide < coax < microstrip at a given frequency.
- Power handling**: waveguide (kW–MW) $\gg$ coax $\gg$ microstrip (W).
- Bandwidth**: TEM lines are broadband; waveguide is limited by cut-off and the next higher mode.
- Cost and integration**: microstrip is photolithographic and near-free; waveguide is machined and expensive.

Low-frequency line vs microwave line

Aspect	Conventional (LF)	RF / Microwave
Model	Ideal short; length irrelevant.	Distributed R', L', G', C' per metre.
Analysis	KVL / KCL, lumped.	Telegrapher's equations, Maxwell.
V, I	Uniform along the line.	$v(z, t), i(z, t)$; standing waves.
Loss	Ohmic ($I^2 R$) only.	Conductor ($\propto \sqrt{f}$, skin effect), dielectric ($\tan \delta$), and radiation.
Reflection	None considered.	At every discontinuity; measured by Γ , VSWR.
Typical lines	Wire, twisted pair, PCB trace.	Coax, waveguide, microstrip, stripline.
Termination	Matching not needed.	Must terminate in Z_0 or match with a tuner.
Joints	Any solder joint.	Precision connectors (SMA, N, APC-7); a bad joint is a reflector.

The planar family, in cross-section



(a) stripline, (b) microstrip, (c) slot line, (d) coplanar strips, with E (solid) and H (dashed). Only (a) has one homogeneous dielectric round the conductor, so only (a) is pure TEM. Das & Das Fig 3.27.

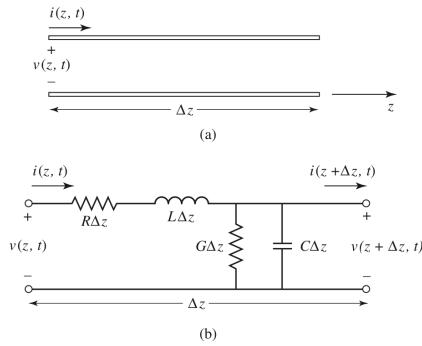
→ For microstrip vs stripline (asked 71 Ma), see §1.6 — geometry, quasi-TEM vs pure TEM, dispersion, shielding and fabrication are compared there and are not repeated.

2.2 Transmission Line Theory

Derive the transmission line (telegrapher's) equations and obtain the characteristic impedance, reflection coefficient, VSWR and input impedance **4 6 8**

HOT 3/24 69 Bh, 79 Ch, 71 Ma · the toolkit every numerical in this chapter uses

Distributed model and the telegrapher's equations



(a) v, i on a length Δz ; (b) its equivalent cell. Draw (b) before writing KVL and KCL.

Pozar Fig 2.1.

- A length Δz of line is modelled by series $R'\Delta z, L'\Delta z$ and shunt $G'\Delta z, C'\Delta z$, all **per unit length**.
- Applying KVL and KCL to that cell and letting $\Delta z \rightarrow 0$:

$$\frac{\partial v}{\partial z} = -R'i - L'\frac{\partial i}{\partial t}, \quad \frac{\partial i}{\partial z} = -G'v - C'\frac{\partial v}{\partial t}$$

- In phasor form these decouple into the **wave equations**

$$\frac{d^2 V}{dz^2} = \gamma^2 V, \quad \frac{d^2 I}{dz^2} = \gamma^2 I$$

- **Propagation constant and characteristic impedance:**

$$\gamma = \alpha + j\beta = \sqrt{(R' + j\omega L')(G' + j\omega C')}$$

$$Z_0 = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}}$$

- Solution: $V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{+\gamma z}$ – a forward and a reflected travelling wave.

Lossless line ($R' = G' = 0$) – used for every PYQ in this chapter

- $\alpha = 0, \quad \beta = \omega\sqrt{L'C'}, \quad Z_0 = \sqrt{L'/C'}$ (purely real).
- $\lambda = 2\pi/\beta = v_p/f, \quad v_p = 1/\sqrt{L'C'} = c/\sqrt{\epsilon_r}$.

Distortionless line

- Condition: $R'C' = L'G'$. Then $\alpha = \sqrt{R'G'}$ is frequency-independent and $\beta = \omega\sqrt{L'C'}$ is linear in ω , so all frequencies travel at the same v_p and the pulse shape survives.

Terminated line: the four quantities every problem needs

- **Load reflection coefficient**

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{z_L - 1}{z_L + 1}, \quad z_L = \frac{Z_L}{Z_0}$$

- **At a distance d toward the generator**, only the phase changes on a lossless line:

$$\Gamma(d) = \Gamma_L e^{-j2\beta d}, \quad |\Gamma(d)| = |\Gamma_L|$$

This is why the Smith chart works: moving along the line is a **rotation at constant radius**.

- **VSWR**

$$S = \frac{V_{\max}}{V_{\min}} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}, \quad 1 \leq S \leq \infty$$

- **Input impedance**

$$Z_{in}(d) = Z_0 \frac{Z_L + jZ_0 \tan \beta d}{Z_0 + jZ_L \tan \beta d}$$

- **Return loss** = $-20 \log_{10} |\Gamma|$ dB; **mismatch loss** = $-10 \log_{10} (1 - |\Gamma|^2)$ dB.

Special line lengths – memorise these

Case	Input impedance
Matched $Z_L = Z_0$	$Z_{in} = Z_0$ for any length; $\Gamma = 0, S = 1$.
Short $Z_L = 0$	$Z_{in} = jZ_0 \tan \beta d$ – purely reactive, inductive for $d < \lambda/4$.
Open $Z_L = \infty$	$Z_{in} = -jZ_0 \cot \beta d$ – purely reactive, capacitive for $d < \lambda/4$.
$d = \lambda/2$	$Z_{in} = Z_L$ – the line repeats every half wavelength.
$d = \lambda/4$	$Z_{in} = Z_0^2/Z_L$ – the quarter-wave transformer ; it inverts , so a short looks open.
$d = \lambda/8$	$Z_{in} = Z_0 \frac{Z_L + jZ_0}{Z_0 + jZ_L}$ ($\tan \beta d = 1$).

A shorted or open stub is therefore a *pure reactance whose value you choose by cutting it to length* – that single fact is the whole basis of stub matching.

2.3 The Smith Chart

Explain the construction of the Smith chart. What does each family of curves represent, and how is it read? 4 5 10

HOT 5/24 72 Ash, 75 Bh, 72 Ma, 81 Ba, 80 Ch · counted together with the immittance chart in §2.4

What the chart is

- A plot of **normalised impedance** $z = r + jx$ mapped onto the complex Γ plane by the bilinear transform

$$\Gamma = \frac{z - 1}{z + 1}, \quad z = \frac{1 + \Gamma}{1 - \Gamma}$$

- Because $|\Gamma| \leq 1$ for a passive load, the entire right half of the z plane folds into the **unit circle** – an infinite plane on one sheet of paper.

The two families of curves

- Constant resistance r :** circles centred on $\left(\frac{r}{1+r}, 0\right)$ with radius $\frac{1}{1+r}$. All pass through the open-circuit point $\Gamma = +1$.
- Constant reactance x :** arcs centred on $\left(1, \frac{1}{x}\right)$ with radius $\left|\frac{1}{x}\right|$. Upper half = $+x$ (**inductive**), lower half = $-x$ (**capacitive**).
- The two families are **orthogonal** everywhere they cross.

Landmarks

- Centre** $z = 1$: matched, $\Gamma = 0$, $S = 1$.
- Left** $z = 0$: short circuit, $\Gamma = -1$.
- Right** $z = \infty$: open circuit, $\Gamma = +1$.
- Rim** $|\Gamma| = 1$: pure reactance, total reflection, $S = \infty$.
- Horizontal axis: $x = 0$, i.e. purely resistive. To the right of centre lies $V_{\max}$ (and $r = S$); to the left lies $V_{\min}$ (and $r = 1/S$).

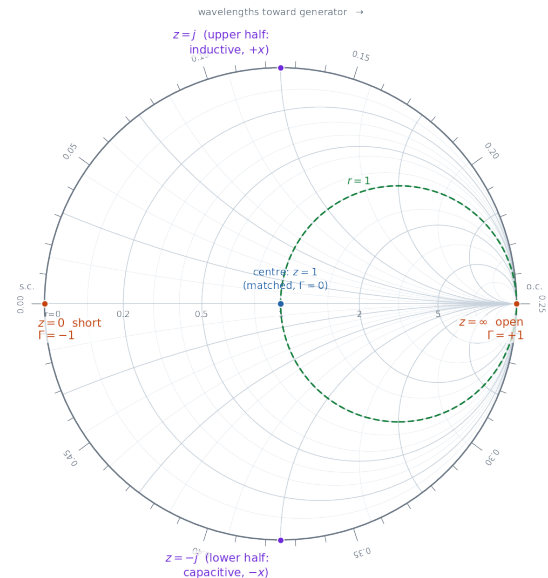
Reading it – the five rules

- Normalise: $z_L = Z_L/Z_0$. Plot it.
- $|\Gamma_L|$ = distance from centre $\div$ chart radius; its angle is read on the **angle of reflection coefficient** scale.
- Draw the **constant- $|\Gamma|$ (SWR) circle** through the point. Read S where it cuts the right-hand real axis.
- Move toward the generator = clockwise**; toward the load = anticlockwise. **One full lap is $\lambda/2$** , so a quarter turn is $\lambda/8$.
- Read distances on the **WTG** (wavelengths toward generator) rim scale, which runs $0 \rightarrow 0.5$ clockwise starting at the short-circuit point.

Why the chart is worth using at all

- It turns the awkward Z_{in} formula into a **rotation**, and $z \rightarrow y$ into a **half-turn** – both done with a compass instead of complex arithmetic.
- Γ , S , Z_{in} , Y_{in} , return loss and the positions of $V_{\max}$ and $V_{\min}$ are all read off one diagram.
- It shows *graphically* which matching solutions exist, and lets you pick the one with the shortest stub or the widest bandwidth – something a formula does not show you.

The Smith chart: constant- r circles and constant- x arcs



every point is one normalised impedance $z = r + jx$; $|\Gamma|$ is its distance from the centre, so one lap of the rim is $\lambda/2$ of line

The impedance chart. Constant- r circles (all through the open point) and constant- x arcs; the rim carries the WTG scale, on which one complete lap is half a wavelength of line.

2.4 The Immittance (Admittance) Chart

Sketch an immittance chart. List the duality parameters and compare the two scales **2 4 5 10**

HOT 5/24 72 Ash, 75 Bh, 72 Ma, 81 Ba, 80 Ch · short, self-contained, and asked five times

The admittance chart

- Since $y = 1/z$, and

$$\Gamma_y = \frac{y-1}{y+1} = -\frac{z-1}{z+1} = -\Gamma_z$$

the admittance chart is **the impedance chart rotated through 180°** (a $\lambda/4$ line length).

- So on a plain Smith chart you convert $z \rightarrow y$ by drawing a diameter through the point and reading the diametrically opposite intersection.

The immittance chart

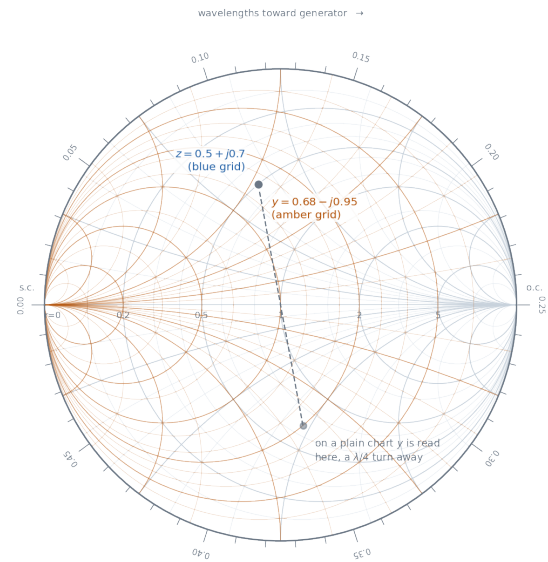
- Both grids printed on one chart**, usually in two colours: the impedance grid and the admittance grid superimposed.
- A single location then carries **two readings at once** — z on one grid, y on the other. The $\lambda/4$ half-turn is built into the printing.
- Why it matters:** stub matching is done in *admittance* (shunt stubs add susceptance) but loads are quoted as *impedance*. Without an immittance chart every problem costs an extra half-turn, and each half-turn is a chance to misread.

Duality parameters

Impedance grid	Admittance grid
$z = r + jx$	$y = g + jb$
Resistance r	Conductance g
Reactance x	Susceptance b
$+x$ inductive (upper)	$+b$ capacitive (upper)
$-x$ capacitive (lower)	$-b$ inductive (lower)
Short $z = 0$ at $\Gamma = -1$	Open $y = 0$ at $\Gamma = -1$
Open $z = \infty$ at $\Gamma = +1$	Short $y = \infty$ at $\Gamma = +1$
Series elements add	Shunt elements add
$r = 1$ circle	$g = 1$ circle

Trap. The sign convention flips: a point in the *upper* half is inductive read as an impedance but *capacitive* read as an admittance. Both are the same physical point.

Immittance chart = impedance grid + admittance grid



one location, two readings: the $\lambda/4$ half-turn that converts $z \rightarrow y$ is printed into the second grid, so no rotation is needed

Immittance chart. The amber grid is the blue grid turned through 180° ; one location reads $z = 0.5 + j0.7$ on the impedance grid and $y = 0.68 - j0.95$ on the admittance grid. The faint point is where y would have to be read on a plain chart — a $\lambda/4$ turn away.

2.5 Impedance Matching: Why, and Which Technique

Why is impedance matching necessary? Compare the different impedance matching techniques 4 5

6

HOT 3/24 78 Ch, 74 Ma, 77 Ch · the theory half of several double-stub questions

Why match

- **Maximum power transfer:** a matched load absorbs all incident power; the fraction reflected is $|\Gamma|^2$.
- **Protect the source:** reflected power returning to a high-power amplifier or a magnetron causes heating, and can destroy it.
- **Remove standing waves:** high VSWR means voltage peaks that cause dielectric breakdown and arcing, and current peaks that cause I^2R loss.
- **Preserve receiver sensitivity:** mismatch at an LNA input degrades the noise figure directly.
- **Avoid distortion:** multiple reflections on a long line arrive delayed, producing ghosting and intersymbol interference.

The matching network must be lossless

- A resistive pad would “match” but dissipate the very power being matched.
- So matching uses only **reactive** elements: line lengths, stubs, lumped L and C .



The general problem: a lossless network between line Z_0 and load Z_L , chosen so the line sees Z_0 . Every technique in the table is one way to build the box. Pozar Fig 5.1.

Comparison of techniques

Technique	How it works	Trade-offs
Quarter-wave transformer	$\lambda/4$ of line with $Z_{01} = \sqrt{Z_0 Z_L}$.	Simplest, but needs a real Z_L – a complex load must first be walked to a $V_{\max}$ or $V_{\min}$. Narrow-band.
L-section (lumped)	Two reactances, series + shunt.	Compact; fine below ~ 1 GHz. Above that, component parasitics (\$1.3) dominate.
Single stub	One shunt (or series) stub at a computed distance d .	Matches any load; only line, so no lumped parts. But d must move if the load changes – awkward to build as an adjustable tuner.
Double stub	Two stubs at fixed spacing; only the lengths are tuned.	Positions are preset, so it makes a practical adjustable tuner. But a range of loads cannot be matched (the forbidden region).
Triple stub	Three fixed stubs.	Removes the forbidden region entirely; used in laboratory slide-screw tuners.

Advantages of double stub over single stub

- Stub **positions are fixed**; matching is done by adjusting lengths only, so it can be built as a mechanical tuner with two sliding shorts.
- No need to break into the line at a computed distance – the connection points are decided at manufacture.
- Retunes for a new load without physically relocating anything.
- **Cost:** a set of load admittances (the forbidden region) is unmatchable, and two stubs mean a slightly narrower bandwidth.

2.6 Single-Stub Tuning

Explain the design of a single-stub shunt tuner, stating all steps with reasoning. When is a short-circuited stub preferred over an open one? 8 10 2+8

HOT 6/24 72 Ash, 74 Bh, 72 Ma, 76 Bh, 79 Ch, 81 Bh · worked instances in the numerical companion

Principle

- Two degrees of freedom are needed to cancel a complex mismatch, and a single stub supplies exactly two: **the distance d from the load to the stub**, and **the stub length ℓ** .
- Work in **admittance**, because a shunt stub *adds* its susceptance:

$$y_{in} = y(d) + y_{stub} = 1 + jb + (-jb) = 1$$

- Step 1 — choose d** so the line admittance has unit conductance: $y(d) = 1 + jb$. Geometrically, walk round the SWR circle until it cuts the $g = 1$ circle. It always cuts it **twice**, so there are always two solutions.
- Step 2 — choose ℓ** so the stub presents $-jb$, cancelling the residual susceptance and leaving $y_{in} = 1$, i.e. $Z_{in} = Z_0$.

Design procedure on the chart

- Normalise $z_L = Z_L/Z_0$ and plot it.
- Half-turn to y_L (or read it on the admittance grid). Note its WTG reading.
- Draw the SWR circle through y_L .
- Move **clockwise** to each intersection with the $g = 1$ circle; call them $y_1 = 1 - jb$ and $y_2 = 1 + jb$. The WTG difference from y_L gives d_1 and d_2 .
- The stub must supply $+jb$ and $-jb$ respectively.
- For the stub length, start at the **termination point of the stub** — the open-circuit point ($y = 0$) for an open stub, the short-circuit point ($y = \infty$) for a shorted stub — and run **clockwise round the rim** to the required susceptance. That arc, read on the WTG scale, is ℓ .

Analytic check (use it to verify the chart)

- With $t = \tan \beta d$ and $Z_L = R_L + jX_L$, the distance satisfies

$$t = \frac{X_L \pm \sqrt{R_L[(Z_0 - R_L)^2 + X_L^2]}/Z_0}{R_L - Z_0}$$

reducing to $t = -X_L/2Z_0$ when $R_L = Z_0$.

- Then $d/\lambda = \frac{1}{2\pi} \tan^{-1} t$ (add 0.5 if negative).
- Stub length from the required susceptance b_s :**

$$\frac{\ell_{open}}{\lambda} = \frac{1}{2\pi} \tan^{-1}(b_s), \quad \frac{\ell_{short}}{\lambda} = \frac{-1}{2\pi} \tan^{-1}\left(\frac{1}{b_s}\right)$$

always reduced modulo 0.5λ .

Short-circuited or open-circuited stub?

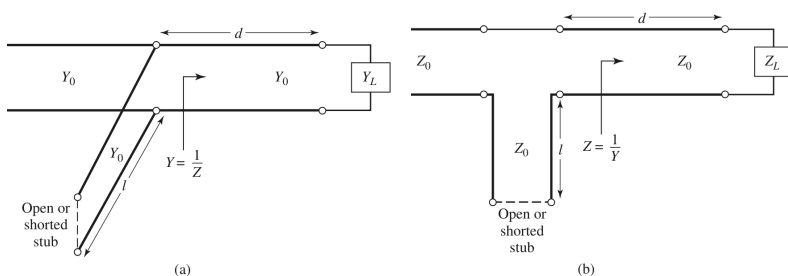
Stub	When it is preferred
Short	Coax and waveguide. A shorting plate is exact and radiates nothing; an open coax end radiates and has fringing capacitance, so its electrical length is not its physical length. Also easier to make adjustable (sliding short).
Open	Microstrip and other planar lines. Ending the strip is free; a true short needs a via drilled and plated through the substrate, which adds inductance and cost.

The two lengths differ by exactly $\lambda/4$: $\ell_{short} = \ell_{open} \pm \lambda/4$.

Which of the two solutions to build

- Prefer the **smaller d** : less line between load and stub means a shorter high-VSWR section, hence lower loss and **wider bandwidth**.
- Prefer the **shorter stub** for the same reason, and for compactness.
- A series stub is used when the line cannot be shunted (some waveguide and coaxial geometries); the construction is identical but done on the *impedance* grid against the $r = 1$ circle.

The circuit, which is what to draw



- (a) Shunt stub:** at distance d from the load the line admittance is $Y = 1/Z = Y_0 + jB$; the stub of length ℓ , open or shorted, adds $-jB$.
- (b) Series stub:** same two unknowns d and ℓ , but the stub is in series, so work in impedance: $Z = Z_0 + jX$ at d , stub supplies $-jX$.
- Mark d from the **load** and ℓ from the stub's **open or shorted end** on the sketch: those are the two answers the question asks for.
- Pozar Fig 5.4.

2.7 Double-Stub Tuning

Explain the steps for impedance matching of a load using a double-stub network. State the advantages over single-stub matching and the limitation of the technique **8 10 12 2+8**

TOP 17/24 69 Bh, 70 Bh, 70 Ma, 71 Bh, 73 Bh, 73 Ma, 74 Ma, 75 Bh, 77 Ch, 78 Ch, 80 Ch, 80 Ba, 80 Bh, 81 Ba, 79 Bh, 82 Ba, 82 Bh · **the single most asked item in the subject**

Principle

- The two stubs sit at **fixed** positions a distance d apart – almost always $\lambda/8$ or $3\lambda/8$ in the papers. Only the two lengths ℓ_1 and ℓ_2 are adjusted, and those are the two degrees of freedom.
- **Stub 1** cannot change the conductance it sees – a shunt susceptance moves the point *along its own constant- g circle*. Its job is to land the point where the intervening d of line will carry it onto the $g = 1$ circle.
- **Stub 2** then simply cancels whatever susceptance is left, exactly as in single-stub matching.

The spacing circle – the key construction

- Define the **spacing circle** as the $g = 1$ circle rotated d toward the load (anticlockwise).
- Any point on it is carried *onto* $g = 1$ by d wavelengths of line.
- So stub 1 must move the point from y_1 to the intersection of its own constant- g circle with the spacing circle. There are generally **two** intersections, hence two solutions.

Design procedure

1. Normalise z_L , plot, half-turn to y_L .
2. If the first stub is offset from the load by d_1 , rotate y_L clockwise by d_1 to get y_1 , the admittance *just before* stub 1.
3. Draw the spacing circle ($g = 1$ rotated d toward the load).
4. Follow the **constant- g circle through** y_1 to its intersections with the spacing circle: y_a and y'_a .
5. $b_1 = \text{Im}(y_a) - \text{Im}(y_1)$ – the susceptance stub 1 must add. Read ℓ_1 off the rim from the stub's termination point.
6. Rotate y_a clockwise by d along its SWR circle; it lands on $g = 1$ at $y_2 = 1 + jb'_2$.
7. $b_2 = -b'_2$; read ℓ_2 off the rim the same way. The network is matched.

Analytic check

- With $y_1 = g_1 + jb_L$ and $t = \tan \beta d$, requiring $\text{Re}(y_2) = 1$ after the spacing gives

$$\text{Re}(y_2) = \frac{g_1(1+t^2)}{(1-tB)^2 + t^2g_1^2} = 1$$

$$\Rightarrow B = \frac{1 \pm \sqrt{g_1(1+t^2) - g_1^2t^2}}{t}, \quad b_1 = B - b_L$$

where B is the total susceptance after stub 1. **Note the leading 1 in the numerator, not g_1 – a common slip.**

- Then $y_a = g_1 + jB$, and $y_2 = \frac{y_a + jt}{1 + jt y_a}$, with $b_2 = -\text{Im}(y_2)$.

The forbidden region – always state this

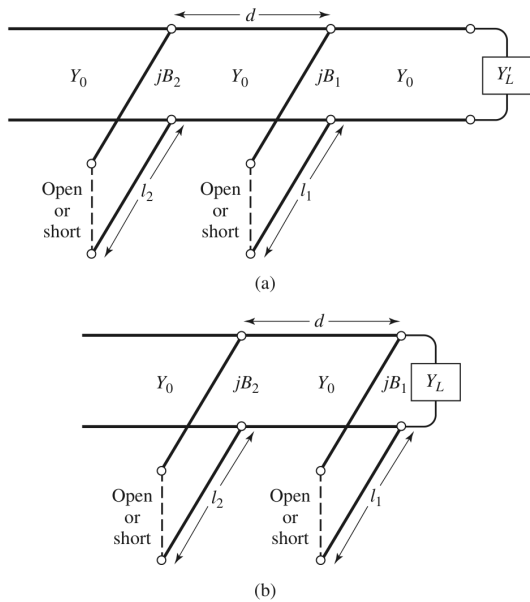
- The square root is real only if

$$g_1 \leq \frac{1}{\sin^2 \beta d}$$

- For $d = \lambda/8$ ($\beta d = 45^\circ$): $g_1 \leq 2$.
For $d = 3\lambda/8$: $g_1 \leq 2$ likewise.
For $d = \lambda/4$: $g_1 \leq 1$.
- Loads whose conductance at the first stub exceeds this **cannot be matched** at all with that spacing. Graphically, their constant- g circle lies wholly inside the spacing circle and never cuts it.
- **The fix:** add a length of line before the first stub (which moves g_1), or use a third stub.
- This is why $\lambda/4$ spacing is avoided in practice and $\lambda/8$ or $3\lambda/8$ is used – and why the examiner keeps specifying $3\lambda/8$.

Trap. Stub spacings near $\lambda/2$ make the two stubs nearly coincident and the tuner hopelessly narrowband, so practical spacings stay between $\lambda/8$ and $3\lambda/8$.

The circuit, which is what to draw



- (a) load an arbitrary distance from the first stub: stub 1 (jB_1 , length ℓ_1) nearer the load, stub 2 (jB_2 , ℓ_2) a fixed d further toward the generator.
- (b) the same tuner with that offset line absorbed into the load, now called Y_L' . This is step 2 of the procedure: rotate y_L by d_1 first, then design as if stub 1 sat at the load.
- Only ℓ_1 and ℓ_2 are adjusted; d is fixed at manufacture, which is the whole advantage over one stub.
- Pozar Fig 5.7.

2.8 S-Matrix of a Matched Network

Prepare the scattering matrix of the network before and after matching 2 2+2 4 6

TOP 10/24 76 Bh, 77 Ch, 78 Ch, 80 Ba, 80 Bh, 81 Ba, 81 Bh, 79 Bh, 82 Ba, 82 Bh · free marks: two lines of work, tacked onto ten stub questions

Before matching — a one-port

- An unmatched load seen from the line is a **one-port**, so its scattering matrix is the single entry

$$[S] = [\Gamma_L] = \left[\frac{Z_L - Z_0}{Z_L + Z_0} \right]$$

- Quote it in polar form, $|\Gamma_L| \angle \theta^\circ$, and give the VSWR alongside — the examiner usually wants both.

After matching — an ideal two-port

- A perfectly matched, lossless, reciprocal two-port has **no reflection at either port**, so $S_{11} = S_{22} = 0$, and all power passes through, so $|S_{12}| = |S_{21}| = 1$:

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- Keeping the electrical length $\theta = \beta \ell$ of the through path:

$$[S] = \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

Either answer earns the marks; state which convention you are using.

Justifying each property (this is where the marks are)

- $S_{11} = S_{22} = 0$: that is the definition of a match — the tuner was designed to make $Z_{in} = Z_0$ looking in at either port.
- **Reciprocity**, $S_{12} = S_{21}$: the network contains only passive, isotropic media (line and stubs), no ferrite and no active device, so $[S]$ is symmetric, $[S] = [S]^T$.
- **Lossless**, so $[S]$ is **unitary**: $[S]^{*T}[S] = [I]$, giving $\sum_k |S_{ki}|^2 = 1$ for every column. With $S_{11} = 0$ this forces $|S_{21}| = 1$.
- **Reactive elements only** means no power is dissipated, only redirected — so all incident power appears at the far port.

Worked shape of the answer

- “Before: $\Gamma_L = (Z_L - Z_0)/(Z_L + Z_0) = 0.45 \angle 60^\circ$, so $[S] = [0.45 \angle 60^\circ]$ and $S = 2.64$.”
- “After: the tuner makes $Z_{in} = Z_0$, so $S_{11} = S_{22} = 0$; the network is reciprocal and lossless, so $S_{12} = S_{21}$ with $|S_{21}| = 1$, giving $[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.”

S-parameter theory in full — properties, derivation, shifting reference planes — is Chapter 3. This band covers only the rider attached to stub questions.

2.9 Microstrip Realisation of a Stub Tuner

Sketch the physical diagram of the matched network using microstrips 2 2+2 4

HOT 4/24 72 Ash, 72 Ma, 77 Ch, 82 Ba · turns λ answers into millimetres

From electrical length to millimetres

- Microstrip is **quasi-TEM**, so the wave sees an effective permittivity between air and substrate:

$$\epsilon_e = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \frac{1}{\sqrt{1 + 12h/W}}$$

- Guide wavelength, and hence every physical length:

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_e}} = \frac{c}{f\sqrt{\epsilon_e}}, \quad \ell_{\text{mm}} = \frac{\ell}{\lambda} \times \lambda_g$$

- Strip width** for a wanted Z_0 (Hammerstad synthesis), with $B = \frac{377\pi}{2Z_0\sqrt{\epsilon_r}}$ and $W/h > 2$:

$$\frac{W}{h} = \frac{2}{\pi} \left[B - 1 - \ln(2B - 1) + \frac{\epsilon_r - 1}{2\epsilon_r} \left(\ln(B - 1) + 0.39 - \frac{0.61}{\epsilon_r} \right) \right]$$

Layout rules the examiner looks for

- Stubs are drawn as **T-junctions off the main line**, at the computed spacing.
- Open stubs** are drawn as a plain truncated strip; **shorted stubs** need a **via to the ground plane**, drawn as a filled circle.
- Label W for the Z_0 line, the stub lengths ℓ_1, ℓ_2 in mm, and the spacing d in mm.
- State the substrate: e.g. **RT/duroid 5880** ($\epsilon_r = 2.2$), **FR-4** ($\epsilon_r = 4.4$), **alumina** ($\epsilon_r = 9.8$), with h .

Worked conversion — carry this shape into the exam

- Take $Z_0 = 50 \Omega$ on FR-4, $\epsilon_r = 4.4$, $h = 1.6$ mm, $f = 3$ GHz.
- Synthesis gives $W/h \approx 1.9$, so $W \approx 3.0$ mm, and $\epsilon_e \approx 3.33$.
- $\lambda_0 = c/f = 100$ mm, so $\lambda_g = 100/\sqrt{3.33} \approx 54.8$ mm.
- A stub of 0.146λ is then $0.146 \times 54.8 \approx \mathbf{8.0}$ mm, and a $3\lambda/8$ spacing is $0.375 \times 54.8 \approx \mathbf{20.5}$ mm.

Practical corrections worth a mark

- Open-end fringing**: an open stub behaves slightly longer than it is, by $\Delta\ell \approx 0.3\text{--}0.4h$. Shorten the drawn stub by that much.
- T-junction discontinuity**: the junction adds shunt susceptance; the reference plane is not exactly the strip centreline.
- Via inductance**: a shorted stub's via is not a perfect short, which is the practical reason open stubs dominate in microstrip.
- Radiation and dispersion** rise with frequency; above about 10 GHz on FR-4 the losses make the design impractical, so low-loss substrates are used.

2.10 PYQ Mapping

Theory questions — covered in this document

- Types of transmission lines; LF line vs microwave line** — §2.1. 69 Bh 4, 79 Ch 10
- Microstrip vs stripline** — §1.6. 71 Ma 5
- Smith chart construction and reading** — §2.3.
- Immittance / admittance chart, duality, scale comparison** — §2.4. 72 Ash 2, 75 Bh 10, 72 Ma 5, 81 Ba 4, 80 Ch 5
- Impedance matching techniques compared; advantages of double over single stub** — §2.5. 78 Ch 4+6, 74 Ma 2+8, 77 Ch 2+8
- Single-stub design steps and reasoning** — §2.6. 79 Ch 10, 81 Bh 5+3+2, 76 Bh 10
- Double-stub design steps; forbidden region** — §2.7. 74 Ma 2+8, 78 Ch 4+6
- S-matrix before and after matching** — §2.8. 10 papers, see the band
- Microstrip realisation of the matched network** — §2.9. 72 Ash, 72 Ma, 77 Ch, 82 Ba

Numerical questions — solved in the numerical companion

- Smith chart reading** (Γ_L , VSWR, Z_{in} , first resistive point): 79 Ch 10 — Problem 1.
- Single stub (6)**: 72 Ash, 74 Bh, 72 Ma, 76 Bh, 79 Ch, 81 Bh — Problems 2–7.
- Double stub (17)**: 69 Bh, 70 Bh, 70 Ma, 71 Bh, 73 Bh, 73 Ma, 74 Ma, 75 Bh, 77 Ch, 78 Ch, 80 Ch, 80 Ba, 80 Bh, 81 Ba, 79 Bh, 82 Ba, 82 Bh — Problems 8–24.

Syllabus coverage check

- Types of transmission lines** — §2.1 ✓

- *Transmission line theory* — §2.2 ✓
- *Smith Chart analysis* — §2.3, §2.4 ✓
- *Impedance transformations and matching analysis* — §2.5–§2.9 and the numerical companion ✓

3. RF and M/W Network Theory and Analysis

4 hrs · asked in **18 of 24 papers** · 2 syllabus sub-topics · **193 marks, 10.1% of the archive** → definitions, one derivation, four matrix properties and one device (the magic tee) that answers half the chapter.

READ THIS FIRST

Chapter 3 is the cheapest 8 to 10 marks in the paper, because the same three asks recur:

1. **Why S-parameters, and how do they differ from h / Z / Y / ABCD?** (13 papers) → one reason list plus one comparison table, §3.1.
2. **Derive the S-parameters of a two-port network** (7 papers) → the same six lines every time, §3.2. Learn it as a script.
3. **Identify the passive device from this S-matrix** (5 papers) → **in every one of the five it is the magic tee**, in four different disguises. Learn the fingerprint in §3.7 and the matrix in §3.6.

Two more traps worth knowing before the paper starts:

- The radar questions (“two transmitters, twice the power”, “half the power to the antenna via a duplexer”, 4 papers) are the **same magic tee** asked as a design problem, §3.8. Nothing new has to be learnt for them.
- An examiner who writes “3-port network” almost always wants the **impossibility theorem** (§3.5), not just a 3×3 matrix. Stating that a 3-port cannot be lossless, reciprocal and matched at the same time, and then naming the three escapes, is what separates a 4 from an 8.

3.1 Why S-parameters: the Limits of Z, Y, h and ABCD

Why are S-parameters used in microwave network analysis? Compare S-parameters with h-parameters

2 3 4 4+4 5+5

TOP 13/24 69 Bh, 70 Bh, 70 Ma, 73 Ma, 73 Bh, 74 Ma, 75 Bh, 78 Ch, 79 Bh, 80 Ba, 80 Bh, 80 Ch, 81 Bh · **the single most repeated ask in the chapter**

What the low-frequency parameter sets need

- All four classical sets relate **total voltage and total current** at the terminals:
 - $[V] = [Z][I]$, $Z_{ij} = V_i/I_j$ with all other ports **open**, $I_k = 0$.
 - $[I] = [Y][V]$, $Y_{ij} = I_i/V_j$ with all other ports **shorted**, $V_k = 0$.
 - h-parameters: mixed, h_{11} and h_{21} need a **short** on port 2, h_{12} and h_{22} need an **open** on port 1.
 - ABCD: cascade form, still defined on V and I .
- Two things are therefore assumed:
 - V and I are **unique** at a terminal pair (true only when $d \ll \lambda$).
 - Ideal **open** and **short** terminations are available.
- Both assumptions fail above roughly 1 GHz.

What S-parameters do instead

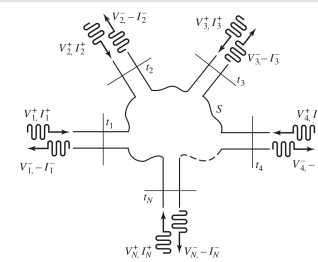
- Defined on **travelling waves**, not on totals:
 - a_i = wave incident on port i , b_i = wave leaving port i .
 - Normalised so $|a_i|^2$ and $|b_i|^2$ **are powers**, in watts.
- Every port is terminated in Z_0 , a **matched load**:
 - Easy to build and broadband, unlike open or short.
 - No reflection back into the device → no oscillation, no damage, bias undisturbed.
- Measured with **directional couplers** plus a vector network analyser, which separate incident from reflected wave and read magnitude and phase directly.
- Every practical figure of merit falls straight out: gain, return loss, insertion loss, isolation, VSWR, stability (§3.4).
- Valid at **any** frequency, so the same formalism covers DC to mmWave.
- Cascadable after conversion to the transfer (T) form, so a chain of measured blocks can be multiplied out.

The five reasons they break at microwave

1. **No unique V and I :**
 - Circuit size is comparable to λ , so voltage varies along the conductor.
 - In a waveguide there is no unique pair of conductors at all, so “the voltage” has no single definition.
2. **Open and short cannot be realised:**
 - A physical short has series lead inductance, an open has fringing capacitance.
 - A short is a short at **one frequency and one reference plane** only: $\lambda/4$ away it looks like an open.
3. **Total reflection is destructive:**
 - $|\Gamma| = 1$ sends all the power back into the device.
 - Transistors and negative-resistance diodes **oscillate or burn out** under open or short test conditions.
4. **No instrument measures RF V and I :**
 - Probes load the circuit and cannot follow GHz waveforms.
 - **Power** is what a microwave detector actually reads.
5. **Bias shifts:**
 - Open and short terminations change the DC operating point and the junction capacitance of active devices, so the measurement is not of the device in use.

h-parameters vs S-parameters

Aspect	h-parameters	S-parameters
Variables	Total V, I	Incident, reflected waves a, b
Termination Measured	Short and open Voltage, current	Matched Z_0 Power (magnitude + phase)
Instrument	Curve tracer, bridge	Vector network analyser
On active devices	Oscillation, damage	Safe, bias preserved
Frequency	Below ≈ 100 MHz	Any, chosen for RF / M/W
Broadband	No, termination drifts	Yes
Cascading	Awkward	Via T-matrix



An N -port as S-parameters see it: at each reference plane t_n an **incident** V_n^+ and a **reflected** V_n^- , not one total V and I . Pozar Fig 4.5.

Answer skeleton for the 4-mark version 4

- One line on what fails: no unique V / I , open and short unrealisable, active devices damaged, no RF voltmeter.
- One line on the fix: normalised travelling waves measured against matched Z_0 loads with directional couplers.
- One line on the payoff: gain, RL, IL, VSWR and stability read straight off $[S]$.
- Close with $[b] = [S][a]$ and the two-port matrix. Four sentences, full marks.

3.2 Definition and Derivation for a Two-Port Network

Using a two-port network, define and derive the S-parameters 4 5 6 10

TOP 7/24 69 Bh, 70 Bh, 70 Ma, 78 Ch, 80 Ba, 80 Bh, 80 Ch · write it as a script: figure, normalisation, matrix, four conditions

Step 1: the wave variables

- At port i the line carries a forward wave V_i^+ and a backward wave V_i^- . Normalise both by $\sqrt{Z_0}$:

$$a_i = \frac{V_i^+}{\sqrt{Z_0}}, \quad b_i = \frac{V_i^-}{\sqrt{Z_0}}$$

- The normalisation is chosen so that the squared magnitudes are powers:

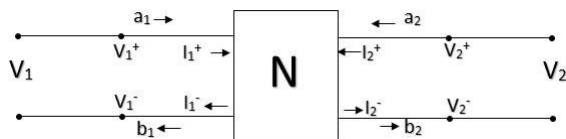
$$P_i^{\text{inc}} = \frac{1}{2}|a_i|^2, \quad P_i^{\text{ref}} = \frac{1}{2}|b_i|^2$$

(the $\frac{1}{2}$ disappears if a, b are taken as rms).

- In terms of the total quantities:

$$V_i = \sqrt{Z_0}(a_i + b_i), \quad I_i = \frac{a_i - b_i}{\sqrt{Z_0}}$$

so nothing is lost, only re-expressed.



Port 1 and port 2 of a general two-port, each carrying an incident wave a and an emerging wave b . Source: RF Pulchowk Ch3 deck.

Step 2: the defining matrix

- The network is linear, so each outgoing wave is a weighted sum of all the incoming ones:

$$b_1 = S_{11}a_1 + S_{12}a_2, \quad b_2 = S_{21}a_1 + S_{22}a_2$$

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \quad \text{i.e.} \quad [b] = [S][a]$$

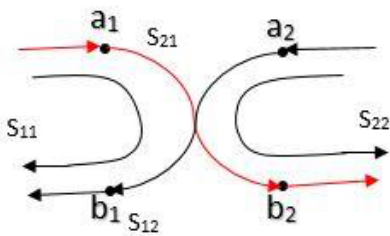
- Subscript order:** in S_{ij} the **second** index j is the port driven, the **first** index i is the port observed. S_{21} is “out of 2, in at 1”.
- Written out, the matrix is the complete input-output description of the black box: knowing four numbers replaces solving for every internal voltage and current.

Step 3: isolate each parameter

- Terminate the unused port in Z_0 . A matched load reflects nothing, so its incident wave vanishes and one column of the matrix survives at a time.

Parameter	Condition	Meaning
$S_{11} = \frac{b_1}{a_1}$	$a_2 = 0$	Input reflection coefficient, port 2 matched
$S_{21} = \frac{b_2}{a_1}$	$a_2 = 0$	Forward transmission (gain or insertion loss)
$S_{12} = \frac{b_1}{a_2}$	$a_1 = 0$	Reverse transmission (isolation, feedback)
$S_{22} = \frac{b_2}{a_2}$	$a_1 = 0$	Output reflection coefficient, port 1 matched

- $a_2 = 0$ means “port 2 terminated in Z_0 ”, never “port 2 open”. Writing “open circuit” here loses the mark, because it is exactly the condition S-parameters were invented to avoid.



Signal-flow graph of the same two-port: four branches, one per S-parameter. Useful when the question asks for a “flow diagram”.

Step 4: tie S_{11} back to impedance

- With port 2 matched, port 1 simply sees the input impedance Z_{in} , so

$$S_{11} = \Gamma_{in} = \frac{Z_{in} - Z_0}{Z_{in} + Z_0}$$

- This is the bridge to Chapter 2: a one-port load has the single entry $[S] = [\Gamma_L]$, and a perfectly matched two-port has $[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (§2.8).

Step 5: generalise to n ports

- $b_i = \sum_{j=1}^n S_{ij} a_j$, an $n \times n$ matrix with n^2 entries:

$$S_{ij} = \frac{b_i}{a_j} \Big|_{a_k=0, k \neq j}$$

- Rows and columns both equal the port count, so a 3-port gives 3×3 and a magic tee gives 4×4 .
- **Reference impedance:** every entry is defined against the same Z_0 (usually 50Ω), so a quoted matrix without its Z_0 and its frequency is incomplete.

3.3 Properties of the S-Matrix

State and justify the properties of the scattering matrix 4 4+4 5 6

TOP 8/24 73 Bh, 74 Bh, 75 Bh, 78 Ch, 79 Bh, 80 Bh, 81 Bh, 82 Bh · these four tests are also the tool that identifies a device from its matrix

P1. Zero diagonal $\Leftrightarrow$ matched port

- $S_{ii} = 0$ means a wave arriving at port i is not reflected, so that port is matched to Z_0 .
- “Matched at all ports” means the whole diagonal is zero.

P2. Symmetry $\Leftrightarrow$ reciprocity

- $S_{ij} = S_{ji}$ for all i, j , i.e. $[S] = [S]^T$.
- **Holds** for networks built of passive, linear, isotropic media: lines, stubs, tees, couplers, resistors, capacitors, transformers.
- **Fails** for magnetised ferrite (circulator, isolator, gyrator) and for active devices (amplifier, transistor), where $S_{21} \neq S_{12}$ by design.
- So a matrix that is **not** symmetric can never be a passive isotropic device.

P4. Shifting the reference plane multiplies by a phase

- Move the reference plane of port i outward by an electrical length $\theta_i = \beta \ell_i$. Then

$$S'_{ij} = S_{ij} e^{-j(\theta_i + \theta_j)}$$

- Diagonal terms pick up $e^{-2j\theta_i}$, because the wave travels the added length twice.
- **Magnitudes are unchanged:** $|S'_{ij}| = |S_{ij}|$. Return loss, insertion loss and VSWR do not depend on where the plane is drawn; only phase does.
- This is why an S-parameter measurement is meaningless until the planes are declared, and why a VNA must be calibrated to a stated plane.

P5. Complex, frequency dependent, n^2 entries

- Each S_{ij} carries a magnitude and an angle, and both move with frequency, so a quoted set is meaningless without its f and its Z_0 .
- An n -port has n^2 coefficients; symmetry and structural zeros cut the number of independent ones sharply (the magic tee has 16 entries but only 2 distinct values).

P6. Symmetric and antimetric two-ports

- **Symmetric:** reciprocal and $S_{11} = S_{22}$. The network looks the same from either end.
- **Antimetric:** $S_{11} = -S_{22}$.

P3. Unitary $\Leftrightarrow$ lossless

- A lossless network redirects power but never dissipates it, which forces

$$[S]^* [S] = [I]$$

- Reading that out column by column gives the two tests actually used in exams:

- **Column-sum test** (each column carries unit power):

$$\sum_{k=1}^n |S_{ki}|^2 = 1 \quad \text{for every } i$$

- **Orthogonality test** (distinct columns are orthogonal):

$$\sum_{k=1}^n S_{ki} S_{kj}^* = 0 \quad \text{for every } i \neq j$$

- If a column sums to less than 1, the shortfall is the fraction of incident power the network **dissipates**, which is how “is it lossless?” is answered numerically (§3.4 and Problem 1 of the companion).

The four tests as an exam checklist

Test	Look for	Conclusion
Diagonal	$S_{ii} = 0?$	That port is matched
Transpose	$[S] = [S]^T?$	Reciprocal (passive) or not (ferrite / active)
Column sums	$\sum_k S_{ki} ^2 = 1?$	Lossless, else lossy by the shortfall
Column dots	$\sum_k S_{ki} S_{kj}^* = 0?$	Confirms lossless, ties the phases

Run the four in this order on any matrix the examiner prints: they are enough to name the device (§3.7) and to answer every “judge the condition” rider.

3.4 Return Loss, Insertion Loss and the Other Figures of Merit

Define return loss and insertion loss, and the losses associated with a two-port network 2 1 4

HOT 3/24 72 Ash, 79 Bh, 82 Bh · small marks, but they are the language every other answer is written in

The three powers at a port

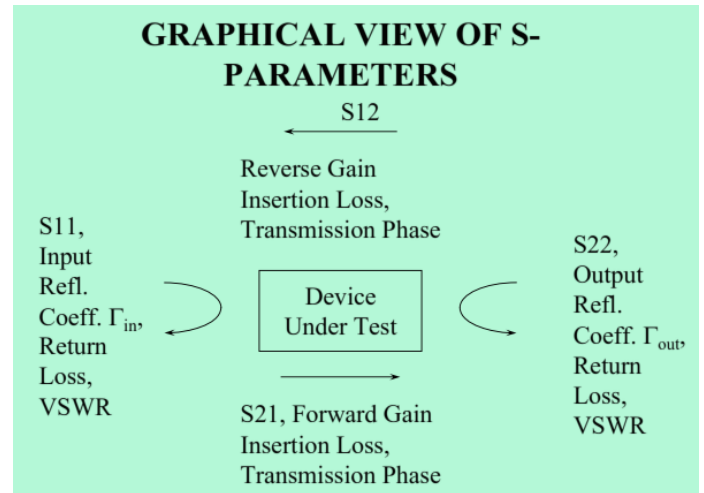
- P_i incident, P_r reflected, P_t transmitted; whatever is left over is dissipated.
- Everything below is a ratio of two of those, expressed in dB and rewritten in terms of $[S]$.

Quantity	In terms of $[S]$	What it tells you
Return loss	$RL = -20 \log_{10} S_{11} $	How well port 1 is matched. Bigger is better
Insertion loss	$IL = -20 \log_{10} S_{21} $	Power lost passing through. 0 dB is ideal
Isolation	$-20 \log_{10} S_{12} $	Leakage backwards. Bigger is better
Forward gain	$G = 20 \log_{10} S_{21} $	Same number as IL , opposite sign, used for amplifiers
VSWR	$\frac{1 + S_{11} }{1 - S_{11} }$	The same match on a 1 to ∞ scale
Reflection (mismatch) loss	$-10 \log_{10} (1 - S_{11} ^2)$	The part of P_i that never enters the network
Dissipated fraction	$1 - \sum_k S_{k1} ^2$	What the network eats. Zero for a lossless network

- **Useful anchors:** $|S_{11}| = 0.5 \rightarrow RL = 6 \text{ dB}$, $VSWR = 3$; $|S_{11}| = 0.1 \rightarrow RL = 20 \text{ dB}$, $VSWR = 1.22$; $|S_{11}| = 0 \rightarrow$ perfect match, $RL = \infty$.

Reading a device off its numbers

- Small $|S_{11}|$, $|S_{22}| \rightarrow$ well matched at both ends.
- $|S_{21}| > 1 \rightarrow$ **active**: the device has gain.
- $|S_{21}| < 1$ with $S_{12} = S_{21} \rightarrow$ **passive and reciprocal**: an attenuator, filter or line.
- $|S_{12}| \ll |S_{21}| \rightarrow$ **unilateral**: an amplifier or isolator; the smaller $|S_{12}|$ is, the more stable the amplifier tends to be.
- This is the whole method for 82 Bh, which prints two amplifier S-parameter sets and asks only for “the basic differences”. Problem 2 of the companion works it.



What each S-parameter of a device under test is called in the lab. Source: RF Pulchowk Ch3 deck.

Sign conventions that cost marks

- Return loss and insertion loss are quoted as **positive** dB for a passive device. The minus sign is already in the definition.
- 20 log for a **ratio of waves** (S is an amplitude ratio), 10 log for a **ratio of powers**. Mixing the two is the commonest slip.

verified/added Correction to the source notes

- The RF Pulchowk deck prints $IL \text{ (dB)} = 10 \log (|a_1|^2 / |b_1|^2)$.
- $|b_1|$ is the wave coming **back out of port 1**, so that expression is the return loss, not the insertion loss.
- Insertion loss compares the incident wave with what leaves the **far** port:

$$IL = 10 \log \frac{|a_1|^2}{|b_2|^2} = 20 \log \frac{1}{|S_{21}|}$$

which is what the deck's own next line prints. Use the $|S_{21}|$ form.

3.5 Three-Port Networks and Why the Ideal One Cannot Exist

Analyse a three-port network using S-parameters. Write the properties of a 3-port network and define its characteristic parameters 4 4+4 2+6 5+2

HOT 6/24 72 Ash, 73 Ma, 73 Bh, 74 Ma, 80 Bh, 82 Bh · the theorem is the answer, the matrix alone is half marks

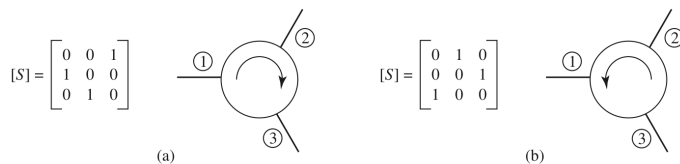
The general 3-port

- Nine coefficients:

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix}$$

- Characteristic parameters** an examiner expects you to define:
 - Matching:** S_{ii} , hence return loss and VSWR at each port.
 - Coupling / division ratio:** $|S_{ij}|^2$, the fraction of the input power delivered to port i .
 - Isolation:** $-20 \log |S_{ij}|$ for the pair meant to be decoupled.
 - Phase relation:** $\arg S_{13}$ against $\arg S_{23}$, i.e. whether the two output arms are in phase or in antiphase.
 - Insertion loss** and **bandwidth** over which all of the above hold.

The non-reciprocal escape, drawn



(a) clockwise, (b) counterclockwise circulator with its $[S]$: zero diagonal (matched), one unit entry per column (lossless), and $[S] \neq [S]^T$. Pozar Fig 7.2.

Theorem: no 3-port is lossless, reciprocal and matched at once

- Assume all three. Matched gives $S_{11} = S_{22} = S_{33} = 0$, reciprocal gives $S_{ij} = S_{ji}$, so

$$[S] = \begin{bmatrix} 0 & S_{12} & S_{13} \\ S_{12} & 0 & S_{23} \\ S_{13} & S_{23} & 0 \end{bmatrix}$$

- Apply the orthogonality test to each pair of columns:

$$S_{13}^* S_{23} = 0, \quad S_{12}^* S_{23} = 0, \quad S_{12}^* S_{13} = 0$$

so at least two of S_{12} , S_{13} , S_{23} must be zero.

- Apply the column-sum test:

$$|S_{12}|^2 + |S_{13}|^2 = 1, \quad |S_{12}|^2 + |S_{23}|^2 = 1, \quad |S_{13}|^2 + |S_{23}|^2 = 1$$

which requires all three to be non-zero.

- The two demands **contradict**, so no such network exists.

Give up exactly one of the three properties.

The three escapes, and the device each one gives

Give up	Device	Consequence
Lossless	Resistive divider, matched 3-port	All ports matched and reciprocal, but power is burnt in the resistors
Reciprocal	Circulator (ferrite)	Matched and lossless, but $S_{ij} \neq S_{ji}$; power circulates $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$
Matched at every port	E-plane tee, H-plane tee, T-junction	Lossless and reciprocal, but only the side arm can be matched

- Ideal circulator**, matched and lossless but not reciprocal:

$$[S] = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Terminate port 3 in a matched load and it becomes an **isolator**.

- A **4-port** escapes the trap entirely: a directional coupler and a magic tee are matched, lossless *and* reciprocal all at once, which is exactly why the exam favourite is a 4-port (§3.6).

E-plane tee: the worked 3-port (§3.6 has the full matrix)

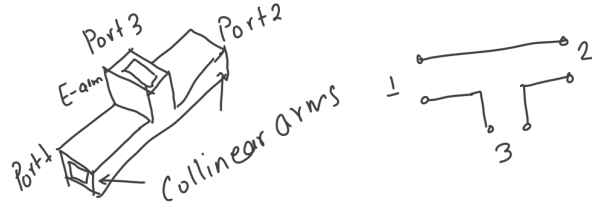
- Ports 1 and 2 collinear, port 3 the E-arm, assumed matched to the junction.
- Three physical facts start the derivation:
 - An input at the E-arm splits into two outputs that are **180° out of phase**: $S_{23} = -S_{13}$.
 - Port 3 is matched: $S_{33} = 0$.
 - The junction is passive and isotropic: $S_{ij} = S_{ji}$.
- Four unknowns are left, and the unitary property fixes all four (§3.6, and Problem 5 of the companion works every line).
- Ports 1 and 2 cannot be matched**, and indeed the answer gives $S_{11} = S_{22} = 1/2$, which is the theorem showing up in the arithmetic.

3.6 S-Matrix Catalogue of the Standard Devices

Prepare the S-matrices of a perfectly working E-plane Tee, H-plane Tee and a Duplexer, and explore their uses 4+4 8 3+7

HOT 4/24 78 Ch, 81 Ba, 80 Bh, 82 Bh · memorise these five matrices and half the chapter is already answered

E-plane tee (series tee)

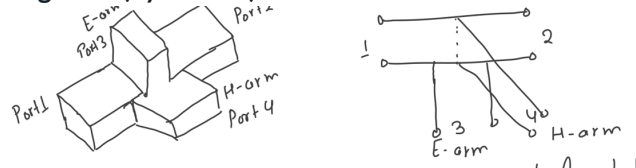


- Side arm in the plane of the **E** field, so it excites the collinear arms in **antiphase**.

$$[S] = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

- Fingerprint:** the $-1/\sqrt{2}$ in the third row / column.
- Use:** power splitting with a phase reversal; the difference arm of a magic tee.
- Ports 1 and 2 are not matched** ($S_{11} = S_{22} = 1/2$), which is the 3-port theorem of §3.5 showing up in the answer.

Magic tee (hybrid tee): the exam favourite

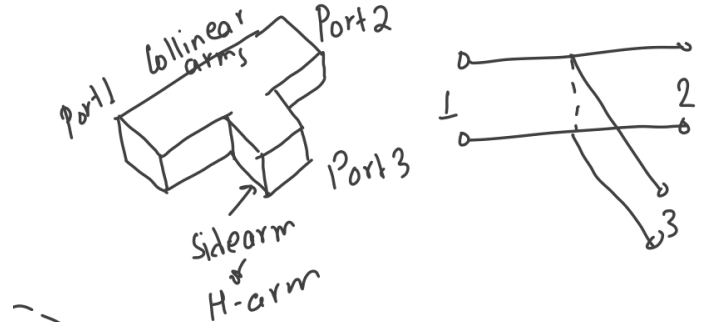


- Ports 1 and 2 collinear, port 3 the **E-arm** (difference), port 4 the **H-arm** (sum). Four physical facts:
 - E-arm drive gives antiphase outputs: $S_{23} = -S_{13}$.
 - H-arm drive gives in-phase outputs: $S_{24} = +S_{14}$.
 - E-arm and H-arm are decoupled: $S_{34} = S_{43} = 0$.
 - Ports 3 and 4 matched: $S_{33} = S_{44} = 0$; and the collinear arms are decoupled, $S_{12} = S_{21} = 0$.
- The unitary property then forces $S_{11} = S_{22} = 0$ as well, so the tee ends up matched at **all four ports**:

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

- Fingerprint:** an all-zero 2×2 block on each diagonal, one column with a sign flip (E-arm), one without (H-arm).

H-plane tee (shunt tee)



- Side arm in the plane of the **H** field, so the collinear arms come out **in phase**.

$$[S] = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

- Fingerprint:** the minus sign has moved to S_{12} ; the side-arm column is all positive.
- Use:** in-phase power divider or combiner; the sum arm of a magic tee.
- E-tee vs H-tee in one line:** same geometry, same magnitudes, opposite phase relation. The E-tee subtracts, the H-tee adds.

Circulator, isolator, gyrator

- Circulator:** $[S]$ as in §3.5, matched and lossless but **not** symmetric.
- Isolator:** a circulator with port 3 loaded, $[S] = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$; forward passes, reverse is absorbed.
- Gyrator:** $[S] = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, a 180° phase difference between the two directions.

Directional coupler

- Through arm α , coupled arm $j\beta$, isolated arm 0, with $\alpha^2 + \beta^2 = 1$:

$$[S] = \begin{bmatrix} 0 & \alpha & j\beta & 0 \\ \alpha & 0 & 0 & j\beta \\ j\beta & 0 & 0 & \alpha \\ 0 & j\beta & \alpha & 0 \end{bmatrix}$$

- A 3 dB coupler has $\alpha = \beta = 1/\sqrt{2}$, and is then a magic tee in quadrature form.

Duplexer

- **Circulator duplexer** (the usual answer): port 1 transmitter, port 2 antenna, port 3 receiver. Ideally lossless, so **all** the transmitter power reaches the antenna and **all** the echo reaches the receiver, with the transmitter isolated from the receiver.
- **Magic-tee duplexer**: transmitter on the H-arm, antenna and a matched dummy load on the collinear arms, receiver on the E-arm. Splits the transmitter power in **half** and keeps transmitter and receiver isolated. This is the version **76 Bh** asks for (§3.8).

verified/added **Magic tee with both E- and H-arms short-circuited** 4+4

PIN 1/24 82 Bh · not in any source note, derived here and checked in `sparam.py`

- Shorts at the junction reference plane give $a_3 = -b_3$ and $a_4 = -b_4$.
- Substituting into the magic-tee equations collapses the 4-port to the two-port seen from ports 1 and 2:

$$[S]_{\text{shorted}} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Amplifier, attenuator, amplitude modulator

- Matched amplifier: $[S] = \begin{bmatrix} 0 & S_{12} \\ S_{21} & 0 \end{bmatrix}$ with $|S_{21}| > 1 \gg |S_{12}|$. Not reciprocal, not lossless.
- Matched attenuator of L dB: $[S] = \begin{bmatrix} 0 & \alpha \\ \alpha & 0 \end{bmatrix}$, $\alpha = 10^{-L/20}$. Reciprocal but lossy.
- **Amplitude modulator (78 Ch)**: a matched, reciprocal two-port whose transmission is **driven by the modulating signal**,

$$[S] = \begin{bmatrix} 0 & \alpha(t) \\ \alpha(t) & 0 \end{bmatrix}, \quad \alpha(t) = \alpha_0 (1 + m \cos \omega_m t)$$

Problem 6 of the companion derives it and evaluates the numbers.

- **Behaviour**: each collinear arm reflects **all** of its incident power with a 180° phase reversal, and the two arms stay **completely isolated** from each other.
- The E-arm and H-arm reflections add in the driven arm and cancel in the other one, which is why S_{21} stays zero.
- Still lossless (each column sums to 1), still reciprocal. The tee has become a **pair of independent short circuits**, useful as a reflective phase shifter or a reference short.

3.7 Identifying a Device from its S-Matrix

Identify and explain, with a neat diagram, the passive microwave device described by the given S-matrix **5 6 8 10**

HOT 5/24 74 Bh, 75 Bh, 78 Ch, 81 Bh, 82 Bh · in all five papers the printed matrix is a magic tee

The six-step method

- Count the ports:** the order of the matrix is the port count. 4×4 narrows it to coupler, magic tee or hybrid ring immediately.
- Read the diagonal:** zeros mark the matched ports. An all-zero diagonal on a 4-port means matched everywhere.
- Test the transpose:** symmetric means passive and isotropic; asymmetric means ferrite or active.
- Test the columns:** sums of 1 and orthogonal pairs mean lossless. A shortfall is dissipation.
- Map the zeros:** an off-diagonal zero is an **isolated pair**. Which pairs are decoupled is the strongest single clue.
- Read the signs and the j 's:** a sign flip down a column means an **antiphase (difference, E) arm**; matching signs mean an **in-phase (sum, H) arm**; a factor j means a **quadrature coupled arm**.

Answer skeleton

- Name the device in the first line. Do not narrate the tests first.
- Draw it: the 3D waveguide sketch with all arms labelled, plus the schematic.
- Justify with the four tests, one line each.
- Close with two uses and one limitation (the tee needs matching irises at the collinear arms to reach the ideal matrix in practice).

The four matrices the papers actually print

- 78 Ch 10**, printed without its normalising factor:

$$[S] = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

Zero diagonal, symmetric, the (1, 2) and (3, 4) blocks zero, column 3 sign-flipped: **magic tee**. It is unitary only after dividing by $\sqrt{2}$, and saying so earns the “judge the condition” mark.

- 81 Bh 5** and **74 Bh 8**, printed symbolically:

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{13} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix}$$

Column 3 in phase (H-arm), column 4 antiphase (E-arm), $S_{33} = S_{44} = 0$: the **magic tee before the unitary property is applied**. Finish the job and $S_{11} = S_{22} = S_{12} = 0$, $S_{13} = S_{14} = 1/\sqrt{2}$.

Fingerprint table

Pattern	Device
2×2 , $S_{11} = S_{22} = 0$, $S_{12} = S_{21} = 1$	Matched lossless line or ideal matched network
2×2 , $S_{12} = 0$, $S_{21} = 1$	Isolator
2×2 , $ S_{21} > 1$, $ S_{12} $ tiny	Amplifier (active, not reciprocal)
3×3 , cyclic ones, not symmetric	Circulator
3×3 with a $\pm 1/\sqrt{2}$ column	E-plane tee (–) or H-plane tee (+)
4×4 , zero diagonal, two zero 2×2 blocks, one column sign-flipped	Magic tee
4×4 , zero diagonal, j on the coupled arm, one zero per row	Directional coupler

- 75 Bh 6**, the same device with the collinear isolation already written in:

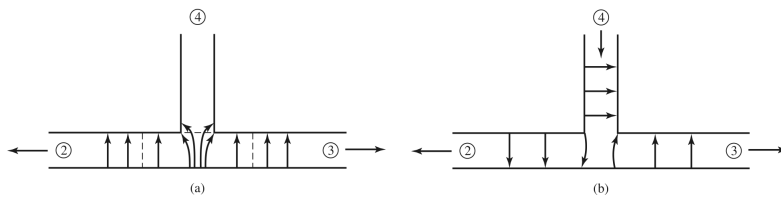
$$[S] = \begin{bmatrix} S_{11} & 0 & S_{13} & S_{14} \\ 0 & S_{22} & -S_{13} & S_{14} \\ S_{13} & -S_{13} & 0 & 0 \\ S_{14} & S_{14} & 0 & 0 \end{bmatrix}$$

Here column 3 is the antiphase E-arm and column 4 the in-phase H-arm, the mirror of the previous labelling.

Read the signs, not the port numbers.

- 82 Bh 4** prints two *amplifier* sets instead, so steps 3 and 4 answer it: $S_{12} \neq S_{21}$ and $|S_{21}| > 1$ make both of them active and neither reciprocal (§3.4, Problem 2).

Why the signs fall where they do



E in the collinear arms (2), (3) of a magic tee. (a) fed from the H-arm (Pozar's port 1, the sum port): the arms are driven **in phase**, the matching-sign column. (b) fed from the E-arm (4): **antiphase**, the sign-flipped column. Pozar numbers the arms differently from the papers, so label by E and H, not by number. Pozar Fig 7.50.

3.8 Power Splitting and Combining: the Radar Questions

Two identical radar transmitters must deliver twice the power to one antenna; and an ASR needs half the transmitter power at the antenna through a duplexer. Solve both with an S-matrix **6 8 10**

HOT 4/24 73 Ma, 73 Bh, 74 Ma, 76 Bh · **one device, two directions: the magic tee combines and splits**

Combining: two transmitters into one antenna

- **Device:** magic tee (or a 3 dB hybrid coupler).
- **Connection:** the two transmitters drive the collinear arms 1 and 2 **in phase and at equal amplitude**; the antenna hangs on the H-arm (port 4); the E-arm (port 3) carries a **matched dummy load**.
- **Mathematics**, with $a_1 = a_2 = a$ and $a_3 = a_4 = 0$:

$$b_4 = \frac{1}{\sqrt{2}}(a_1 + a_2) = \sqrt{2}a \Rightarrow |b_4|^2 = 2|a|^2$$

$$b_3 = \frac{1}{\sqrt{2}}(a_1 - a_2) = 0$$

- **Result:** the antenna receives $2P$, twice what either transmitter can deliver, and the dummy load receives nothing.
- **Why the tee and not a plain T-junction:** $S_{12} = 0$, so the two transmitters are **isolated** and cannot pull or damage each other; and all four ports stay matched, so neither transmitter sees a VSWR.
- **Practical riders** worth a mark each:
 - Phase must be equalised, so the two feed lines are cut to the same electrical length.
 - Amplitude imbalance Δ shows up as $|b_3|^2 \neq 0$: the dummy load absorbs exactly the mismatch, which is what protects the transmitters.
 - The dummy load must be rated for the full unbalanced power.

Splitting: half the transmitter power to the antenna

- **Device:** the same magic tee, used as a duplexer, driven from the H-arm.
- **Connection:** transmitter on port 4 (H-arm), antenna on port 1, matched load on port 2, receiver on port 3 (E-arm).
- **Mathematics**, with $a_4 = a$ and all other inputs zero:

$$b_1 = \frac{1}{\sqrt{2}}a, \quad b_2 = \frac{1}{\sqrt{2}}a \Rightarrow |b_1|^2 = |b_2|^2 = \frac{1}{2}|a|^2$$

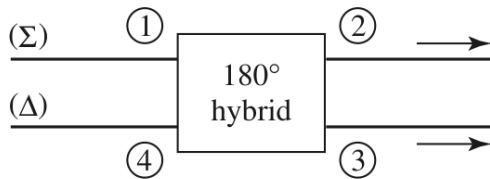
$$b_3 = 0$$

- **Result:** the antenna gets exactly $P/2$, as **76 Bh** requires, and the receiver on the E-arm gets **none** of the transmitter power, which is the whole point of a duplexer.
- **On receive**, the echo arrives at port 1 alone and divides between the E-arm and the H-arm, so half of it reaches the receiver. The 3 dB penalty in each direction is the price of the magic-tee duplexer, and naming it shows you understand the device rather than reciting it.
- **Better answer for a real ASR: a ferrite circulator** duplexer routes the full power $1 \rightarrow 2$ and the full echo $2 \rightarrow 3$ with no 3 dB loss. Quote it as the improvement whenever the question does not force the half-power split.

Answer skeleton for both

- Name the device, draw it with every arm labelled and every termination named.
- Write $[S]$ for the magic tee.
- Write the drive vector $[a]$ the connection creates.
- Multiply out, and read the answer off $|b|^2$.
- Close with the isolation and matching argument: that is the "sufficient explanation" the marks are for.

The schematic to draw beside the tee



- A magic tee is a 180° hybrid: **sum** port Σ (the H-arm) and **difference** port Δ (the E-arm), with outputs (2) and (3).
- **Combining**: equal in-phase inputs at (2), (3) add at Σ and cancel at Δ , so the dummy load on Δ sees nothing.
- **Splitting**: drive Σ and (2), (3) each get half, in phase; Δ stays isolated.
- Pozar Fig 7.41.

3.9 PYQ Mapping

Theory questions, covered in this document

- **Why S-parameters; importance in microwave network analysis; vs h-parameters** → §3.1. **The most repeated topic in the chapter, 13 papers.** 69 Bh 4, 70 Bh 4, 70 Ma 4, 80 Ch 5, 81 Bh 3
- **Derive S-parameters for a two-port network** → §3.2. 69 Bh 6, 70 Bh 4, 80 Ba 6, 80 Ch 5
- **Properties of the S-matrix** → §3.3. The four tests are also the identification tool of §3.7. 80 Bh 4
- **Return loss and insertion loss** → §3.4. 72 Ash 2, 79 Bh 1
- **Three-port network: analysis, properties, characteristic parameters** → §3.5, with the impossibility theorem and its three escapes. 72 Ash 5, 73 Ma 4, 73 Bh 4, 74 Ma 6, 82 Bh 4
- **Three-port parameters using an E-plane tee example** → §3.5 for the parameters, §3.6 for the matrix, companion Problem 5 for the derivation. 80 Bh 4
- **S-matrices of E-plane tee, H-plane tee and a duplexer, and their uses** → §3.6. 81 Ba 8
- **Magic tee with both E- and H-arms shorted** → §3.6, **verified/added** derived here because no source note carries it. 82 Bh 4
- **Identify the passive device from a printed S-matrix** → §3.7, with all four printed matrices worked. 74 Bh 8, 75 Bh 6, 78 Ch 10, 81 Bh 5
- **Radar power combining and duplexer power splitting** → §3.8. 73 Ma 8, 73 Bh 8, 74 Ma 6, 76 Bh 10
- **Evaluate the S-matrix of an amplitude modulator** → §3.6 and companion Problem 6. **verified/added** 78 Ch 7

Numerical content

- **This chapter has a numerical companion, Ch3 - RF and MW Network Theory and Analysis - Numerical Problems and Solutions.** It carries every S-parameter calculation the archive asks for: the 79 Bh reciprocal / lossless / return-loss / reflected-power question, the 82 Bh comparison of two amplifier sets, the four device-identification matrices, the E-plane, H-plane and magic-tee derivations, and the two radar power problems.
- **Not in this chapter**: amplifier stability (K , Δ , μ) and the unilateral / bilateral maximum gain calculations. Those ride on S-parameters but the archive files them under **RF Design Practices**, so they belong to Chapter 6. §3.4 gives only the reading of the numbers that 82 Bh asks for.

Syllabus coverage check

- *Scattering matrix and its properties* → §3.3, §3.5, §3.6 ✓
- *S-parameter derivation and analysis* → §3.1, §3.2, §3.4, §3.7, §3.8 ✓

Cross-references, so nothing is said twice

- **Chapter 1** §1.2 justifies S-parameter analysis as part of the lumped-vs-distributed answer. The full case is §3.1 here.
- **Chapter 2** §2.8 carries the two-line S-matrix rider tacked onto ten stub-tuning questions ($[\Gamma_L]$ before matching, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ after). The theory behind it is §3.2 and §3.3.
- **Chapter 4** covers the physical construction of the tees, couplers, circulators and waveguide junctions whose matrices are catalogued in §3.6.

4. RF/Microwave Components and Devices

8 hrs · asked in **all 24 papers** · 11 syllabus sub-topics · **317 marks, 16.7% of the archive, 52 questions** → one derivation (the waveguide) worth up to 10 marks, one device (the magic tee) asked 9 times, and a ring of 4 to 6 mark short notes.

READ THIS FIRST

Chapter 4 has the biggest question count in the paper, but three scripts answer most of it:

- Field equations of a rectangular waveguide, TE or TM** (14 papers, up to **10 marks**) → §4.1. **One six-step script covers both modes**: only the boundary condition in step 4 changes. Learn TM first; TE is the same page with sin and cos swapped.
- Mode questions** (9 papers, 1 to 5 marks each): dominant, degenerate, “prove TM_{01} cannot exist”, “show TM_{11} is the lowest TM mode” → §4.2. Every one is **the cut-off formula plus one sentence**.
- Magic tee** (9 papers) → §4.5 for construction and the derivation, §3.6 for the matrix. The combinations (two tees joined, arms terminated) are worked in the numerical companion.

Two more things worth knowing before the paper starts:

- Circular waveguide** (4 papers, 7 to 10 marks) is the rectangular script again in cylindrical coordinates, with Bessel functions in place of sines. The only new fact to memorise is the **root table** of §4.3.
- Short notes** rotate among circulator (3 papers), directional coupler (3), cavity resonator (2), Gunn diode, MASER, probe/loop coupling and microstrip (1 each). Each gets a ready answer below.

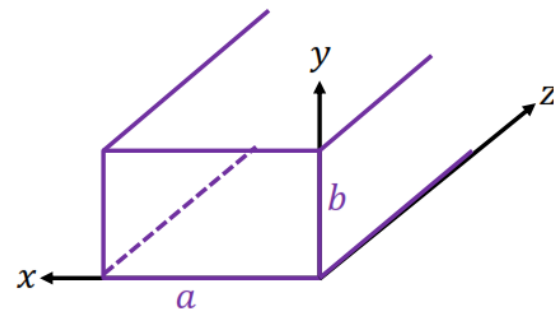
4.1 Rectangular Waveguide: TE and TM Field Equations

Derive the field equations and characteristic parameters of a rectangular waveguide operating in TE / TM mode **4 5 7+3 8 8+2 10**

TOP 14/24 69 Bh, 70 Bh, 70 Ma, 71 Bh, 72 Ma, 74 Ma, 74 Bh, 76 Bh, 77 Ch, 79 Ch, 80 Ch, 80 Bh, 81 Ba, 81 Bh · **the most repeated ask in the chapter:** **TE 6 papers, TM 7, modes in general 1**

Set-up

- Hollow metal pipe, inner broad wall a along x , narrow wall b along y , propagation along $+z$, $a > b$.
- Lossless dielectric (μ, ϵ) inside, perfectly conducting walls.
- Every field component carries $e^{-\gamma z}$, $\gamma = \alpha + j\beta$.
- Notation used throughout (Gangaju):
 - $k = \omega\sqrt{\mu\epsilon}$, the free-space wave number of the filling.
 - $h^2 = \gamma^2 + k^2 = k_x^2 + k_y^2$, the cut-off wave number squared (Pozar writes k_c^2).
- No TEM mode.** With $E_z = H_z = 0$ every transverse component below is zero. Physically, a closed H loop in the cross-section needs an axial current, and a single hollow conductor has no centre conductor to carry it.



Co-ordinates of the rectangular guide. Source: Gangaju, Ch-4 deck, slide 12.

Step 1: transverse fields from the longitudinal ones (Maxwell's curl equations)

- Write $\nabla \times E = -j\omega\mu H$ and $\nabla \times H = j\omega\epsilon E$ in components with $\partial/\partial z \rightarrow -\gamma$, and solve the four transverse equations for E_x, E_y, H_x, H_y :

$$\begin{aligned} E_x &= -\frac{\gamma}{h^2} \frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y}, & E_y &= -\frac{\gamma}{h^2} \frac{\partial E_z}{\partial y} + \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x}, \\ H_x &= -\frac{\gamma}{h^2} \frac{\partial H_z}{\partial x} + \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial y}, & H_y &= -\frac{\gamma}{h^2} \frac{\partial H_z}{\partial y} - \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial x}. \end{aligned}$$

- This block is the whole trick**: find E_z (TM) or H_z (TE) and the other four components follow by differentiation.

Step 2: the wave equation for the longitudinal component

- Helmholtz: $\nabla^2 E_z + k^2 E_z = 0$, i.e. $\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \frac{\partial^2 E_z}{\partial z^2} + \omega^2 \mu \epsilon E_z = 0$.
- The same equation holds for H_z .

Step 3: separation of variables

- Put $E_z = X(x) Y(y) Z(z)$ and divide by XYZ : $\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} = -k^2$.
- Each ratio must be a constant:
 - $X''/X = -k_x^2 \Rightarrow X = C_1 \cos k_x x + C_2 \sin k_x x$.
 - $Y''/Y = -k_y^2 \Rightarrow Y = C_3 \cos k_y y + C_4 \sin k_y y$.
 - $Z''/Z = \gamma^2 \Rightarrow Z = C_5 e^{\gamma z} + C_6 e^{-\gamma z}$; a wave along $+z$ only $\Rightarrow C_5 = 0$.
- Adding the constants back: $-k_x^2 - k_y^2 + \gamma^2 = -k^2$, so $\boxed{h^2 = k_x^2 + k_y^2 = \gamma^2 + k^2}$.
- General solution: $E_z = (A_1 \cos k_x x + A_2 \sin k_x x)(A_3 \cos k_y y + A_4 \sin k_y y) e^{-\gamma z}$.
- H_z has the same form with its own constants B_1, B_2, B_3, B_4 ; the working below reuses the letters A for brevity.

Step 4 (TM): $H_z = 0$, apply $E_z = 0$ on the walls

- E_z is tangential to every wall, so it must vanish there.
- $x = 0$: $E_z = 0 \Rightarrow A_1 = 0$.
- $y = 0$: $E_z = 0 \Rightarrow A_3 = 0$.
- $x = a$: $\sin k_x a = 0 \Rightarrow k_x = m\pi/a$.
- $y = b$: $\sin k_y b = 0 \Rightarrow k_y = n\pi/b$.

$$E_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-\gamma z}$$

Step 4 (TE): $E_z = 0$, apply $E_x, E_y = 0$ on the walls

- With $E_z = 0$: $E_x \propto \partial H_z / \partial y$, $E_y \propto \partial H_z / \partial x$.
- $x = 0$: $E_y = 0 \Rightarrow \partial H_z / \partial x = 0 \Rightarrow A_2 = 0$.
- $y = 0$: $E_x = 0 \Rightarrow \partial H_z / \partial y = 0 \Rightarrow A_4 = 0$.
- $x = a$: $\sin k_x a = 0 \Rightarrow k_x = m\pi/a$.
- $y = b$: $\sin k_y b = 0 \Rightarrow k_y = n\pi/b$.

$$H_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-\gamma z}$$

Deck slide 18 prints $k_y = n\pi/a$ in the last line; it is $n\pi/b$.

Step 5: substitute into Step 1 (write $e^{-\gamma z}$ on every line in the exam)

TM_{mn} ($H_z = 0$)

$$\begin{aligned} E_x &= -\frac{\gamma}{h^2} \frac{m\pi}{a} E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \\ E_y &= -\frac{\gamma}{h^2} \frac{n\pi}{b} E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} \\ H_x &= \frac{j\omega\epsilon}{h^2} \frac{n\pi}{b} E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} \\ H_y &= -\frac{j\omega\epsilon}{h^2} \frac{m\pi}{a} E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \end{aligned}$$

TE_{mn} ($E_z = 0$)

$$\begin{aligned} E_x &= \frac{j\omega\mu}{h^2} \frac{n\pi}{b} H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \\ E_y &= -\frac{j\omega\mu}{h^2} \frac{m\pi}{a} H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} \\ H_x &= \frac{\gamma}{h^2} \frac{m\pi}{a} H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} \\ H_y &= \frac{\gamma}{h^2} \frac{n\pi}{b} H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \end{aligned}$$

All eight components are checked in `src/wg.py` against both curl equations and against tangential $E = 0$ on all four walls; a single flipped sign fails the check.

Step 6: the characteristic equations (the “+2” or “+3” of every version)

Quantity	Expression	Note
Propagation constant	$\gamma = \sqrt{h^2 - k^2} = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2\mu\epsilon}$	real below cut-off, imaginary above
Cut-off frequency	$f_c = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$	set $\gamma = 0$; air: $\frac{c}{2}\sqrt{\cdot}$
Cut-off wavelength	$\lambda_c = \frac{2}{\sqrt{(m/a)^2 + (n/b)^2}}$	TE ₁₀ : $\lambda_c = 2a$
Phase constant	$\beta = \omega\sqrt{\mu\epsilon}\sqrt{1 - (f_c/f)^2}$	above cut-off
Guide wavelength	$\lambda_g = \frac{\lambda}{\sqrt{1 - (f_c/f)^2}}$	$\frac{1}{\lambda^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}$
Phase velocity	$v_p = \frac{\omega}{\beta} = \frac{v}{\sqrt{1 - (f_c/f)^2}}$	$> v$, carries no energy
Group velocity	$v_g = \frac{d\omega}{d\beta} = v\sqrt{1 - (f_c/f)^2}$	$< v$; $v_p v_g = v^2$
Wave impedance, TE	$Z_{TE} = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{\omega\mu}{\beta} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}}$	$> \eta$
Wave impedance, TM	$Z_{TM} = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{\beta}{\omega\epsilon} = \eta\sqrt{1 - (f_c/f)^2}$	$< \eta$

- Three regimes from $\gamma^2 = h^2 - \omega^2\mu\epsilon$:
 - $f < f_c$: $\gamma = \alpha$, real → **evanescent**, the field decays and carries no power. A waveguide is a **high-pass filter**.
 - $f = f_c$: $\gamma = 0$, cut-off.
 - $f > f_c$: $\gamma = j\beta$ → **propagates**.
- $v = 1/\sqrt{\mu\epsilon}$, $\eta = \sqrt{\mu/\epsilon}$; for air $v = c$ and $\eta_0 = 120\pi \approx 377 \Omega$.

Answer skeleton for the 4 to 5 mark short-note version 4 5

- Figure of the guide with a, b, z (1 line).
- Step 1 block, then the one boundary-condition result ($E_z = E_0 \sin \sin$ for TM, $H_z = H_0 \cos \cos$ for TE).
- The four transverse components from Step 5, without re-deriving them.
- f_c and one of Z_{TE} / Z_{TM} . For the 10-mark version add Steps 2 to 4 in full and the whole Step 6 table.

4.2 Modes: Dominant, Degenerate, and the Short Proofs

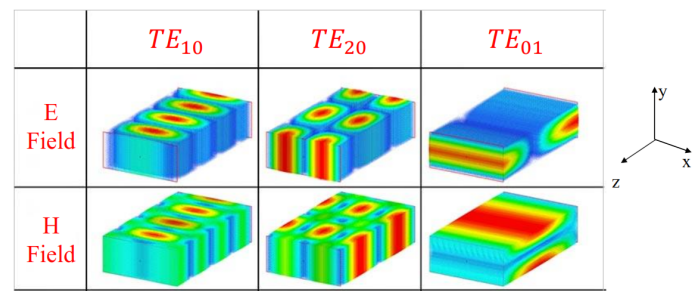
Dominant and degenerate modes; show which modes cannot exist; compare modes by cut-off 1+1+2

2 2+2 3 5

TOP 9/24 70 Bh, 70 Ma, 71 Bh, 72 Ma, 72 Ash, 80 Bh, 81 Bh, 82 Ba, 82 Bh · every one is the f_c formula plus one sentence

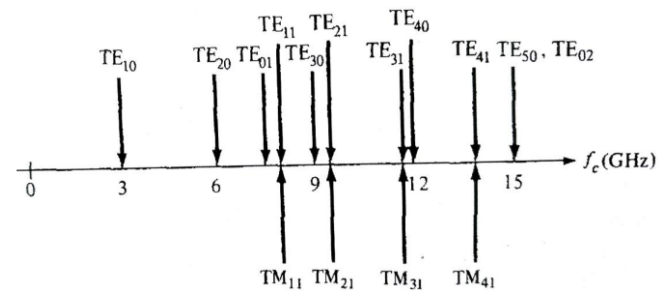
Definitions

- Mode**: one allowed field pattern, labelled by the integers m, n = number of half-cycle variations across a and across b .
- Dominant mode**: the mode with the **lowest cut-off frequency**.
 - Rectangular guide ($a > b$): **TE₁₀**, $f_c = c/2a$, $\lambda_c = 2a$.
 - Circular guide: **TE₁₁**, $f_c = 1.841 c/2\pi r$ (§4.3).
 - Between $f_{c,10}$ and the next cut-off only the dominant mode propagates → **single-mode operation**, the reason guides are sized this way.



E and H field of TE₁₀, TE₂₀, TE₀₁: one, two and one half-cycles across the walls.
Source: Gangaju, slide 20.

- **Degenerate modes: different field patterns with the same cut-off frequency.**
 - TE_{mn} and TM_{mn} for every $m, n \geq 1$: same f_c formula.
 - Square guide ($a = b$): TE_{10} and TE_{01} as well.
 - Circular guide: TE_{01} and TM_{11} (both root 3.832).
 - Unwanted, because power can couple between them at any discontinuity.



Cut-off ladder, Gangaju slide 22 (from Sadiku). The slide's caption says $a = 2.5$ cm, $b = 1$ cm; the arrows (TE_{10} at 3 GHz, TE_{20} at 6, TE_{50} and TE_{02} at 15) are those of a $5 \text{ cm} \times 2 \text{ cm}$ guide. TM arrows below the line sit on their TE twins: degeneracy.

verified/added

Prove that TM_{01} and TM_{10} modes do not exist 3

70 Ma

- TM: $H_z = 0$ and $E_z = E_0 \sin(m\pi x/a) \sin(n\pi y/b) e^{-\gamma z}$.
- $m = 0$ or $n = 0 \Rightarrow \sin 0 = 0 \Rightarrow E_z = 0$ everywhere.
- Every transverse component is a derivative of E_z (Step 1 with $H_z = 0$) $\Rightarrow E_x = E_y = H_x = H_y = 0$.
- All six components vanish $\Rightarrow$ **no field, no mode**. TM_{mn} needs $m \geq 1$ and $n \geq 1$.

Prove that the lowest cut-off TM mode is TM_{11} 3

82 Bh

- From above, TM needs $m, n \geq 1$.
- $f_c = \frac{c}{2} \sqrt{(m/a)^2 + (n/b)^2}$ grows with both m and n .
- Smallest admissible pair is $m = n = 1$: $f_{c, TM_{11}} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$, for any a, b .

Show that the dominant TE mode is TE_{10} ; compare TE_{10} with TE_{20} 2 3

71 Bh, 81 Bh

- TE: $H_z = H_0 \cos \cos$ survives with one index zero ($\cos 0 = 1$); only TE_{00} is excluded (constant H_z , all transverse components zero).
- Two candidates for the lowest f_c : $f_{c,10} = \frac{c}{2a}$, $f_{c,01} = \frac{c}{2b}$.
- $a > b \Rightarrow f_{c,10} < f_{c,01} \Rightarrow$ **TE_{10} is dominant**. It also beats TM_{11} , since $\frac{1}{a^2} < \frac{1}{a^2} + \frac{1}{b^2}$.
- 81 Bh prints "when $b > a$ ". With $b > a$ it is TE_{01} that is dominant; TE_{10} needs $a > b$. The paper has the inequality reversed. Prove the $a > b$ case and state this.

verified/added

	TE_{10}	TE_{20}
f_c	$c/2a$	$c/a = 2f_{c,10}$
λ_c	$2a$	a
Variation across a	one half-cycle	two half-cycles
Field components	E_y, H_x, H_z only	same three
Status	dominant	higher-order

- Single-mode band: $c/2a < f < c/a$ provided $a \geq 2b$ (otherwise TE_{01} arrives first at $c/2b$).

Dominant vs degenerate modes; for $a = 3b$ find the dominant mode among $TM_{01}, TM_{10}, TM_{11}, TM_{21}, TM_{12}, TM_{02}, TM_{20}$ 2+2 3

72 Ma, 82 Ba · full working in the companion, Problem 4

- With $a = 3b$: $f_c = \frac{c}{2a} \sqrt{m^2 + 9n^2}$.
- $TM_{01}, TM_{10}, TM_{02}, TM_{20}$ **do not exist** (an index is zero).
- TM_{11} : $\sqrt{10} = 3.162$; TM_{21} : $\sqrt{13} = 3.606$; TM_{12} : $\sqrt{37} = 6.083$ (in units of $c/2a$).
- **Dominant among those listed: TM_{11}** . (Overall, TE_{10} at 1 unit.)

Cut-off checks: does the guide pass the signal? 1+1+2 3 5

70 Bh, 72 Ash, 80 Bh · worked in the companion, Problem 3

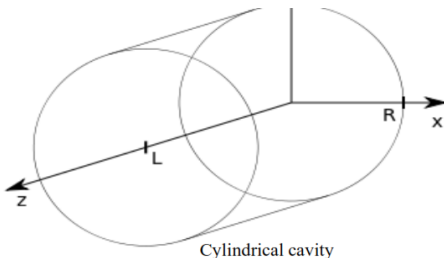
- Rule: compute f_c of the **dominant** mode; the signal propagates only if $f > f_c$.
- **70 Bh**: wall separation 5 cm, TE_{10} : $f_c = 3 \text{ GHz} > 1 \text{ GHz} \Rightarrow$ **1 GHz cannot propagate**; it decays at 59.2 Np/m.
- **80 Bh**: width 2.254 cm: $f_c = 6.65 \text{ GHz} > 6 \text{ GHz} \Rightarrow$ **6 GHz is not supported**.

4.3 Circular Waveguide**Field equations of a circular waveguide in TE / TM mode; TE vs TM; dominant mode** 7 8+2 10**HOT 4/24** 71 Ma, 75 Bh, 79 Bh, 82 Ba · the rectangular script in cylindrical co-ordinates**Set-up**

- Hollow circular pipe of inner radius a , co-ordinates (ρ, ϕ, z) , fields $\propto e^{-j\beta z}$; $k_c^2 = k^2 - \beta^2$, with $k = \omega\sqrt{\mu\epsilon}$.
- **Step 1**, the cylindrical version of the transverse equations (Waveguide.pdf p8, Pozar §3.4):

$$E_\rho = -\frac{j}{k_c^2} \left(\beta \frac{\partial E_z}{\partial \rho} + \frac{\omega\mu}{\rho} \frac{\partial H_z}{\partial \phi} \right), \quad E_\phi = -\frac{j}{k_c^2} \left(\frac{\beta}{\rho} \frac{\partial E_z}{\partial \phi} - \omega\mu \frac{\partial H_z}{\partial \rho} \right),$$

$$H_\rho = \frac{j}{k_c^2} \left(\frac{\omega\epsilon}{\rho} \frac{\partial E_z}{\partial \phi} - \beta \frac{\partial H_z}{\partial \rho} \right), \quad H_\phi = -\frac{j}{k_c^2} \left(\omega\epsilon \frac{\partial E_z}{\partial \rho} + \frac{\beta}{\rho} \frac{\partial H_z}{\partial \phi} \right).$$



Cylindrical cavity

Cylindrical co-ordinates. The figure is a circular cavity (a guide closed at both ends, §4.8); the guide itself is the same cylinder, open along z .

Source: Gangaju, slide 93.

TE_{nm}: $E_z = 0$

- $H_z = (A \sin n\phi + B \cos n\phi) J_n(k_c \rho) e^{-j\beta z}$.
- **Boundary**: $E_\phi(\rho = a) = 0$. Since $E_\phi \propto \partial H_z / \partial \rho$:
 $J'_n(k_c a) = 0 \Rightarrow \boxed{k_c = p'_{nm}/a}$, $p'_{nm} = m\text{-th root of } J'_n$.

$$E_\rho = \frac{-j\omega\mu n}{k_c^2 \rho} (A \cos n\phi - B \sin n\phi) J_n(k_c \rho)$$

$$E_\phi = \frac{j\omega\mu}{k_c} (A \sin n\phi + B \cos n\phi) J'_n(k_c \rho)$$

$$H_\rho = \frac{-j\beta}{k_c} (A \sin n\phi + B \cos n\phi) J'_n(k_c \rho)$$

$$H_\phi = \frac{-j\beta n}{k_c^2 \rho} (A \cos n\phi - B \sin n\phi) J_n(k_c \rho)$$

$$- f_c = \frac{p'_{nm}}{2\pi a \sqrt{\mu\epsilon}}, \quad Z_{TE} = \frac{E_\rho}{H_\phi} = \frac{\omega\mu}{\beta} = \frac{\eta k}{\beta}.$$

Step 2 and 3: separate the Helmholtz equation

- $\frac{\partial^2 h_z}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial h_z}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 h_z}{\partial \phi^2} + k_c^2 h_z = 0$.
- $h_z = R(\rho)P(\phi)$:
 - $P'' + n^2 P = 0 \Rightarrow P = A \sin n\phi + B \cos n\phi$; n an integer because the field repeats every 2π .
 - $\rho^2 R'' + \rho R' + (k_c^2 \rho^2 - n^2) R = 0$: **Bessel's equation**
 $\Rightarrow R = C J_n(k_c \rho) + D Y_n(k_c \rho)$.
 - $Y_n \rightarrow \infty$ at $\rho = 0 \Rightarrow D = 0$.

TM_{nm}: $H_z = 0$

- $E_z = (A \sin n\phi + B \cos n\phi) J_n(k_c \rho) e^{-j\beta z}$.
- **Boundary**: $E_z(\rho = a) = 0$ directly:
 $J_n(k_c a) = 0 \Rightarrow \boxed{k_c = p_{nm}/a}$, $p_{nm} = m\text{-th root of } J_n$.

$$E_\rho = \frac{-j\beta}{k_c} (A \sin n\phi + B \cos n\phi) J'_n(k_c \rho)$$

$$E_\phi = \frac{-j\beta n}{k_c^2 \rho} (A \cos n\phi - B \sin n\phi) J_n(k_c \rho)$$

$$H_\rho = \frac{j\omega\epsilon n}{k_c^2 \rho} (A \cos n\phi - B \sin n\phi) J_n(k_c \rho)$$

$$H_\phi = \frac{-j\omega\epsilon}{k_c} (A \sin n\phi + B \cos n\phi) J'_n(k_c \rho)$$

$$- f_c = \frac{p_{nm}}{2\pi a \sqrt{\mu\epsilon}}, \quad Z_{TM} = \frac{E_\rho}{H_\phi} = \frac{\beta}{\omega\epsilon} = \frac{\eta\beta}{k}.$$

- **Sign**: Waveguide.pdf p11 prints E_ϕ with a +; it is -.
 wg.py fails the curl equation with the +. verified/added

Root table and mode order (all a circular answer needs)

Mode	Root	λ_c	Order
TE ₁₁	$p'_{11} = 1.841$	$3.41a$	1st, dominant
TM ₀₁	$p_{01} = 2.405$	$2.61a$	2nd, lowest TM
TE ₂₁	$p'_{21} = 3.054$	$2.06a$	3rd
TE ₀₁	$p'_{01} = 3.832$	$1.64a$	4th (low-loss mode)
TM ₁₁	$p_{11} = 3.832$	$1.64a$	degenerate with TE ₀₁

$\lambda_c = 2\pi a/p$. Roots from `scipy` in `wg.py`. Worked example: $a = 2$ cm gives TE₁₁ at 4.40 GHz and TM₀₁ at 5.74 GHz (Waveguide.pdf p13).

TE vs TM in a circular guide (71 Ma)

	TE _{nm}	TM _{nm}
Longitudinal	$E_z = 0, H_z \neq 0$	$H_z = 0, E_z \neq 0$
Boundary	$J'_n(k_c a) = 0$	$J_n(k_c a) = 0$
Roots	p'_{nm}	p_{nm}
Dominant	TE ₁₁ , 1.841	TM ₀₁ , 2.405
Impedance	$\eta k/\beta > \eta$	$\eta\beta/k < \eta$
$m = 0$?	TE _{n0} absent	TM _{n0} absent
Special	TE ₀₁ : loss falls with f	TM ₀₁ : symmetric, rotary joints

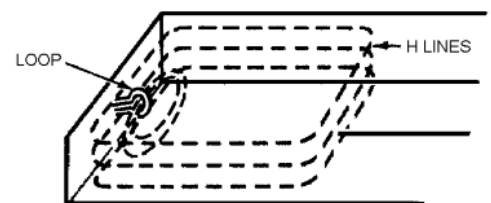
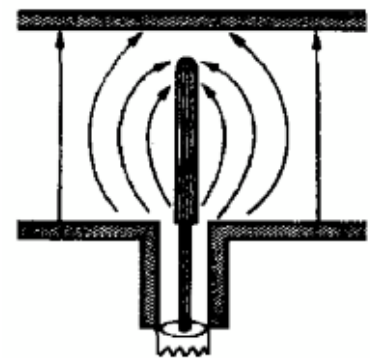
- **75 Bh:** dominant TE mode = TE₁₁, dominant TM mode = TM₀₁, and TE₁₁ is dominant overall since $1.841 < 2.405$.

4.4 Coupling Energy into a Waveguide: Probes, Loops and Slots

Probe coupling vs loop coupling 5

PIN 1/24 69 Bh · also the answer to "how is a cavity or klystron excited"

	Probe	Loop
Couples to	E field	H field
Acts as	quarter-wave monopole antenna	small magnetic dipole
Best position	centre of broad wall a , parallel to E lines, $\lambda_g/4$ from the shorted end (E maximum)	where H is strongest, e.g. at the shorted end; loop plane normal to H lines
Less coupling	shorten, move off centre, shield	rotate, or move to fewer H lines
Bandwidth	grows with probe diameter	grows with wire size
Power	door-knob probe: more area, more power	grows with loop diameter
Removal	same probe, reverse action	same loop, induced current



Probe (top) launches E lines; loop (bottom) links H lines.

Source: Gangaju, slides 3 and 6.

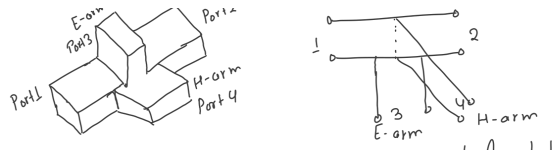
- **Slot / aperture:** a hole in the wall where the field is strong; gives **loose** coupling, used between a guide and a cavity or a second guide (directional coupler, §4.6).
- Choice depends on: voltage breakdown near an antinode, ease of adjustment, constancy of coupling under mechanical change, not disturbing an electron beam, matching.

4.5 Waveguide Tee Junctions and the Magic Tee

Explain the magic tee with a neat diagram and derive its S-matrix; E-plane vs H-plane tee 3+3 4 5 6

8

TOP 9/24 71 Ma, 71 Bh, 72 Ash, 72 Ma, 79 Ch, 79 Bh, 80 Ch, 80 Bh, 82 Bh · the matrices are catalogued in §3.6; this band adds construction, derivation and use



Magic tee: collinear ports 1, 2; E-arm 3; H-arm 4.

Construction

- A **hybrid** of the two simple tees on one junction:
 - **E-plane (series) tee**: side arm on the **broad** wall, its axis parallel to the main guide's E field. Port 3.
 - **H-plane (shunt) tee**: side arm on the **narrow** wall, in the plane of H. Port 4.
- Collinear arms 1 and 2 are symmetric about both side arms.
- A **post and iris** inside the junction match ports 3 and 4; without them the junction is simply a "hybrid tee" and is poorly matched.
- It is called **magic** because a matched 4-port with this symmetry turns out to be matched at **all** ports and isolates the two pairs, which nothing in its shape makes obvious.

Derivation of the S-matrix, the 4-mark version asked in 80 Bh and 72 Ma

1. 4 ports $\Rightarrow$ a 4×4 matrix $[S]$; write all sixteen S_{ij} out, i, j from 1 to 4.
2. **H-arm symmetry**: input at 4 reaches 1 and 2 equally and **in phase** $\Rightarrow S_{14} = S_{24}$.
3. **E-arm antisymmetry**: input at 3 reaches 1 and 2 equally and in **antiphase** $\Rightarrow S_{13} = -S_{23}$.
4. **E and H arms are orthogonal**: an E-arm wave cannot excite the H-arm mode $\Rightarrow S_{34} = S_{43} = 0$.
5. **Matched side arms** (post and iris) $\Rightarrow S_{33} = S_{44} = 0$.
6. **Reciprocity** $\Rightarrow S_{ij} = S_{ji}$.
7. **Unitary, column 3**: $|S_{13}|^2 + |S_{23}|^2 = 1$, with $|S_{13}| = |S_{23}| \Rightarrow S_{13} = 1/\sqrt{2}$. Same for column 4: $S_{14} = 1/\sqrt{2}$.
8. **Unitary, column 1**: $|S_{11}|^2 + |S_{12}|^2 + \frac{1}{2} + \frac{1}{2} = 1 \Rightarrow S_{11} = S_{12} = 0$; column 2 gives $S_{22} = 0$.
9. Result: $[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$, matched at all four ports, lossless, reciprocal.

Properties (read straight off the matrix)

- Into **H-arm** (4): equal halves at 1 and 2, **in phase**; nothing at 3.
- Into **E-arm** (3): equal halves at 1 and 2, **antiphase**; nothing at 4.
- Into **collinear** port 1: halves at 3 and 4; **nothing** at 2.
- Equal in-phase signals at 1 and 2: **sum** emerges at 4, nothing at 3.
- Equal antiphase signals at 1 and 2: **difference** emerges at 3, nothing at 4.
- All four ports matched: no reflection anywhere.

Applications

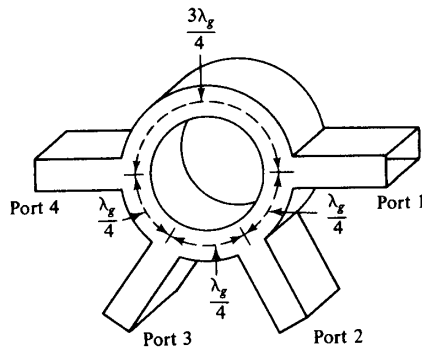
- Power combiner, e.g. two radar transmitters into one antenna (§3.8).
- Duplexer: transmitter and receiver on the isolated E and H arms (§3.6).
- Balanced mixer: signal on one side arm, local oscillator on the other.
- Impedance bridge / measurement: unknown and standard on 1 and 2, null detector on 3.
- Monopulse radar comparator: sum and difference channels.

E-plane tee vs H-plane tee

	E-plane (series)	H-plane (shunt)
Side arm on	broad wall, $\parallel$ E	narrow wall, $\parallel$ H
Side-arm input	antiphase at 1, 2	in phase at 1, 2
Relation	$S_{13} = -S_{23}$	$S_{13} = S_{23}$
Equivalent	series junction	shunt junction
In-phase at 1, 2	cancel in side arm	add in side arm
Matrix	§3.6, $-1/\sqrt{2}$ in row 3	§3.6, $-1/2$ at S_{12}

- Both are 3-ports and **cannot** be lossless, reciprocal and matched at once (§3.5): ports 1 and 2 stay mismatched with $|S_{11}| = 1/2$.

69 Bh 5, 80 Ba 4



Hybrid ring (rat race)

- Annular line, circumference $1.5\lambda_g$, four arms at $\lambda_g/4$, $\lambda_g/4$, $\lambda_g/4$ and $3\lambda_g/4$.
- Into port 1: the two ways round to port 3 differ by $\lambda_g/2$ (180°) → cancel; to 2 and 4 they add. Same behaviour as a magic tee, in planar form, but only at the design frequency.

Tee combinations: arms terminated, two tees joined, a 3-port coupler 4 4+4+2 5 8 10

HOT 5/24 70 Ma, 75 Bh, 77 Ch, 78 Ch, 82 Ba · each solved by connecting or terminating ports in `wg.py`; full working in the companion, Problems 6 to 8

- **Hybrid tee with a terminated arm** (82 Ba, 70 Ma): a matched load simply deletes that row and column.
 - E-arm terminated $\Rightarrow$ a matched, reciprocal **in-phase divider** from the H-arm; collinear ports isolated; lossy only when driven from a collinear arm (half goes to the load).
 - H-arm terminated $\Rightarrow$ the same but **antiphase** from the E-arm.
 - Both collinear arms terminated $\Rightarrow [S] = [0]$: E and H arms matched and completely isolated.
 - For **shorted** arms (82 Bh) see §3.6: $[S] = -[I]$.
- **Two magic tees joined E-arm to E-arm** (77 Ch): a matched, lossless, reciprocal 6-port. The **difference** signal of one tee crosses to the other and re-appears in antiphase; the **sum** signal stays in its own H-arm. Joined H to H (78 Ch), the roles swap.
- **3-port directional coupler** (75 Bh): the 4-port coupler with its isolated port loaded inside the casing. Matched and reciprocal, so by §3.5 it **must** be lossy, and it is: power driven backwards into the through port loses its coupled share in the internal load.

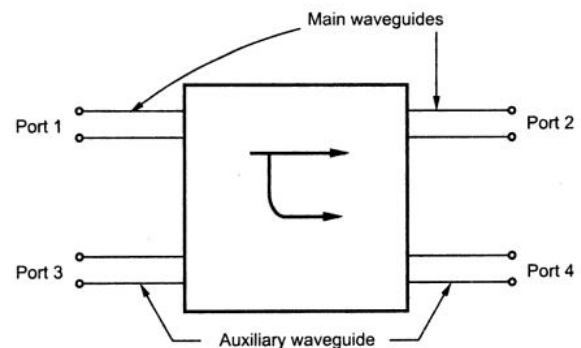
4.6 Directional Couplers

Explain directional couplers and the four parameters used to describe their performance 4 5 6

HOT 3/24 73 Bh, 79 Ch, 80 Ba

What it is

- A **4-port** junction that samples power travelling in **one direction** only.
- Main guide: port 1 (input) to port 2 (through).
- Auxiliary guide: port 4 (coupled) and port 3 (isolated), Gangaju's numbering.
- Ideal, all ports matched:
 - $1 \rightarrow 2$ passes almost all the power.
 - $1 \rightarrow 4$ a known small fraction.
 - $1 \rightarrow 3$ nothing.
- Uses: **power monitoring**, **reflectometers** and **VSWR** measurement (one coupler each way), network analysers, signal injection, and as a 3 dB hybrid.
- S-matrix: §3.6. Note §3.6 labels the coupled port 3; the numbering varies by text, the physics does not.



Block view, Gangaju slide 54.

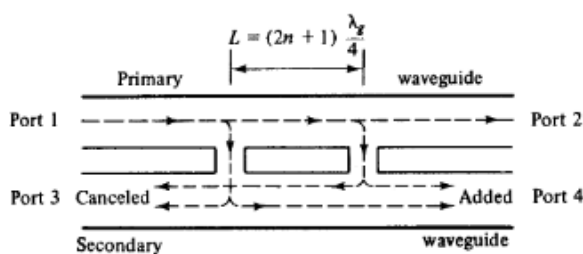
The four parameters (P_i = power at port i , input at 1)

Parameter	Definition	Measures
Coupling C	$10 \log \frac{P_1}{P_4}$ dB	how much is sampled
Directivity D	$10 \log \frac{P_4}{P_3}$ dB	forward vs backward discrimination
Isolation I	$10 \log \frac{P_1}{P_3}$ dB	leakage to isolated port
Insertion loss	$10 \log \frac{P_1}{P_2}$ dB	loss on the main line

- $I = C + D$. Example: $P_1 = 10$ mW, $P_4 = 1$ mW, $P_3 = 0.1$ μ W $\Rightarrow C = 10$ dB, $D = 40$ dB, $I = 50$ dB.
- Good coupler: D of 30 to 40 dB, VSWR near 1.

Two-hole coupler: why it is directional

- Two small holes in the common wall, spaced $d = (2n + 1)\lambda_g/4$.
- Each hole radiates a small wave into the auxiliary guide, both ways.
- **Forward** waves (towards port 4) travel equal total paths $\rightarrow$ **in phase** $\rightarrow$ add.
- **Backward** waves (towards port 3) differ by $2d = (2n + 1)\lambda_g/2 \rightarrow (2n + 1)\pi$ out of phase $\rightarrow$ **cancel**.
- Exact only at one frequency; more holes with tapered sizes widen the band.



Two-hole coupler, Gangaju slide 57.

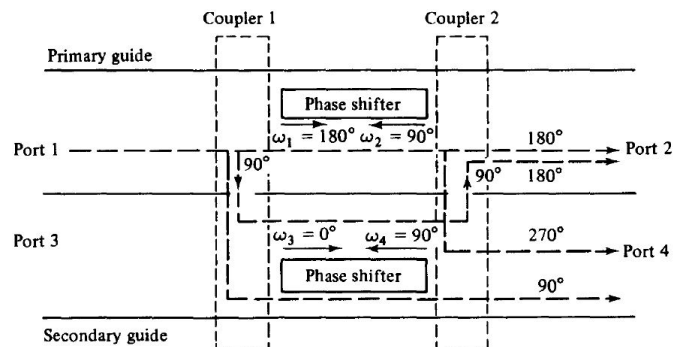
4.7 Ferrite Devices: Circulators, Isolators, Phase Shifters

Explain the microwave circulator 5

HOT 3/24 70 Bh, 72 Ash, 73 Ma · asked word for word each time

Circulator

- A **non-reciprocal** multi-port in which power entering port n leaves only by port $n + 1$: $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ (3-port), or round four ports.
- Ideal: **matched** at every port and **lossless**, but $S_{ij} \neq S_{ji}$.
3-port form (matrix of §3.5): $[S] = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$.
- Non-reciprocity comes from a **magnetised ferrite**.
- **Faraday rotation**: a plane wave along a ferrite rod in an axial magnetic field has its polarisation rotated by an angle set by the field strength and the rod length, and in the **same absolute sense** whichever way it travels. A return trip therefore doubles the rotation instead of undoing it.
- **Four-port construction** (slide 62): two 3 dB side-hole couplers, a primary and a secondary guide, and a non-reciprocal ferrite phase shifter in each.
 - Port 1 $\rightarrow$ port 2: both paths arrive with the same phase $\rightarrow$ add.
 - Port 1 $\rightarrow$ port 4: the paths differ by $180^\circ \rightarrow$ cancel.
 - The phase shifters behave differently in the reverse direction, so port 2 feeds port 3, port 3 feeds 4, port 4 feeds 1.



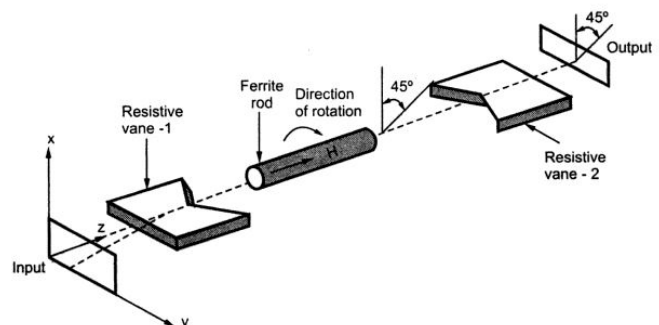
Four-port circulator: two 3 dB couplers plus ferrite phase shifters. Gangaju slide 62.

Applications

- **Duplexer**: transmitter port 1, antenna 2, receiver 3 (§3.6).
- Reflection amplifiers (tunnel diode, parametric): separate input from output.
- An **isolator**, by loading one port.

Isolator (uniline)

- Passes power one way, **absorbs** it the other way: $[S] = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$.
- Circulator with port 3 matched-terminated, or a Faraday-rotation isolator:
 - Input E vertical; ferrite rotates it 45° ; output guide twisted 45° to accept it.
 - Reflection from the load is rotated a further 45° in the same sense $\rightarrow 90^\circ$ from the input orientation $\rightarrow$ absorbed by a resistive vane.
- Use: between a klystron or magnetron and its load, so a mismatch cannot reflect back and pull the oscillator frequency.



Phase shifter (syllabus, not yet asked)

- Changes the phase of the transmitted wave by $\Delta\phi = 2\pi l/\lambda_g$, either by changing the **length** l or the **guide wavelength** λ_g .
- Mechanical: telescoping coaxial line; **dielectric vane** moved across the guide (more dielectric in the high-E centre, smaller λ_g , more phase).
- Electrical: **ferrite** under a variable bias field; PIN-diode switched lines.
- Uses: phased-array beam steering, phase discriminators, power-amplifier linearisation.

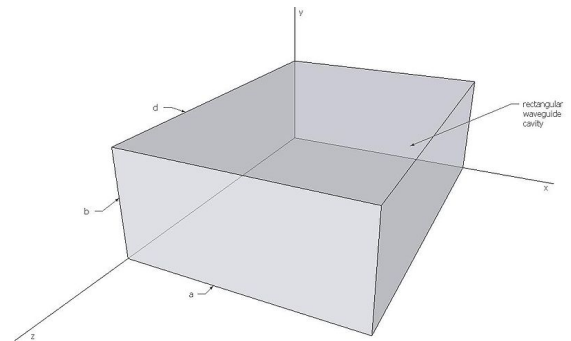
4.8 Cavity Resonators

Explain microwave cavity resonators 4 6

PIN 2/24 74 Ma, 74 Bh

What and why

- A closed metal enclosure that stores EM energy at discrete resonant frequencies: a section of waveguide shorted at **both** ends.
- Microwave replacement for the LC tank: lumped L and C fail above a few hundred MHz (radiation, skin effect, lead parasitics).
- Circuit analogue: stored E energy $\rightarrow$ C, stored H energy $\rightarrow$ L, wall loss $\rightarrow$ R.
- Types: **rectangular**, **circular**, **re-entrant** (the klystron cavity, Ch 5).
- Infinitely many modes; the lowest is the **dominant** mode.



Rectangular cavity $a \times b \times d$. Gangaju slide 90.

Rectangular cavity: resonant frequency

- Standing wave along z : the guide length must hold p half guide-wavelengths, $d = p\lambda_g/2 \Rightarrow \beta = p\pi/d$.
- $k^2 = k_c^2 + \beta^2$ gives

$$f_{mnp} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

- Allowed indices:
 - TE_{mnp}: $p \geq 1$, m and n not both zero.
 - TM_{mnp}: $m, n \geq 1$, $p \geq 0$.
- Dominant for $b < a < d$: **TE₁₀₁**, $f_{101} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{d^2}}$.
Example $2 \times 1 \times 3$ cm: 9.01 GHz, next TE₁₀₂ at 12.5 GHz.
- Circular cavity, radius a , length d :

$$f = \frac{c}{2\pi} \sqrt{\left(\frac{p'_{nm}}{a}\right)^2 + \left(\frac{l\pi}{d}\right)^2} \text{ (TE), with } p_{nm} \text{ for TM.}$$

Quality factor

- $Q = \omega \frac{\text{energy stored}}{\text{power lost}}$.
- Cavities reach 10^4 to 10^6 ; a lumped LC circuit about 10^2 .
- Losses add as reciprocals:

$$\frac{1}{Q_L} = \frac{1}{Q_c} + \frac{1}{Q_d} + \frac{1}{Q_{ext}}$$

- Q_c : wall conduction loss.
- $Q_d = 1/\tan\delta$: dielectric filling.
- Q_{ext} : power taken by the coupling; Q_L is the loaded Q.
- Coupling (probe, loop, aperture) is **under-, critically or over-coupled**; at critical coupling the input is matched.

Applications

- Klystron and magnetron cavities; wavemeters; filters; oscillator stabilisation; TR cells in duplexers.

4.9 Solid-State and Quantum Devices: Gunn Diode, MASER, Microwave Transistors

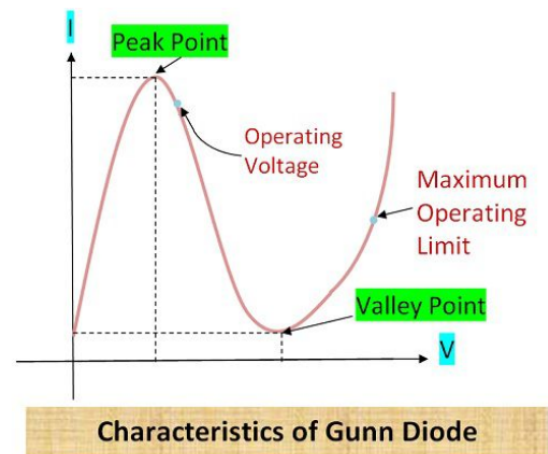
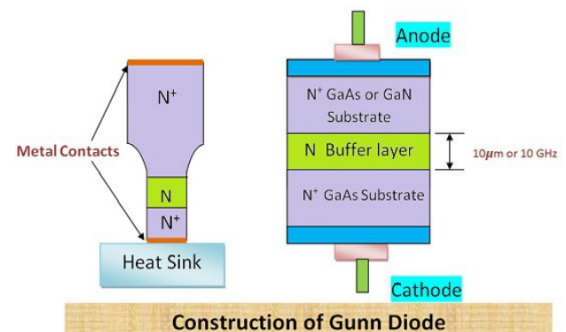
Gunn diode 4

PIN 1/24 79 Bh · short note

- **Transferred-electron device (TED):** bulk n-type GaAs (or InP), **no p-n junction**; “diode” only because it has two terminals.
- **Construction:** n⁺ contact / lightly doped n active layer (epitaxial, about 6 to 18 μm) / n⁺ substrate, metal contacts, heat sink.
- **Two-valley model** (Ridley-Watkins-Hilsum):
 - GaAs conduction band has a **lower valley** (light electrons, high mobility) and an **upper satellite valley** about 0.36 eV higher (heavy, low mobility).
 - Low field: electrons stay in the lower valley, current rises with voltage.
 - Above a threshold field (about 3.2 kV/cm) electrons gain enough energy to **transfer** to the upper valley → average mobility falls → drift velocity **falls as field rises**.
 - That is **negative differential resistance** between the peak and the valley of the I-V curve.
- **Oscillation:** a high-field **domain** forms at the cathode, drifts to the anode at $v_d \approx 10^5$ m/s, and a new one forms when it arrives:

$$f = \frac{v_d}{L} \quad \text{e.g.} \quad \frac{10^5}{10 \mu\text{m}} = 10 \text{ GHz}$$

- **Correction to the deck** (slide 72): it says the low-field current rises because electrons move from the valence band to the conduction band. It does not; the carriers are already conduction electrons in the lower valley. verified/added



Construction and the peak-valley characteristic. Gangaju slides 69 and 73.

(+) and (-)

- (+) small, cheap, low noise, wide band, reliable, 1 to 100+ GHz.
- (-) low efficiency (a few %), poor temperature stability, high bias current.

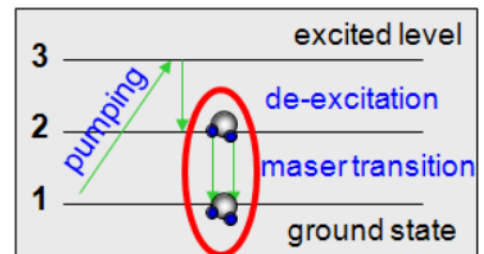
Applications

- Local oscillators, police and door-opening radar sensors, low-power transmitters, microwave relay links.

MASER 4

PIN 1/24 80 Ba · short note

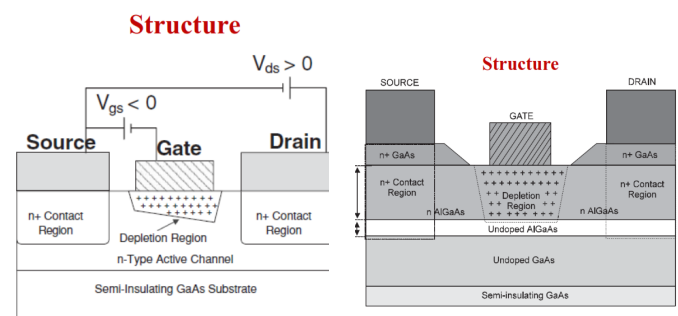
- **Microwave Amplification by Stimulated Emission of Radiation:** the microwave ancestor of the laser (Townes, 1954).
- A **quantum** device working on energy levels of atoms or molecules. The deck's "semiconductor device" is wrong. verified/added
- **Three-level principle:**
 - **Pump** at f_{13} lifts particles from level 1 to 3.
 - They decay without radiating to the **metastable** level 2 and accumulate → **population inversion** between 2 and 1.
 - An incoming signal at $f_{21} = (E_2 - E_1)/h$ **stimulates** emission of identical photons (same frequency, phase, direction) → coherent **amplification**.
- **Types:** ammonia gas maser (23.87 GHz, the first), hydrogen maser (1.42 GHz, frequency standard), solid-state ruby maser (Cr^{3+} in Al_2O_3 , tuned by a magnetic field, cooled to a few kelvin).
- **Key property:** extremely **low noise**.
- **Applications:** radio-telescope and deep-space receivers, satellite ground stations, atomic clocks.



Pump to level 3, relax to 2, stimulated transition 2 to 1.
Gangaju slide 78.

Microwave transistors (syllabus, not yet asked)

- Differ from low-frequency parts in **geometry** and **packaging**: interdigitated, overlay and matrix layouts cut lead inductance and junction capacitance.
- **Si BJT**: up to a few GHz. **HBT** (heterojunction, e.g. SiGe, InGaP/GaAs): higher f_T , amplifiers and MMICs.
- **GaAs MESFET**: Schottky-barrier gate on an n-channel; high electron mobility → well above 5 GHz; low noise, depletion-mode.
- **HEMT**: AlGaAs/GaAs heterojunction confines a 2-D electron gas in undoped GaAs, away from donor scattering → even higher mobility, beyond 20 GHz, lowest noise.



MESFET (left) and HEMT (right) cross-sections. Gangaju slides 85 and 86.

4.10 Attenuators, Terminations, Windows, Microstrip

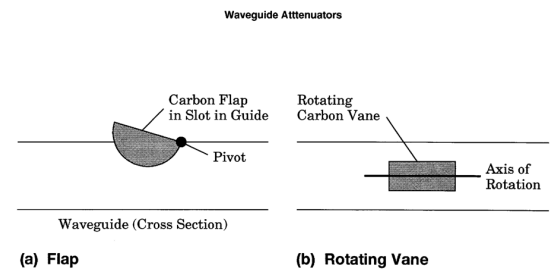
Microstrips 5

PIN 1/24 69 Bh · geometry and microstrip vs strip-line are in §1.6; stub realisation in §2.9

- Strip of width W on a substrate of thickness h and permittivity ϵ_r , ground plane underneath, air above.
- **Quasi-TEM**: the field lives partly in air and partly in the substrate, so it sees an effective permittivity $\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \frac{1}{\sqrt{1 + 12h/W}}$, $1 < \epsilon_{\text{eff}} < \epsilon_r$.
- $v_p = c/\sqrt{\epsilon_{\text{eff}}}$, $\lambda_g = \lambda_0/\sqrt{\epsilon_{\text{eff}}}$; Z_0 falls as W/h rises.
- (+) planar, cheap photolithography, components mount on top, MMIC friendly.
- (-) radiation at bends, dispersion, lower Q and power than waveguide, crosstalk.

Attenuators and terminations (syllabus, not yet asked)

- **Attenuator:** reduces power by a set amount with low VSWR; a resistive (carbon) vane or flap in the guide absorbs power through induced currents.
 - **Fixed:** lossy vane, or lossy walls.
 - **Variable:** **flap** lowered through a slot in the centre of the broad wall (deeper = more loss, since E is strongest there), **vane** moved from side wall to centre, or **rotary** vane in a circular guide.
- **Matched termination:** absorbs all incident power; a guide section ending in a **tapered** absorbing wedge so the impedance changes gradually and nothing reflects.
- Reference **short:** a metal plate; **sliding short:** movable, sets a reactance.



Flap and rotating-vane attenuators. Gangaju slide 39.

Windows, posts and screws (waveguide matching)

- **Inductive window:** diaphragms from the side walls → shunt L .
- **Capacitive window:** from top and bottom → shunt C ; limits power (breakdown).
- **Resonant window:** both → parallel LC , transparent at resonance.
- **Post / screw:** part-way in → capacitive; right through → inductive; $\lambda/4$ deep → series resonance.
- Waveguide equivalent of a stub (§2.6).

4.11 PYQ Mapping

Theory questions, covered in this document

- Rectangular waveguide field equations, TE mode → §4.1. 70 Bh 7, 74 Ma 4, 77 Ch 8, 79 Ch 10, 80 Ch 10, 80 Bh 4
- Rectangular waveguide field equations, TM mode → §4.1. **Most repeated topic of the chapter, with the TE list: 14 papers.** 70 Ma 7, 71 Bh 8, 72 Ma 5, 74 Bh 10, 76 Bh 10, 81 Ba 8, 81 Bh 8
- Rectangular waveguide: modes of propagation and critical parameters → §4.1 Step 6 and §4.2. 69 Bh 8
- Prove TM_{01} / TM_{10} do not exist → §4.2. 70 Ma 3
- Lowest TM mode is TM_{11} → §4.2. 82 Bh 3
- TE_{10} dominant; TE_{10} vs TE_{20} → §4.2, with the 81 Bh inequality defect. 71 Bh 2, 81 Bh 3
- Dominant and degenerate modes; $a = 3b$ → §4.2 and companion Problem 4. 72 Ash 5, 72 Ma 2+2, 82 Ba 3
- Circular waveguide, TE / TM field equations, TE vs TM, dominant mode → §4.3. 71 Ma 10, 75 Bh 8+2, 79 Bh 10, 82 Ba 7
- Probe vs loop coupling → §4.4. 69 Bh 5
- Magic tee: principle, S-parameters, derivation → §4.5 and §3.6. 71 Ma 5, 71 Bh 5, 72 Ash 3+3, 72 Ma 6, 79 Ch 5, 79 Bh 4, 80 Ch 5, 80 Bh 4
- Magic tee with both arms shorted → §3.6. 82 Bh 4
- E-plane vs H-plane tee → §4.5. 69 Bh 5, 80 Ba 4
- Hybrid tee; terminated E, H, co-planar arms → §4.5 and companion Problem 7. 70 Ma 8, 82 Ba 4+4+2
- Two magic tees joined at E-arms / in H-plane → §4.5 and companion Problems 6 and 8. 77 Ch 10, 78 Ch 5
- 3-port directional coupler, S-parameter analysis → §4.5, §3.5. 75 Bh 4
- Directional couplers; four performance parameters → §4.6. 73 Bh 6, 79 Ch 5, 80 Ba 4
- Circulators → §4.7. 70 Bh 5, 72 Ash 5, 73 Ma 5
- Cavity resonators → §4.8. 74 Ma 4, 74 Bh 6
- Gunn diode → §4.9. 79 Bh 4
- MASER → §4.9. 80 Ba 4
- Microstrips → §4.10 and §1.6. 69 Bh 5

Numerical content

- This chapter has a numerical companion, Ch4 - RF Microwave Components and Devices - Numerical Problems and Solutions: rectangular-guide parameters (80 Ba, 77 Ch), cut-off checks (70 Bh, 80 Bh), mode ranking (72 Ma), circular-guide cut-offs (75 Bh), and the junction problems (77 Ch, 78 Ch, 82 Ba).

Syllabus topics never asked in 24 papers

- Termination, phase shifter, attenuators, microwave transistor. Each has a short band above (§4.7, §4.9, §4.10) sized for a 4-mark short note.

Errors in the sources, corrected here

- Deck slide 18: $k_y = n\pi/a$ should be $n\pi/b$.
- Deck slide 22: figure caption gives the wrong guide size (it is $5\text{ cm} \times 2\text{ cm}$).
- Deck slide 72: Gunn low-field current is not a valence-to-conduction transition.
- Deck slide 77: a MASER is a quantum device, not a semiconductor device.
- Waveguide.pdf p11: TM E_ϕ has a minus sign.
- Paper 81 Bh Q3a: TE₁₀ is dominant when $a > b$, not $b > a$.

5. Microwave Generators

5 hrs · asked in **23 of 24 papers** · 5 syllabus sub-topics · **166 marks, 8.7% of the archive** → two tubes (klystron, magnetron) answer most of it; one idea (bunching) is the 2-mark rider on almost every question.

READ THIS FIRST

Chapter 5 is descriptive, with no numerical PYQ in 24 papers, and the questions come in a fixed shape: **a 2-mark definition, then a 6 to 8 mark “construction and working”**.

1. **The 2-mark riders** (bunching, density modulation, transit time, cross-field effect, slow-wave structure) → each has a boxed two-line answer below. Bunching alone is asked in 9 papers.
2. **Klystrons** (two-cavity, multicavity, reflex; 9 papers) → §5.3 and §5.4. One figure (the Applegate diagram) explains all three.
3. **Magnetron** (9 papers) → §5.5. The “45° / 60° / 90° phase shift between adjacent cavities” versions are the same answer plus one formula, $\phi = 2\pi n/N$.

Two things worth knowing before the paper starts:

- Every tube here does the same three things: **velocity-modulate** a beam, let it **bunch**, and **decelerate the bunches** in a field that takes their energy. Say this in the first line of any tube answer and the structure of the answer follows.
- Linear-beam (O-type) tubes: klystron, reflex klystron, TWT, BWO. Crossed-field (M-type) tubes: magnetron, crossed-field amplifier. An examiner who writes “field effect” or “cross-field” wants the M-type family.

5.1 Conventional Tubes at Microwaves and the Transit-Time Effect

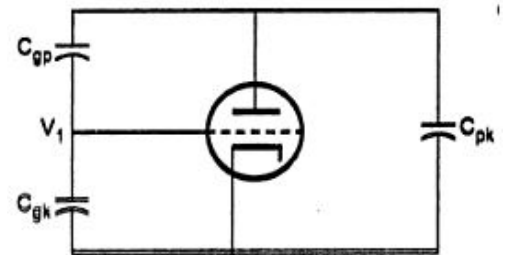
How is the output of conventional tubes reduced at microwaves? What is transit-time effect? **2 2+6**

2+8

HOT 3/24 70 Ma, 77 Ch, 79 Bh · always the 2-mark opener to a tube question

Why triodes and pentodes fail above about 1 GHz

- **Inter-electrode capacitance** C_{gp}, C_{gk}, C_{pk} :
 - Reactance $1/\omega C$ falls with frequency → shorts the input and output → gain falls.
 - Cure (spread electrodes, smaller area) raises transit time: a **compromise**.
- **Lead inductance**: ωL grows → degenerative feedback; cure: short leads (disk-seal tubes).
- **Transit-time effect** (below).
- **Cathode emission**: more current needs a bigger cathode, which adds capacitance; filament voltage and temperature are limited.
- **Power losses**: skin-effect resistance grows as $\sqrt{f}$; dielectric loss in the glass envelope; radiation from leads.



Inter-electrode capacitances of a triode. Karkee, Ch-5 deck, slide 4.

Transit-time effect: the 2-mark answer **2**

- **Transit time** = time an electron takes from cathode to anode. **Transit-time effect**: when this becomes a significant fraction of one RF cycle, the grid voltage changes while the electron is in flight → plate current lags the grid voltage → efficiency and gain collapse.
- Numbers: $\tau = 1$ ns is 0.001 cycle at 1 MHz but a **whole cycle** at 1 GHz. Above about 0.1 cycle the loss is serious.
- Estimate: 100 V gives $v = \sqrt{2eV/m} = 5.93 \times 10^6$ m/s; a 1 mm gap then takes 3.4×10^{-10} s, i.e. 0.34 cycle at 1 GHz.
- Cure in a conventional tube: closer electrodes and higher voltage (fights the capacitance). **Cure in a microwave tube: exploit it.** Klystrons, TWTs and magnetrons turn transit time into velocity modulation and bunching.

5.2 Velocity Modulation, Density Modulation and Bunching

What is bunching effect (density modulation)? Explain velocity modulation and bunching 2 5 10

TOP 9/24 70 Bh, 71 Bh, 72 Ash, 72 Ma, 73 Ma, 75 Bh, 80 Ba, 80 Bh, 81 Bh · the most repeated ask in the chapter

2-mark answer. Velocity modulation: an RF voltage across a gap speeds up electrons crossing in one half-cycle and slows those crossing in the other. **Bunching (= density modulation):** in the field-free drift space beyond, fast electrons catch up with slow ones sent earlier, so the uniform beam gathers into **bunches** once per RF cycle. The beam current is now modulated at the RF frequency and can drive a second cavity.

The equations (for the 5 and 10 mark versions)

- Beam accelerated by V_0 : $v_0 = \sqrt{2eV_0/m} = 0.593 \times 10^6 \sqrt{V_0}$ m/s.
- Buncher gap voltage $V_1 \sin \omega t$. Leaving at t_1 :

$$v(t_1) = v_0 \left[1 + \frac{\beta_i V_1}{2V_0} \sin \omega t_1 \right]$$

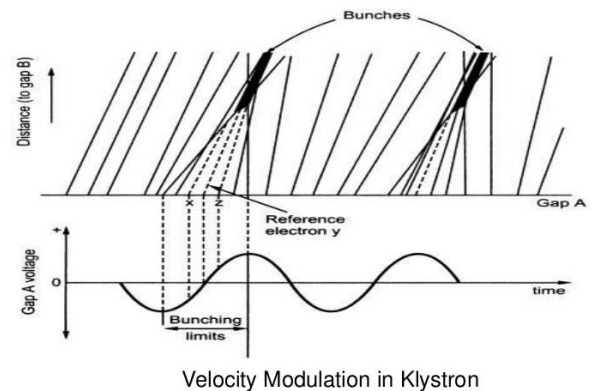
$\beta_i = \frac{\sin(\theta_g/2)}{\theta_g/2}$, the **beam coupling coefficient**; $\theta_g = \omega d/v_0$ the gap transit angle.

- Three reference electrons: #2 crosses at zero voltage (unchanged), #3 on the rising half (accelerated), #1 before it (decelerated). #1 and #3 converge on #2.
- Arrival at the catcher, drift length L : $t_2 = t_1 + L/v(t_1)$.
- Bunching parameter**

$$X = \frac{\beta_i V_1}{2V_0} \theta_0, \quad \theta_0 = \frac{\omega L}{v_0}$$

- Catcher current: $i_2 = I_0 + \sum_n 2I_0 J_n(nX) \cos n\omega(t_2 - \tau_0)$.
- Fundamental $2I_0 J_1(X)$ peaks at $X = 1.841$, where it is $1.164 I_0$ → optimum drift length.

Two cavity klystron



Applegate diagram: each line is one electron's distance vs time; steep lines are fast. Lines converge (a bunch) around the electron that crossed at zero gap voltage. Karkee deck, PDF p22.

- Checked by simulation in tubes.py:** electrons launched with this velocity law and drifted ballistically give a fundamental of exactly $2J_1(X)$.
- Too short a drift ($X < 1.84$): weak bunches. Too long: electrons overtake and the bunch spreads again (**debunching**).

5.3 Two-Cavity and Multicavity Klystrons

Describe the construction and operation of a two-cavity / multicavity klystron amplifier or oscillator

4 5 2+7 2+8

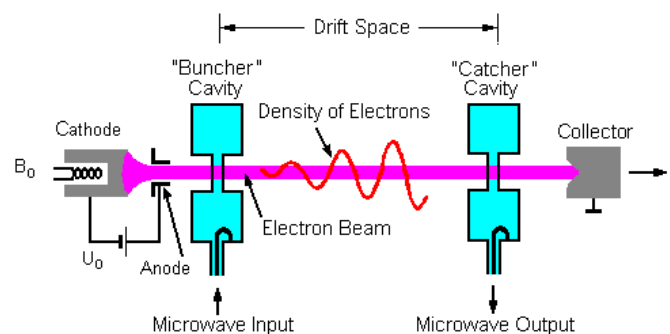
TOP 7/24 69 Bh, 70 Ma, 71 Bh, 72 Ash, 78 Ch, 80 Ch, 82 Ba · 70 Ma two-cavity; the rest mostly multicavity

Construction (two-cavity amplifier)

- Electron gun:** heated cathode, focusing electrode, anode at $+V_0$ forms a narrow beam; an axial magnetic field keeps it focused.
- Buncher cavity (input):** re-entrant cavity with grids across the beam; RF in via coaxial loop.
- Drift space:** field-free tube of length L .
- Catcher cavity (output):** identical cavity, RF out via loop.
- Collector:** absorbs the spent beam.
- Cavities and collector at the same dc potential, so the beam drifts at constant mean speed.

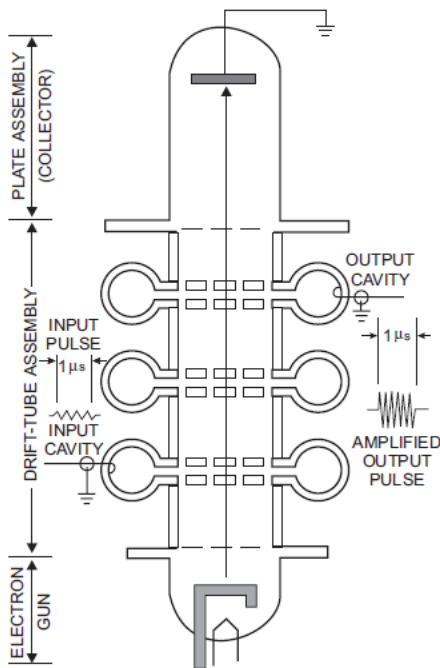
Operation

- Weak RF in the buncher sets up $V_1 \sin \omega t$ across its gap.



Two-cavity klystron. Karkee deck, slide 14.

2. **Velocity modulation:** electrons crossing are accelerated or decelerated (§5.2).
 3. **Bunching** in the drift space: velocity modulation becomes density modulation.
 4. Bunches reach the catcher once per cycle and **induce** RF current in it. Its field builds up to **oppose** the bunches: they are decelerated and give up kinetic energy, while the thin stream between bunches is accelerated and takes back only a little.
 5. Net energy flows from beam to RF → amplified output. The beam leaves slower and is collected.
- **Correction to the source** (PDF p23): it says “all the kinetic energy is converted”. Only part is; the beam still reaches the collector with most of its speed. Theoretical maximum efficiency is $\frac{1}{2} \times 1.164 = 58\%$ (at $X = 1.841$, catcher voltage equal to V_0); practical about 40%. verified/added
 - Gain of two cavities is low (about 10 dB) → multicavity.



Multicavity klystron. Karkee deck, slide 19.

Multicavity klystron (amplifier)

- One or more **intermediate cavities** between buncher and catcher, not externally driven.
- Weak bunches from cavity 1 excite cavity 2 → its larger gap voltage re-modulates the beam → tighter bunches → and so on.
- Each stage adds gain: **30 to 70 dB** overall.
- **Stagger tuning** the cavities slightly apart widens the bandwidth.
- Higher efficiency and power: up to MW pulsed, kW CW.
- Uses: UHF TV and troposcatter transmitters, satellite ground stations, radar transmitters, linear accelerators.

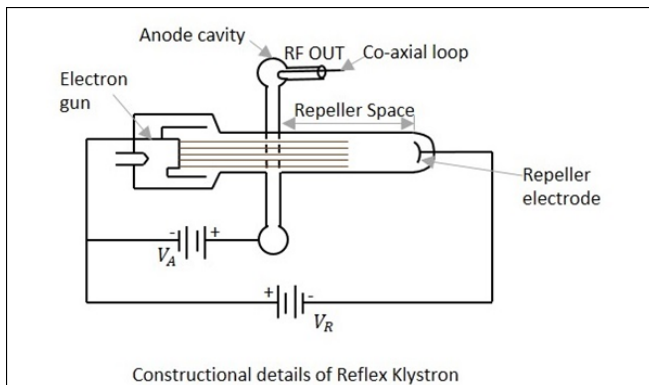
Klystron oscillator

- Feed part of the catcher output **back** to the buncher with the right phase (feedback loop, slide 18) → self-sustained oscillation. Multicavity oscillator: the same loop around a multicavity tube (72 Ash).
- Tuned by the cavities; for a single-cavity, voltage-tuned oscillator use the reflex klystron (§5.4).

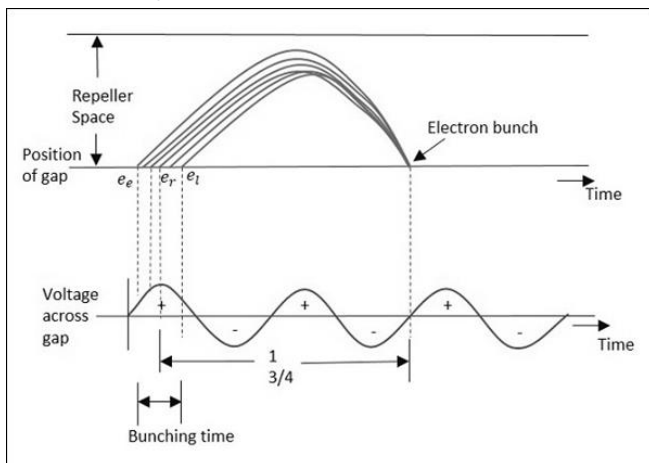
5.4 Reflex Klystron

Explain velocity modulation and bunching in a reflex klystron with neat diagrams and equations 5 10

PIN 2/24 72 Ma, 75 Bh · 75 Bh prints “multicavity reflex klystron”; a reflex klystron has one cavity



Karkee deck, PDF p29.



Electrons e_e , e_r , e_l return together: the bunch arrives $n + \frac{3}{4}$ cycles later. PDF p30.

- **Electronic tuning:** changing V_R changes T' and pulls the frequency a few percent; modes are the different n .
- **Efficiency:** maximum $\frac{2X'J_1(X')}{2\pi n + 3\pi/2}$ with $X' = 2.405$: 53% for $n = 0$ (not practical), **22.7% for $n = 1$** . The deck's 22.78% is this, with X' rounded to 2.408. Practical 10 to 20%.
- **Performance:** 4 to 200 GHz, 1 mW to 2.5 W. **Uses:** local oscillator in receivers, signal source for bench tests, portable links, pump for parametric amplifiers.

5.5 Cavity Magnetron and the Cross-Field Effect

Describe the construction and working of a cavity magnetron; magnetron with 45 / 60 / 90 degree phase shift between adjacent cavities 5 2+6 8 2+8 10

TOP 9/24 70 Bh, 71 Ma, 72 Ma, 74 Ma, 77 Ch, 79 Ch, 80 Ba, 81 Ba, 81 Bh · 81 Bh's "high-power oscillator using bunching and field effects" is this tube

2-mark answer: cross-field effect (72 Ma, 78 Ch, 81 Bh). An electron in **perpendicular** dc electric and magnetic fields feels $\vec{F} = -e(\vec{E} + \vec{v} \times \vec{B})$. It does not go straight to the anode: it follows a **cycloidal** path and drifts at $v = E/B$, perpendicular to both fields. Crossed-field (M-type) tubes use this drift to keep electrons in step with an RF wave for a long interaction.

Construction

- Electron gun, **one** re-entrant anode cavity at $+V_0$, and a **repeller** electrode at $-V_R$ beyond it.
- Output via a coaxial loop from the same cavity.

Operation

1. Noise starts a small RF voltage across the cavity gap.
2. Electrons crossing it are **velocity-modulated**.
3. In the retarding repeller field they stop and turn back. A **faster** electron goes deeper and takes **longer**: round trip

$$T' = \frac{2mLv}{e(V_R + V_0)}$$
4. So an early (fast) electron, the reference and a late (slow) one return **together**: the bunch forms in the **repeller space**.
5. The bunch re-crosses the gap. If the gap field then opposes it, it gives energy to the cavity and oscillation grows.

Condition for oscillation

- The bunch forms round the electron that left at zero gap voltage (going positive), and must return when the field most strongly retards it:

$$T' = \left(n + \frac{3}{4}\right) \text{ cycles}, \quad n = 0, 1, 2$$

- Power returned to the gap $\propto -J_1(X) \sin \theta_0$: positive for θ_0 in the second half-cycle, largest at $\frac{3}{4}$. `tubes.py` scans θ_0 with simulated electrons and finds the peak at 1.750 cycles.

Construction

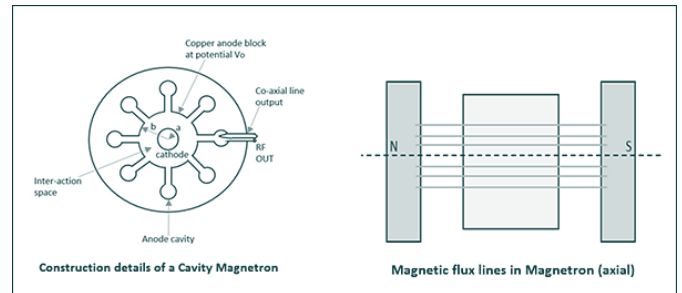
- Thick cylindrical **cathode** on the axis (radius a).
- Copper **anode block** (radius b) with N (typically 8) **resonant cavities** cut into it, opening onto the **interaction space**.
- Radial** dc field from anode voltage V_0 ; **axial** magnetic field from a permanent magnet, so $E \perp B$.
- Output: coupling loop in one cavity to a coaxial line or waveguide.
- Straps** join alternate anode segments to hold the tube in the π -mode.

Electron paths and the Hull cut-off

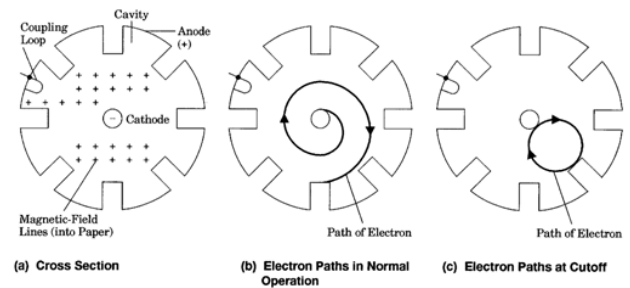
- $B = 0$: straight radial path to the anode.
- Small B : curved path, still reaches the anode.
- $B = B_c$: path just **grazes** the anode (cut-off).
- $B > B_c$: electron returns to the cathode without collecting.
- Hull cut-off field**:

$$B_c = \frac{\sqrt{8mV_0/e}}{b(1 - a^2/b^2)}$$

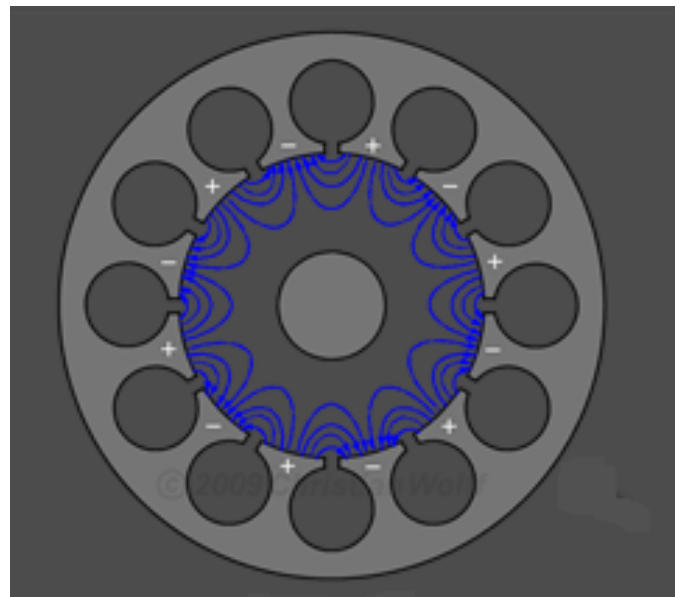
- Checked by integrating electron trajectories in `tubes.py`: for $V_0 = 10$ kV, $a = 5$ mm, $b = 10$ mm the grazing field is 0.0899 T, equal to the formula. verified/added
- The magnetron works just **above** B_c : without RF no electron reaches the anode.



Cavity magnetron cross-section and axial flux. PDF p55.



(b) normal operation, (c) at cut-off. PDF p61.



π -mode spoke pattern. Deck slide 50.

Operation (oscillator)

- Noise excites the cavities; RF fringing fields appear at the vane tips, alternating round the anode.
- Electrons circling the cathode meet these fields. Those in a **retarding** RF field slow down, feel less $v \times B$ force, and **drift outwards**, giving energy to the RF on the way.
- Those in an **accelerating** field curl back to the cathode (back-bombardment heats it, often enough to switch the heater off).
- The result is **phase focusing**: the electron cloud forms rotating **spokes**, each sitting in a retarding field. This is the magnetron's **bunching**.
- On the next half-cycle the field pattern reverses and the spokes rotate one cavity on, staying in step: sustained oscillation. Efficiency 40 to 70%.

The phase-shift versions: 45, 60, 90 degrees between adjacent cavities 6+2 8 3+7

74 Ma, 77 Ch, 80 Ba (45°); 81 Ba (60°); 72 Ma (90°)

- Around a ring of N cavities the total phase must be a whole number of turns:

$$\phi = \frac{2\pi n}{N}, \quad n = 0, \pm 1, \pm 2 \text{ up to } \pm \frac{N}{2}$$

- $n = N/2$: $\phi = 180^\circ$, the **π -mode**, normal operation.
 $n = 0$: zero mode, no RF between anode and cathode, unused.
- The RF pattern repeats every $360^\circ/\phi$ cavities, so there is **one spoke** per that many cavities:
 - 45°: 8 cavities per wavelength (in an 8-cavity tube $n = 1$, one spoke).
 - 60°: 6 cavities per wavelength ($N = 6$, $n = 1$).
 - 90°: 4 cavities per wavelength ($N = 8$ gives $n = 2$, two spokes).
 - 180°: 2 cavities, $N/2$ spokes.

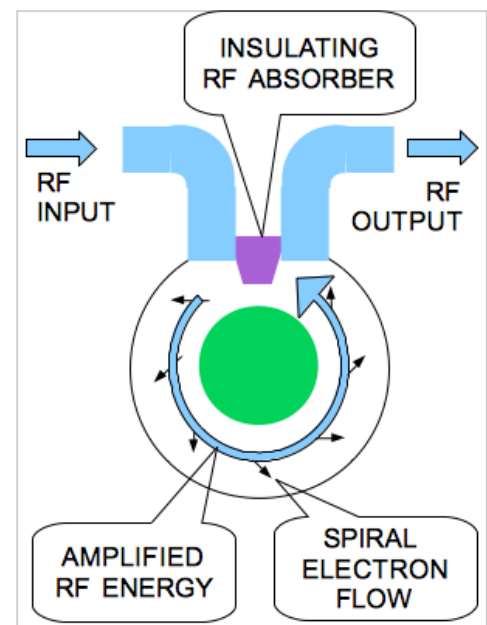
- **Synchronism**: the spokes rotate at the RF pattern's angular velocity ω/n ; set V_0 and B so the electron drift (E/B) matches it.
- **As a power amplifier** (80 Ba, 72 Ma): break the ring, feed RF in at one cavity and take it out several cavities later. The phase shift per cavity sets the wave's speed along the anode, $v_p = \omega p / \phi$ (p = cavity pitch), and the beam drift is matched to it. That is the crossed-field amplifier of §5.6.
- Write the sketch with the cavities marked + / – according to ϕ and draw the spokes in the retarding cavities; that sketch carries most of the marks.
- **Uses**: radar transmitters, microwave ovens (2.45 GHz), industrial heating, linear accelerators.

5.6 Crossed-Field Amplifier

Explain the working principle of a circular-beam type microwave cavity-controlled crossed-field amplifier 3+7 7

PIN 2/24 72 Ma, 82 Bh · the magnetron turned into an amplifier

- A **crossed-field amplifier (CFA)**, also amplatron or platinotron: magnetron geometry, $E \perp B$, but with an RF input and an RF output.
- **Circular-beam (re-entrant beam)**: the electron stream circulates round the cathode continuously, as in a magnetron.
- **Cavity-controlled (non-re-entrant RF circuit)**: the anode cavities form a slow-wave line from input to output; between output and input sits an **RF absorber / insulator**, so the wave cannot circulate and the tube does not oscillate.
- **Working**:
 - The input wave travels along the anode at $v_p = \omega p / \phi$.
 - Electrons drift at E/B ; matched to v_p they **bunch into spokes** in the retarding phase of the wave (§5.5).
 - Spokes move with the wave and give up energy cavity by cavity → the wave grows → amplified output.
 - Left-over electrons carry little RF past the absorber and re-enter the input region.
- **Performance**: high efficiency (up to about 70%), broad band (about 10%), **low gain** (10 to 15 dB), high power.
- **Use**: final power stage of radar transmitters, driven by a TWT.



Circular CFA: input, output, absorber, spiralling electron flow.
Student presentation 2078.

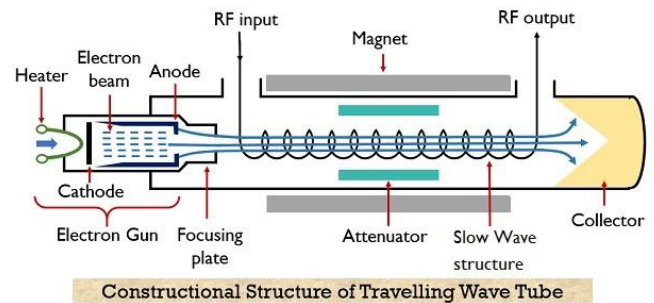
5.7 Travelling Wave Tube

Describe the construction and operation of a travelling wave tube (TWT) 2+6

PIN 2/24 79 Bh, 80 Bh

Construction

- **Electron gun:** cathode, focusing plate, anode.
- **Helix:** a wire coil round the beam, the **slow-wave structure**; RF in at the gun end, out at the collector end.
- **Attenuator** midway along the helix: absorbs reflected waves so the tube cannot oscillate.
- **Focusing magnets** (solenoid or periodic permanent magnets) along the tube.
- **Collector:** collects the spent beam.
- **No resonant cavities** → very wide bandwidth.



TWT construction. Student presentation 2078.

Operation

1. RF on the helix travels round the turns at about c , but **along the axis** only at $v_p \approx c \frac{\text{pitch}}{2\pi r}$ (a few percent of c).
 2. The beam voltage is set so the electrons move slightly **faster** than v_p .
 3. The axial RF field velocity-modulates the beam → bunches form, and because beam and wave travel together the interaction lasts the **whole length**.
 4. Being slightly faster, the bunches sit in the retarding field and give energy to the wave, which grows exponentially → gain 30 to 60 dB.
- **Correction to the source** (PDF p38): the beam does **not** move at the velocity of light; the whole point of the helix is to slow the wave down to the beam. Example (tubes.py): pitch 1 mm, radius 2 mm gives $v_p = 0.079c$, matched by a beam of about 1.6 kV; a 1 kV beam moves at $0.063c$. verified/added
 - **Performance:** 0.5 to 95 GHz, octave bandwidth, 5 to 20% efficiency, mW to 10 MW pulsed. **Uses:** satellite transponders (long life), broadband receivers, radar, troposcatter links, EW jammers.

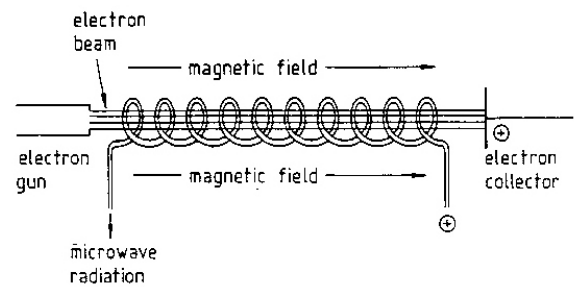
	Two-cavity klystron	TWT
Interaction	at two gaps only	continuous, whole helix
RF circuit	resonant cavities	non-resonant slow-wave helix
Bandwidth	narrow (a few %)	wide (an octave)
Wave	standing wave in cavities	travelling wave
Oscillation risk	needs feedback to oscillate	attenuator needed to prevent it

5.8 Backward Wave Oscillator and Slow-Wave Structures

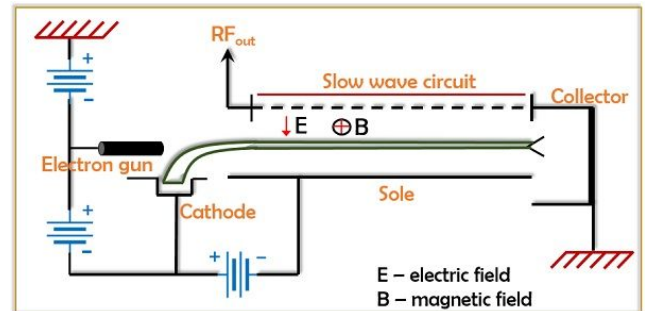
What is bunching effect? Explain the working principle of a BWO with neat diagrams 5 2+8

HOT 3/24 71 Bh, 73 Ma, 74 Bh · 71 Bh and 74 Bh as short notes

- **BWO** (carcinotron): a TWT-family tube used as an **oscillator**, microwave up to terahertz.
- Electron gun, slow-wave structure (helix or folded line), collector, magnetic focusing. **No RF input**.
- The beam interacts with a **backward space harmonic**: its phase velocity is along the beam (so they stay in step) but its **group velocity**, the direction energy flows, is **towards the gun**.
- Bunches give energy to this wave; the energy travels back to the gun end, where it velocity-modulates newly arriving electrons: **built-in feedback** → self-sustained oscillation.
- **Output is taken at the gun end**; the collector end is matched.
- **Frequency is tuned electronically** by the beam voltage (it sets which backward harmonic is in step): very wide tuning range.
- M-type BWO: crossed fields with a negative **sole** electrode (lower figure).
- Uses: swept signal generators, THz spectroscopy and imaging, jammers.
- **Not** a klystron: some student notes describe buncher and catcher cavities and a feedback conductor; a BWO has none.



O-type BWO: output near the gun. Deck slide 40.



Schematic representation of Backward Wave Oscillator

Electronics Desk

M-type BWO with sole. Student presentation 2078.

What do you mean by a slow (backward) wave structure? 2

73 Bh

- A **slow-wave structure** is a periodic RF circuit that makes the wave's axial phase velocity much smaller than c , so an electron beam of practical voltage can keep in step with it: helix, folded-back line, ring-bar, interdigital line, coupled cavities.
- It is **backward** when the space harmonic used has phase and group velocity in **opposite** directions: energy flows against the beam, as the BWO needs.

5.9 Low-Noise Amplifier

Explain the construction and working principle of an LNA 2+6

PIN 1/24 73 Bh · **verified/added** no source note covers it; amplifier design in Ch 6

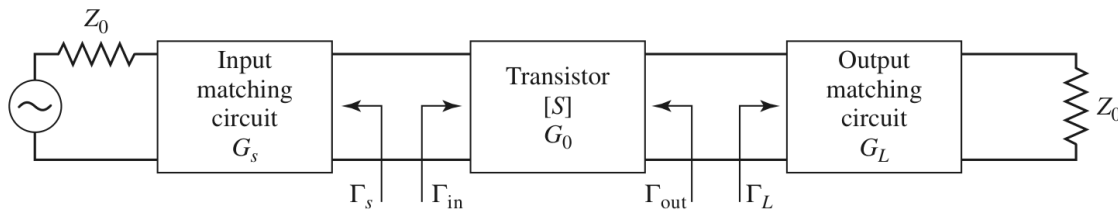
- **Purpose**: the first amplifier after the antenna. Its noise figure sets the whole receiver's, by Friis:

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} \quad (\text{one more term per stage})$$

e.g. a 1 dB, 20 dB-gain LNA before an 8 dB mixer gives 1.18 dB overall; the mixer first would give 8.02 dB (tubes.py).

- **Construction**:
 - **Low-noise device**: GaAs FET / HEMT (historically a parametric amplifier, maser or low-noise TWT).
 - **Input matching network**: presents the source reflection Γ_{opt} for **minimum noise figure**, not conjugate match.
 - **Output matching network**: conjugate match for gain.
 - **Bias network** with RF chokes and bypass capacitors; stabilising resistors so $K > 1$.
- **Working**: the signal is amplified with gain 10 to 20 dB while adding as little noise as possible (NF 0.5 to 2 dB); trade-off between minimum NF and maximum gain drawn as noise circles and gain circles on the Smith chart (Ch 6).

The construction, drawn



Source, **input matching network** (Γ_s , set to Γ_{opt} for an LNA), transistor $[S]$, **output matching network** (Γ_L , conjugate for gain), load. Every amplifier in Ch 6 is this block diagram. Pozar Fig 12.2.

5.10 PYQ Mapping

Theory questions, covered in this document

- Limitations of conventional tubes; transit-time effect → §5.1. 70 Ma 2, 77 Ch 2, 79 Bh 2
- Bunching effect / density modulation → §5.2. **Most repeated ask of the chapter.** 70 Bh 2, 71 Bh 2, 72 Ash 2, 73 Ma 2, 80 Ba 2, 80 Bh 2, 81 Bh 2
- Velocity modulation and bunching in a (multicavity) reflex klystron → §5.2 and §5.4. 72 Ma 5, 75 Bh 10
- Two-cavity klystron amplifier → §5.3. 69 Bh 5, 70 Ma 8
- Multicavity klystron amplifier / oscillator; klystron oscillator → §5.3. 71 Bh 8, 72 Ash 7, 78 Ch 8, 80 Ch 5, 82 Ba 4
- Cross-field effect → §5.5. 72 Ma 3, 78 Ch 2, 81 Bh 2
- Cavity magnetron; oscillator using bunching and field effects → §5.5. 70 Bh 8, 71 Ma 10, 79 Ch 5, 81 Bh 4
- Magnetron with 45 / 60 / 90 degree phase shift → §5.5. 74 Ma 8, 77 Ch 6, 80 Ba 6, 81 Ba 8, 72 Ma 7
- Circular-beam cavity-controlled crossed-field amplifier → §5.6. 82 Bh 7
- TWT construction and operation → §5.7. 79 Bh 6, 80 Bh 6
- Backward wave oscillator → §5.8. 71 Bh 5, 73 Ma 8, 74 Bh 5
- Slow backward wave structure; LNA → §5.8 and §5.9. 73 Bh 2+6

Numerical content

- This chapter has no numerical component: no paper in 24 sets a tube calculation. The formulas above (bunching parameter, Hull cut-off, reflex transit condition, helix velocity) are there to strengthen descriptive answers, and each is checked in `src/tubes.py`.

Corrections, and fixes made to the PYQ documents while building this chapter

- Source PDF p23: a klystron catcher converts only part of the beam energy (58% theoretical maximum), not all of it.
- Source PDF p38: a TWT beam moves far slower than light.
- Deck slide 27: reflex efficiency 22.78% is X' rounded; exact 22.7%.
- Student BWO slides describe klystron cavities; ignored.
- Sorted PYQ documents: 71 Bh's BWO short note was missing; 73 Ma is "bunching + BWO" and 73 Bh "slow backward wave structure + LNA"; 70 Ma asks no bunching question.

6. RF Design Practices

10 hrs · asked in **all 24 papers** · 3 syllabus sub-topics · **426 marks, 22.4% of the archive, the heaviest chapter** → one filter question and one amplifier numerical in almost every paper.

READ THIS FIRST

Chapter 6 is worth about 18 marks a paper and it is **two fixed questions**:

1. **A filter question** (20 of 24 papers): “why the insertion loss method”, “how is a Butterworth / Chebyshev LPF prototyped”, “design steps”, then “sketch it in microstrip”. → §6.1 to §6.3. The theory is one page; the marks are in the **procedure** and the **layout sketch**.
2. **An amplifier numerical** (22 papers, 10 to 18 marks): a transistor’s four S-parameters, “check stability and compare maximum gain for bilateral and unilateral modes”. → theory §6.4, every PYQ set solved in the companion. The papers **attach the formula sheet**; what they mark is the order of the steps and the arithmetic.

Two things worth knowing before the paper starts:

- Of the 13 distinct S-parameter sets in the archive, **three are not unconditionally stable** (the $0.894\angle -60.6^\circ$ set in 4 papers, 76 Bh, 73 Ma). For those a “bilateral maximum gain” does not exist and the answer must say so. The companion shows what to write instead.
- Oscillator and mixer theory are asked in only 2 papers each, as short notes (§6.5, §6.6).

6.1 Microwave Filters and the Insertion Loss Method

Why are microwave filters designed using the insertion loss method? How is a Butterworth / Chebyshev LPF prototyped? 4 5 2+6 3+5 6+2 8 10

TOP 20/24 70 Bh, 70 Ma, 71 Bh, 72 Ash, 72 Ma, 73 Bh, 73 Ma, 74 Bh, 74 Ma, 75 Bh, 76 Bh, 77 Ch, 78 Ch, 79 Bh, 80 Ba, 80 Bh, 81 Ba, 81 Bh, 82 Ba, 82 Bh · **the most repeated ask in the chapter**

Filters, and the two design methods

- **Filter**: two-port that passes a frequency band and attenuates the rest: LPF, HPF, BPF, BSF.
- **Image parameter method** (1930s): cascade simple sections with the right cut-off and attenuation.
 - Cannot specify the **whole** response; needs repeated iteration.
- **Insertion loss method (ILM)**: network synthesis from a specified **power loss ratio** over all frequencies.

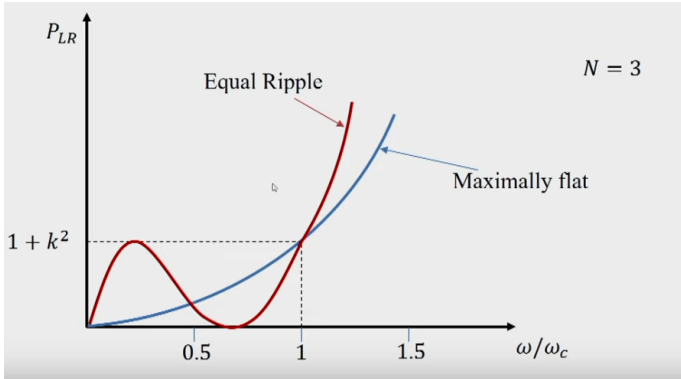
Power loss ratio

$$P_{LR} = \frac{P_{\text{available}}}{P_{\text{load}}} = \frac{1}{1 - |\Gamma(\omega)|^2} = 1 + \frac{M(\omega^2)}{N(\omega^2)}$$

- $|\Gamma|^2$ is even in ω , so M, N are polynomials in ω^2 .
- Insertion loss $IL = 10 \log P_{LR}$ dB; matched both ends:
 $P_{LR} = 1/|S_{21}|^2$.

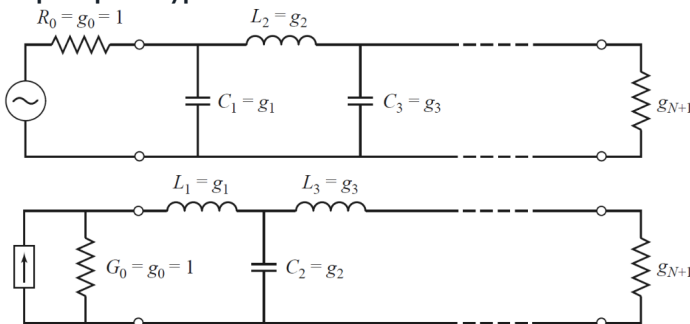
Why the insertion loss method (the 2 to 3 mark answer)

- Response is **completely specified** and met exactly, no trial and error.
- Explicit trade-offs: **binomial** for minimum passband loss, **Chebyshev** for sharpest cut-off, **linear phase** for low distortion.
- Order N follows from the stopband requirement.
- Starts from **normalised low-pass prototypes** ($R_0 = 1 \Omega$, $\omega_c = 1$ rad/s), so one table serves every filter after scaling.
- Designs improve systematically with N ; CAD tools implement it directly.



$N = 3$: equal-ripple is sharper, maximally flat is smoother. Gangaju Ch-6 deck, slide 11.

Low-pass prototype and element values



The two dual ladder forms, $g_0 = 1$ source, g_{N+1} load. Deck slide 7 (Pozar p.403).

- Butterworth** ($k = 1$): $g_k = 2 \sin \frac{(2k-1)\pi}{2N}$, $g_{N+1} = 1$.
- Chebyshev**, ripple L_r dB: $\beta = \ln \coth \frac{L_r}{17.37}$, $\gamma = \sinh \frac{\beta}{2N}$,
 $a_k = \sin \frac{(2k-1)\pi}{2N}$, $b_k = \gamma^2 + \sin^2 \frac{k\pi}{N}$; $g_1 = \frac{2a_1}{\gamma}$,
 $g_k = \frac{4a_{k-1}a_k}{b_{k-1}g_{k-1}}$; $g_{N+1} = 1$ for odd N , $\coth^2 \frac{\beta}{4}$ for even N .
- Order** (Butterworth): smallest N with $10 \log [1 + (\omega/\omega_c)^{2N}] \geq$ required stopband loss.

The three standard responses

- Maximally flat (Butterworth, binomial):**
 $P_{LR} = 1 + k^2(\omega/\omega_c)^{2N}$.
 - Band-edge loss $1 + k^2$; $k = 1$ gives -3 dB.
 - Beyond cut-off rises $20N$ dB/decade; flattest possible passband.
- Equal ripple (Chebyshev):** $P_{LR} = 1 + k^2 T_N^2(\omega/\omega_c)$.
 - Ripple of amplitude $1 + k^2$ in the passband.
 - Also $20N$ dB/decade, but $\frac{2^{2N}}{4}$ times more attenuation than Butterworth well beyond cut-off.
- Linear phase:** $\phi = A\omega[1 + p(\omega/\omega_c)^{2N}]$, maximally flat group delay; poorer selectivity.

Tables (computed in `filt.py`, equal to Pozar's)

N	g_1	g_2	g_3	g_4	g_5	g_6
Maximally flat						
1	2.0000	1.0000				
2	1.4142	1.4142	1.0000			
3	1.0000	2.0000	1.0000	1.0000		
4	0.7654	1.8478	1.8478	0.7654	1.0000	
5	0.6180	1.6180	2.0000	1.6180	0.6180	1.0000
Equal ripple, 0.5 dB						
1	0.6987	1.0000				
2	1.4029	0.7071	1.9841			
3	1.5963	1.0967	1.5963	1.0000		
4	1.6704	1.1925	2.3662	0.8419	1.9841	
5	1.7058	1.2296	2.5409	1.2296	1.7058	1.0000
Equal ripple, 3.0 dB						
1	1.9954	1.0000				
2	3.1014	0.5339	5.8095			
3	3.3489	0.7117	3.3489	1.0000		
4	3.4391	0.7483	4.3473	0.5920	5.8095	
5	3.4815	0.7619	4.5378	0.7619	3.4815	1.0000

- Deck example: 20 dB at 4 GHz for $f_c = 2.5$ GHz. The deck (via Pozar's chart) takes $N = 6$ (24.5 dB); the formula shows $N = 5$ already gives 20.4 dB. $N = 6$ is safe, just not minimal. verified/added

Why are Butterworth and Chebyshev common to prototype two-port filters? 4

70 Bh, 70 Ma

- Both power loss ratios are **polynomials** in ω^2 that give a **realisable lossless LC ladder** with closed-form element values.
- They sit at the two ends of the useful trade-off: **flattest passband** (Butterworth) vs **sharpest cut-off** for a given N (Chebyshev).
- Tabulated, normalised values mean a designer only **scales**; the same table gives LPF, HPF, BPF and BSF by transformation (§6.2).
- Ladders map directly onto transmission-line sections, so both are practical in microstrip (§6.3).

6.2 Filter Design Steps: Scaling and Conversion

Explain the steps of microwave filter design; how is a LPF prototype converted to other filters? 5 6+4

7+3 3+8+3 10

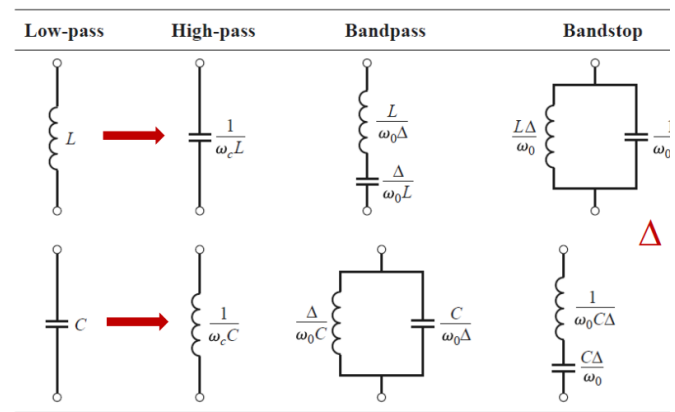
same papers as §6.1 · 71 Bh, 72 Ma, 74 Bh, 80 Bh, 81 Ba, 82 Ba, 82 Bh ask the steps outright

The procedure (write it as four boxes)

- Specifications:** type, cut-off f_c (or band f_1 to f_2), passband loss or ripple, stopband attenuation at a stated frequency, R_0 .
- Low-pass prototype:** choose response, find N , read g_1 to g_{N+1} ($R_0 = 1$, $\omega_c = 1$).
- Scaling and conversion:** impedance-scale by R_0 , frequency-scale to ω_c , and transform LP $\rightarrow$ HP / BP / BS.
- Implementation:** lumped (below about 5 GHz) or distributed microstrip: stepped impedance, Richards' stubs, Kuroda identities (§6.3).

Impedance and frequency scaling (LPF)

$$L'_k = \frac{R_0 L_k}{\omega_c}, \quad C'_k = \frac{C_k}{R_0 \omega_c}, \quad R'_s = R'_L = R_0$$



Element transformations, $\Delta = (\omega_2 - \omega_1)/\omega_0$, $\omega_0 = \sqrt{\omega_1 \omega_2}$. Deck slide 19 (Pozar p.414).

Conversion rules

- LPF $\rightarrow$ HPF**, $\omega \rightarrow -\omega_c/\omega$: series L becomes series C $= \frac{1}{R_0 \omega_c L_k}$; shunt C becomes shunt L $= \frac{R_0}{\omega_c C_k}$.
- LPF $\rightarrow$ BPF**: series L $\rightarrow$ series LC ($L_k R_0 / \omega_0 \Delta$, $\Delta / \omega_0 L_k R_0$); shunt C $\rightarrow$ shunt LC ($\Delta R_0 / \omega_0 C_k$, $C_k / \omega_0 \Delta R_0$).
- LPF $\rightarrow$ BSF**: the reverse pairing (series arms become parallel LC, shunt arms series LC).

Worked scaling check: Butterworth $N = 3$, $f_c = 2.5$ GHz, $R_0 = 50 \Omega$

Prototype	LPF element	HPF element	Check (ABCD in <code>filt.py</code>)
$g_1 = 1$ (shunt)	$C_1 = \frac{1}{50 \times 2\pi \times 2.5\text{e9}} = 1.27 \text{ pF}$	shunt $L = \frac{50}{\omega_c} = 3.18 \text{ nH}$	$ S_{21} = -3.01 \text{ dB at } 2.5 \text{ GHz, both}$
$g_2 = 2$ (series)	$L_2 = \frac{50 \times 2}{\omega_c} = 6.37 \text{ nH}$	series $C = \frac{1}{50 \omega_c \times 2} = 0.637 \text{ pF}$	
$g_3 = 1$ (shunt)	$C_3 = 1.27 \text{ pF}$	shunt $L = 3.18 \text{ nH}$	

6.3 Microstrip Implementation

Implement the filter using microstrip: LPF π / double π , double-pad T-type, double-section shunt-arm or series-arm, HPF, single-section series-arm BPF 2 3 3+5 5+3 7+3 8+2 10

TOP 13/24 72 Ash, 73 Ma, 74 Bh, 74 Ma, 76 Bh, 78 Ch, 79 Bh, 80 Ba, 80 Bh, 81 Ba, 81 Bh, 82 Ba, 82 Bh · the sketch carries the marks; label every section

Why microstrip, and the two methods

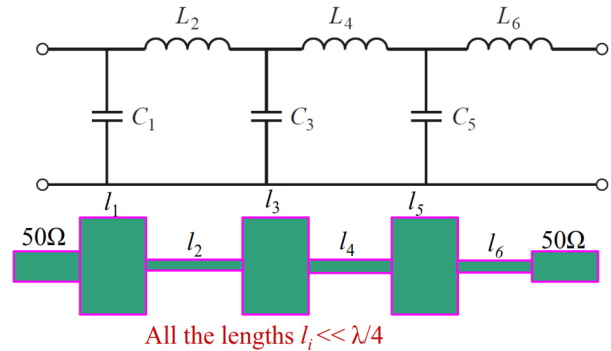
- Lumped L, C work to about 5 GHz; above that their size and spacing are no longer small in wavelengths.
- Microstrip: planar, cheap, light, easy to mount active devices; but lossier, lower power, radiates and couples.
- **Stepped impedance** (Hi-Z / Low-Z): alternate very high and very low impedance short lines.
- **Stubs** (Richards' transformation): inductor $\rightarrow$ short-circuited stub, capacitor $\rightarrow$ open-circuited stub, each $\lambda/8$ long at f_c ; Kuroda identities separate series stubs.

Why a short line is an L or a C

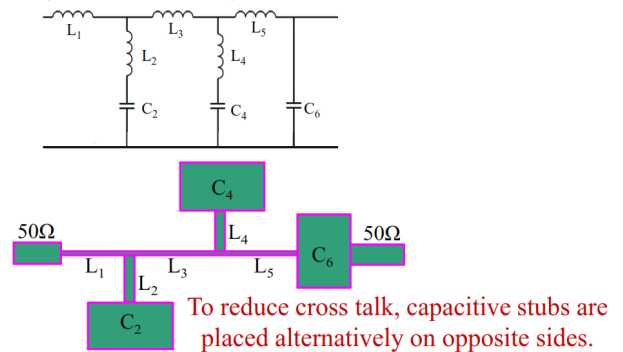
- T-model of a line length l : series $X/2 = Z_0 \tan(\beta l/2)$, shunt $B = \sin(\beta l)/Z_0$.
- **High Z_0 , short**: $X \approx Z_h \beta l$, $B \approx 0 \rightarrow$ a **series inductor**.
- **Low Z_0 , short**: $B \approx \beta l/Z_l$, $X \approx 0 \rightarrow$ a **shunt capacitor**.
- Design lengths:

$$\beta l = \frac{g_k R_0}{Z_h} \text{ (inductor),} \quad \beta l = \frac{g_k Z_l}{R_0} \text{ (capacitor)}$$

Keep $\beta l < 45^\circ$; typically $Z_h \approx 100$ to 120Ω , $Z_l \approx 20 \Omega$.



Stepped-impedance LPF and its layout. Deck slide 26.



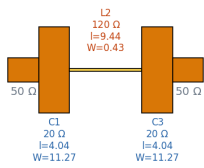
Elliptic LPF with capacitive stubs alternating sides to cut cross-talk. Deck slide 27.

- **Checked:** `filt.py` reproduces the deck's example table ($N = 6$, $Z_h = 120 \Omega$, $Z_l = 20 \Omega$, $\epsilon_r = 4.2$, $h = 1.58$ mm): $\beta l = 11.8, 33.8, 44.3, 46.1, 32.4, 12.3^\circ$, $W = 11.3$ and 0.43 mm.

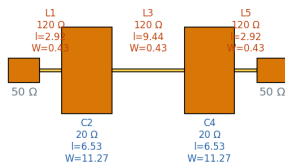
Vocabulary the papers use

- **Pad**: a wide, low- Z section = a **shunt capacitor**.
- **Series arm**: an element in the signal path (series L in an LPF, series C in an HPF). **Shunt arm**: an element to ground (shunt C as a pad or open stub; shunt L as a shorted stub).
- π -**section**: shunt, series, shunt (C-L-C for a LPF). **T-section**: series, shunt, series (L-C-L).
- **Double / triple**: two or three such sections cascaded.

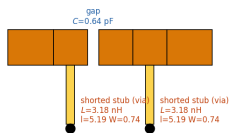
The layouts below are drawn to scale by `filt.py` from computed dimensions for one assumed specification, which is what a paper that says "illustrate an example" expects: Butterworth, $f_c = 2.5$ GHz, $R_0 = 50 \Omega$, $Z_h = 120 \Omega$, $Z_l = 20 \Omega$, $\epsilon_r = 4.2$, $h = 1.58$ mm ($W_{50} = 3.13$ mm).

LPF π -section, $N = 3$ (79 Bh, 73 Ma, 74 Ma, 72 Ash) π -section LPF (C-L-C), $N = 3$ (mm)

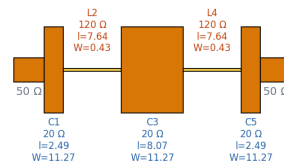
- $g = 1, 2, 1$: pads $\beta l = 22.9^\circ$ (4.04 mm), line 47.7° (9.44 mm).

Double-pad T-type, $N = 5$ (81 Ba; series-arm, 78 Ch)Double-pad T-type LPF (L-C-L-C-L), $N = 5$: two low- Z pads (mm)

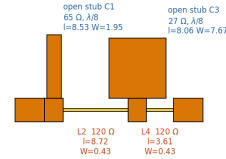
- L-C-L-C-L: the **series arms** are the three high- Z lines, the two **pads** the shunt capacitors. A “double-section series-arm” filter is this layout.

HPF π -section (74 Bh, 80 Bh, 82 Ba)HPF π -section (shunt L - series C - shunt L), $N = 3$, $f_c = 2.5$ GHz (mm)

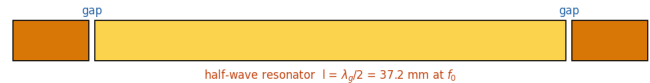
- From the C-L-C prototype: shunt $L = 3.18$ nH, series $C = 0.637$ pF.
- Series capacitor = a **gap** (or interdigital) in the strip; shunt inductor = a **short-circuited stub** through a via: $Z_s \tan \beta l = \omega_c L$ gives 5.19 mm of 100 Ω line.
- **Triple-pad series-arm HPF** (82 Ba): three pads separated by two gaps (the series capacitors) with shorted stubs from the pads.
- 80 Bh’s “second-order π ”: one series gap between two stubbed pads.

Double π -section, $N = 5$ (76 Bh)Double π -section LPF (C-L-C-L-C), $N = 5$ (dimensions in mm)

- $g = 0.618, 1.618, 2, 1.618, 0.618$: three pads, two high- Z lines.

Double-section shunt-arm LPF, $N = 4$ (80 Ba)Shunt-arm LPF: open stubs for shunt C, high- Z lines for series L, $N = 4$ (mm)

- Shunt capacitors as **open stubs**, $\lambda/8$ at f_c with $Z = R_0/g_k$ (Richards): 65 and 27 Ω ; series inductors as high- Z lines.

Single-section series-arm BPF (81 Bh)Single-section series-arm BPF: gap-coupled half-wave resonator, $f_0 = 2.25$ GHz

- The series LC arm of a BPF is a **half-wave resonator** coupled by a gap at each end: resonant at f_0 (series-resonant, passes), capacitively coupled (blocks DC and low frequencies).
- $l = \lambda_g/2$ at f_0 : 37.2 mm on 50 Ω at 2.25 GHz.
- More sections (more resonators) sharpen the skirts.

6.4 RF Amplifier Theory: Gain, Stability and Design Flow

Draw the design flowchart of a microwave amplifier; stability analysis; stability synthesis on the Smith chart; amplifier vs oscillator 4 5 3+8 5+5 6+4 8 10

TOP 10/24 71 Bh, 71 Ma, 73 Bh, 74 Bh, 75 Bh, 76 Bh, 78 Ch, 81 Ba, 81 Bh, 82 Ba · the theory half of the amplifier questions

Reflection coefficients seen by the device

$$\Gamma_{in} = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L} = \frac{S_{11} - \Delta\Gamma_L}{1 - S_{22}\Gamma_L}$$

$$\Gamma_{out} = S_{22} + \frac{S_{12}S_{21}\Gamma_S}{1 - S_{11}\Gamma_S} = \frac{S_{22} - \Delta\Gamma_S}{1 - S_{11}\Gamma_S}$$

$\Delta = S_{11}S_{22} - S_{12}S_{21}$. From $b_2 = S_{21}a_1 + S_{22}\Gamma_L b_2$ (deck slides 33 to 34).

Three power gains

- **Transducer** $G_T = P_L/P_{avs}$ (the one the papers use).
- **Available** $G_A = P_{avn}/P_{avs}$; **operating** $G_P = P_L/P_{in}$.

$$G_T = \frac{1 - |\Gamma_S|^2}{|1 - \Gamma_{in}\Gamma_S|^2} |S_{21}|^2 \frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2}$$

- **Unilateral** ($S_{12} \approx 0$): $G_{TU} = G_S G_0 G_L$ with $G_0 = |S_{21}|^2$; maximum at $\Gamma_S = S_{11}^*$, $\Gamma_L = S_{22}^*$:

$$G_{TU,max} = \frac{1}{1 - |S_{11}|^2} |S_{21}|^2 \frac{1}{1 - |S_{22}|^2}$$

- **Unilateral figure of merit** $M = \frac{|S_{11}||S_{12}||S_{21}||S_{22}|}{(1 - |S_{11}|^2)(1 - |S_{22}|^2)}$; error bound $\frac{1}{(1 + M)^2} < \frac{G_T}{G_{TU}} < \frac{1}{(1 - M)^2}$. Unilateral is acceptable if the bound is a few tenths of a dB.
- **Bilateral maximum** (simultaneous conjugate match, $\Gamma_S = \Gamma_{in}^*$, $\Gamma_L = \Gamma_{out}^*$), only if unconditionally stable:

$$\Gamma_S = \frac{B_1 - \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}, \quad \Gamma_L = \frac{B_2 - \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2, C_1 = S_{11} - \Delta S_{22}^* \text{ (and } 1 \leftrightarrow 2\text{)}. \text{ Then } G_{T,max} = \frac{|S_{21}|}{|S_{12}|} (K - \sqrt{K^2 - 1}).$$

Stability

- **Oscillation** is possible if $|\Gamma_{in}| > 1$ or $|\Gamma_{out}| > 1$ (negative resistance at a port).
- **Unconditionally stable**: $|\Gamma_{in}|, |\Gamma_{out}| < 1$ for every passive source and load. Test (Rollett):

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|} > 1 \text{ and } |\Delta| < 1$$

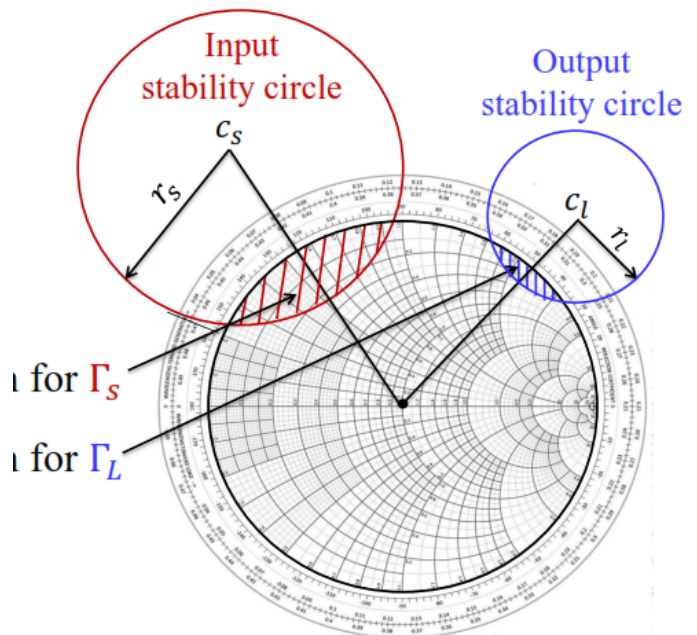
$$\text{or the single test } \mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{12}S_{21}|} > 1.$$

- **Conditionally (potentially) stable**: some Γ_S or Γ_L cause oscillation. Find them with **stability circles**:

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2}, \quad R_L = \left| \frac{S_{12}S_{21}}{|S_{22}|^2 - |\Delta|^2} \right|$$

$$C_S = \frac{(S_{11} - \Delta S_{22}^*)^*}{|S_{11}|^2 - |\Delta|^2}, \quad R_S = \left| \frac{S_{12}S_{21}}{|S_{11}|^2 - |\Delta|^2} \right|$$

- On the **output circle** $|\Gamma_{in}| = 1$; on the **input circle** $|\Gamma_{out}| = 1$. Each is a boundary, not a region.



Deck slides 45 to 46: $S_{11} = 0.65\angle -95^\circ$ etc. at 800 MHz, $K = 0.547$:

$C_S = 1.79\angle 122^\circ$, $R_S = 1.04$; $C_L = 1.30\angle 48^\circ$, $R_L = 0.45$. Hatched overlaps are the unstable Γ . Reproduced exactly in amp.py.

Which side of the circle is stable (synthesis on the Smith chart)

1. Plot C_L , R_L (and C_S , R_S) on the chart.
2. Test the **centre** $\Gamma_L = 0$: there $|\Gamma_{in}| = |S_{11}|$.
3. If $|S_{11}| < 1$ the centre is stable, so the stable region is the side of the circle that contains the centre:
 - $|C_L| > R_L$ (centre outside) $\Rightarrow$ stable **outside** the circle.
 - $|C_L| < R_L$ (centre inside) $\Rightarrow$ stable **inside**.
4. Same for the source circle with $|S_{22}|$.
5. Circle entirely outside the unit chart (or enclosing it on the stable side) $\Rightarrow$ unconditionally stable. Circle cutting the chart $\Rightarrow$ conditionally stable; choose Γ_S , Γ_L well inside the stable part.

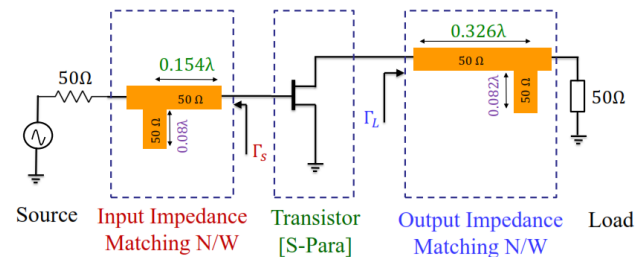
73 Bh, 74 Bh, 75 Bh give a sketched chart; 71 Bh gives the circles directly.

Worked: companion Problem 7.

Amplifier design flowchart (write as boxes joined by arrows)

1. **Specify** f , gain, noise figure, Z_0 ; **choose a transistor** (GaAs FET / HEMT / BJT) and bias; read its S-parameters.
2. **Stability**: compute Δ , K (or μ).
3. $K > 1$, $|\Delta| < 1$? **Yes** $\rightarrow$ step 4. **No** $\rightarrow$ draw stability circles, restrict Γ_S , Γ_L to the stable region (or add resistive loading / feedback), then step 4.
4. **Unilateral check**: compute M and the error bound.
 - Small $\rightarrow \Gamma_S = S_{11}^*$, $\Gamma_L = S_{22}^*$ (or gain circles for a specified gain).
 - Large $\rightarrow$ bilateral: Γ_S , Γ_L from B , C .
5. **Gain**: $G_{TU,max}$ or $G_{T,max}$; compare with the specification.
6. **Matching networks**: realise Γ_S and Γ_L from $50\ \Omega$ with a line plus a stub (Smith chart, Ch 2).
7. **Bias network**, DC blocks, RF chokes; **layout** in microstrip; simulate, build, measure, tune.

Final Amplifier Circuit Diagram



Final amplifier: input network, transistor, output network. Deck slide 58.

Amplifier circuit vs oscillator circuit in terms of stability factor 5

	Amplifier	Oscillator
Wanted	stable: no self-oscillation	unstable: oscillation on purpose
Stability factor	$K > 1$ and $ \Delta < 1$ (or $\mu > 1$), or Γ_S , Γ_L kept in the stable region	$K < 1$: device potentially unstable
71 Ma Port reflection Terminations	$ \Gamma_{in} , \Gamma_{out} < 1$ conjugate match for gain	$ \Gamma_{in} > 1$ (negative resistance) Γ_S chosen inside the unstable region; $\Gamma_{in}\Gamma_S = 1$, $\Gamma_{out}\Gamma_L = 1$
Input signal	required	none; starts from noise
Feedback	avoided, or negative	positive, loop gain $A\beta \geq 1$

Stability and maximum gain numericals (bilateral and unilateral) 5+5 10 12 15 18

TOP 22/24 69 Bh, 70 Bh, 70 Ma, 71 Bh, 71 Ma, 72 Ash, 72 Ma, 73 Ma, 74 Bh, 74 Ma, 76 Bh, 77 Ch, 78 Ch, 79 Bh, 79 Ch, 80 Ba, 80 Bh, 80 Ch, 81 Ba, 81 Bh, 82 Ba, 82 Bh · every set is solved in the companion

- 13 distinct S-parameter sets across 22 papers; the companion works one in full, tabulates the rest with every intermediate, and splits out the cases that need a different answer (conditionally stable, given terminations, full matching design, circles given directly).

6.5 Microwave Oscillator Theory

Microwave oscillator theory 8

PIN 2/24 70 Ma, 71 Ma · 71 Ma as the amplifier-vs-oscillator contrast of §6.4

Feedback view

- Gain A with positive feedback β : $\frac{e_o}{e_i} = \frac{A}{1 - A\beta}$.
- $A\beta = 1 \rightarrow$ finite output with no input: **Barkhausen condition**. To start, $A\beta \approx 1.1$ to 1.2 ; oscillation grows from noise until gain compression brings $A\beta$ back to 1 .

Negative-resistance view (the microwave one)

- A device port with $|\Gamma_{out}| > 1$ has **negative resistance**.
- One-port oscillator**: device $Z_{out} = R_{out} + jX_{out}$ with $R_{out} < 0$, load Z_L . Loop condition $\Gamma_{out}\Gamma_L = 1$ gives

$$R_L + R_{out} = 0, \quad X_L + X_{out} = 0$$

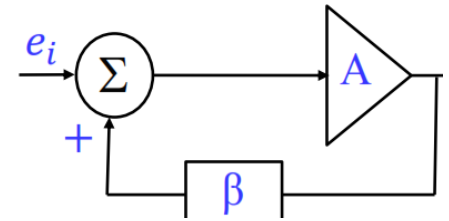
- To start oscillation the circuit must have net negative resistance: $R_L < |R_{out}|$. The deck's rule $|R_{out}| \geq 1.2R_L$ (Pozar uses $R_L = |R_{out}|/3$).
- Two-port oscillator**: transistor with a **generator tuning network** (Γ_S) and a **load network** (Γ_L). Conditions:
 - $K < 1$: device potentially unstable.
 - $\Gamma_{in}\Gamma_S = 1$.
 - $\Gamma_{out}\Gamma_L = 1$ (equivalent to 2, shown by substituting Γ_{in}).

Two-port design steps

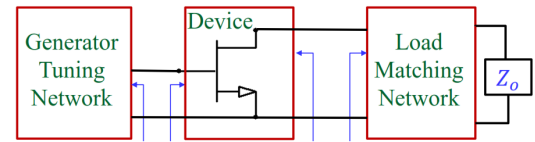
- From the S-parameters find K ; need $K < 1$ (use a common-gate / common-base configuration or series feedback to make it so).
- Draw the **input stability circle**.
- Choose Γ_S **inside the unstable region**, usually on the rim for the largest $|\Gamma_{out}|$; realise it with an inductor or shorted stub.
- $\Gamma_{out} = S_{22} + \frac{S_{12}S_{21}\Gamma_S}{1 - S_{11}\Gamma_S}$, $|\Gamma_{out}| > 1 \rightarrow Z_{out} = R_{out} + jX_{out}$ with $R_{out} < 0$.
- Choose $R_L = |R_{out}|/3$ (start-up margin), $X_L = -X_{out}$; design the load matching network.

Worked: Gunn diode one-port (deck slide 66, checked in amp.py)

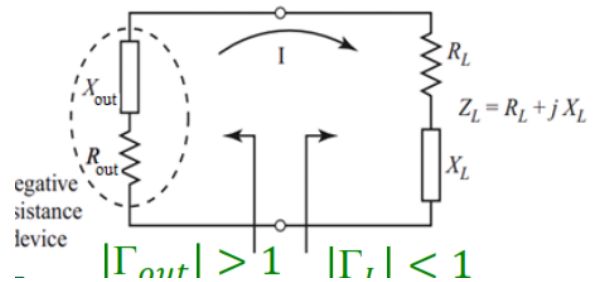
- $\Gamma_{out} = 1.24\angle 30^\circ$ at 10 GHz. Plot $1/\Gamma_{out}^* = 0.806\angle 30^\circ$, read z and negate the real part: $Z_{out} = -68.9 + j159.0 \Omega$ (deck reads $-70 + j160$).
- Resonance: $X_L = -159 \Omega$, a capacitor $C = 1/(2\pi f \cdot 159) = 0.100$ pF.
- Start-up: **the deck picks $R_L = 60 \Omega$, which breaks its own rule**: $|R_{out}| \geq 1.2R_L$ needs $R_L \leq 57.5 \Omega$. Use $R_L = 23 \Omega$ (Pozar's $|R_{out}|/3$) or at most 57Ω . verified/added



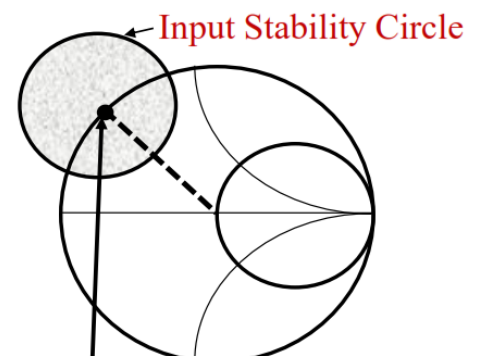
Positive feedback. Deck slide 60.



Two-port oscillator. Deck slide 62.



One-port: negative-resistance device and load. Deck slide 65.



Γ_S chosen inside the input stability circle. Deck slide 67.

Oscillator types and stabilisation

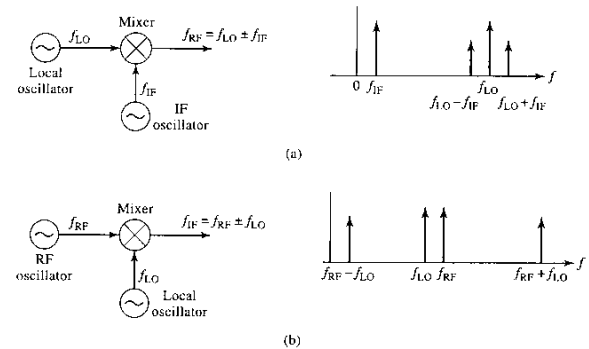
- Transistor (FET, BJT), Gunn, IMPATT (one-port); YIG-tuned (wide tuning), DRO (dielectric resonator, low phase noise).
- Frequency stability: high-Q resonator in the tuning network; isolator at the output (Ch 4).

6.6 Microwave Mixer

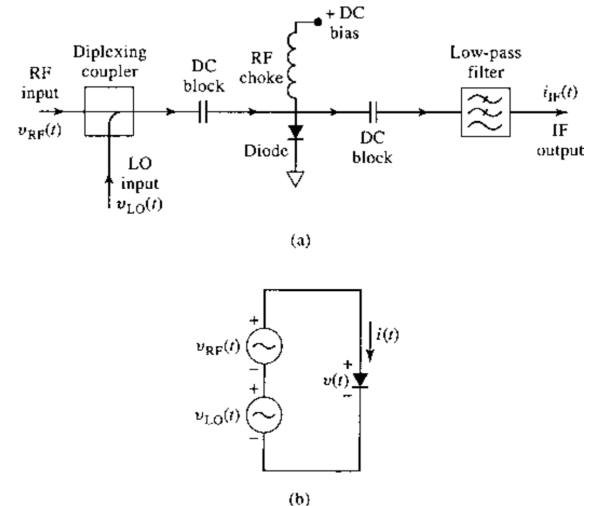
Mixer theory / microwave mixer 5

PIN 2/24 70 Bh, 72 Ma · verified/added the deck has no mixer slides; written from Pozar Ch 13

- **Mixer:** a **nonlinear** three-port (diode or FET) that multiplies an RF signal f_{RF} with a local oscillator f_{LO} to translate it in frequency.
- **Square-law principle:** $i = av^2$ with $v = V_{RF} \cos \omega_{RF}t + V_{LO} \cos \omega_{LO}t$ contains $aV_{RF}V_{LO} \cos(\omega_{RF} \pm \omega_{LO})t$: **sum and difference** (checked by FFT in amp.py).
- **Down-conversion:** keep $f_{IF} = |f_{RF} - f_{LO}|$ with a filter; the **superheterodyne** receiver then amplifies at a fixed IF.
- **Up-conversion** (transmitters): keep $f_{RF} = f_{LO} + f_{IF}$.
- **Image frequency:** $f_{IM} = f_{LO} \pm f_{IF}$ on the other side of the LO also lands on $f_{IF} \rightarrow$ reject it with an RF pre-filter or an image-reject mixer.
- **Figures of merit:** conversion loss $L_c = 10 \log(P_{RF}/P_{IF})$ (typically 4 to 7 dB for diode mixers), noise figure, LO-RF and LO-IF isolation, 1 dB compression, intermodulation.
- **Types:**
 - **Single-ended diode:** one Schottky diode; simple, poor isolation.
 - **Balanced:** two diodes fed through a 90° or 180° hybrid (magic tee, rat race, Ch 4): cancels LO AM noise, better isolation, better input match.
 - **Double-balanced:** diode ring with two baluns: suppresses even harmonics and both RF and LO at the IF port.
 - **FET mixers:** conversion **gain** instead of loss.



(a) up-conversion, (b) down-conversion, each with its output spectrum: both $f_{LO} \pm f_{IF}$ (or $f_{RF} \pm f_{LO}$) appear and a filter keeps one. Pozar Fig 13.24.



(a) single-ended diode mixer: RF and LO combined in a duplexing coupler, DC-biased diode, low-pass filter to the IF port; (b) its ideal equivalent, $v_{RF} + v_{LO}$ across the diode. Pozar Fig 13.25.

6.7 PYQ Mapping

Theory questions, covered in this document

- **Insertion loss method; why it is used** $\rightarrow$ §6.1. 72 Ash 5, 73 Bh 2+6, 73 Ma 8, 75 Bh 5, 80 Ba 3, 81 Bh 8, 82 Bh 3
- **Butterworth and Chebyshev common to prototype filters** $\rightarrow$ §6.1. 70 Bh 4, 70 Ma 4
- **LPF prototyping (Butterworth, Chebyshev); filter parameters** $\rightarrow$ §6.1. 72 Ash 5, 72 Ma 5, 74 Ma 3, 76 Bh 8, 77 Ch 8, 79 Bh 6, 80 Bh 10
- **Filter design steps; conversion to other filter types** $\rightarrow$ §6.2. 71 Bh 5, 74 Bh 6, 80 Bh 10, 81 Ba 7, 82 Ba 7, 82 Bh 8
- **Microstrip implementation (LPF π , double π , double-pad T, shunt-arm, series-arm, HPF, BPF)** $\rightarrow$ §6.3. 72 Ash 3, 73 Ma 2, 74 Bh 4, 74 Ma 5, 76 Bh 2, 78 Ch 10, 79 Bh 2, 80 Ba 5, 80 Bh 10, 81 Ba 3, 81 Bh 2, 82 Ba 3, 82 Bh 3
- **Design steps of a given 5th-order ladder (figure)** $\rightarrow$ companion Problem 10. 82 Bh 3+8+3
- **Amplifier design flowchart** $\rightarrow$ §6.4. 71 Bh 5, 73 Bh 10, 74 Bh 4, 76 Bh 4, 78 Ch 10, 81 Ba 3, 81 Bh 3, 82 Ba 4
- **Stability analysis; synthesis from a sketched Smith chart** $\rightarrow$ §6.4 and companion Problem 7. 73 Bh 10, 74 Bh 8, 75 Bh 5, 81 Ba 4, 82 Ba 4
- **Stability of given circles C_S, R_S, C_L, R_L** $\rightarrow$ companion Problem 7. 71 Bh 5
- **Amplifier vs oscillator in terms of stability factor** $\rightarrow$ §6.4. 71 Ma 5

- **Microwave oscillator theory** → §6.5. 70 Ma 8
- **Mixer theory** → §6.6. 70 Bh 5, 72 Ma 5

Numerical content

- **This chapter has a numerical companion, *Ch6 - RF Design Practices - Numerical Problems and Solutions*:** every amplifier S-parameter set in the archive (22 papers) and the 82 Bh filter design. All values from `src/amp.py` and `src/filt.py`.

Errors found and corrected

- Deck slide 51 (8 GHz example): $|\Delta| = 0.168$, $K = 3.53$; exactly 0.173 and 3.445 (verdict unchanged).
- Deck slide 52: $M = 0.04$ and bounds -0.36 to $+0.37$ dB; exactly $M = 0.034$, -0.29 to $+0.30$ dB.
- Deck slide 66: $R_L = 60\ \Omega$ violates the deck's own start-up rule.
- Deck example 8.6: $N = 5$ already meets 20 dB; $N = 6$ is conservative.
- Sorted PYQ documents: 71 Ma's set and marks, 70 Ma and 80 Ba asks, the missing 75 Bh Smith-chart question and 72 Ma mixer note (fixed before this chapter).

7. Microwave Antennas and Propagation

3 hrs · asked in **23 of 24 papers** · 3 syllabus sub-topics · **162 marks, 8.4% of 24 papers** → almost every mark is on sub-topic (c), radiation hazards; antenna types and propagation have never been examined.

READ THIS FIRST

Chapter 7 is the **one guaranteed question** of the paper: only 72 Ash skips it in 24 sets. It comes as a 4 to 10 mark theory answer, usually two parts glued together.

- Radiation fields / zones** (11 papers) → §7.1. Three zones, two boundary formulas, one figure. The BTS and microwave-oven versions put numbers in them: worked in the **numerical companion**.
- How microwaves are hazardous; HERP, HERO, HERF** (17 papers) → §7.2. Always the same chain: non-ionizing → dielectric heating → no internal heat sensors → poorly cooled organs.
- SAR** (3 papers, one of them for 10 marks) → §7.3.
- Safety standards** (4 papers) and **safety practices** (7 papers) → §7.4 and §7.5. Name the body, the frequency range and one number each.

Two things worth knowing before the paper starts:

- Nearly every question is “part A + part B” from the list above. Answer each part under its own heading and draw the zone figure whenever “fields” appears.
- Every limit is built on one number:** a whole-body SAR of **4 W/kg** warms the body by about 1°C. Divide by 10 for workers (0.4 W/kg), by 50 for the public (0.08 W/kg). Quote this and the standards section writes itself.

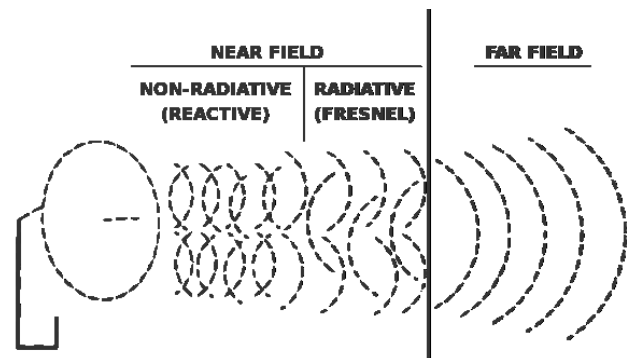
7.1 Microwave Radiation Fields (Radiation Zones)

Name and explain the microwave radiation fields: reactive, radiating near and far field **5 4 3 8 5+5**

TOP 11/24 69 Bh, 71 Ma, 72 Ma, 74 Ma, 76 Bh, 77 Ch, 79 Ch, 80 Ba, 81 Ba, 81 Bh, 82 Ba · oven (74 Ma, 77 Ch) and BTS (81 Ba, 81 Bh, 82 Ba) versions

- Radiation** = energy carried away by EM waves. Round any antenna (or leaking oven, or waveguide opening) the field is split into **three zones** by distance R , antenna largest dimension D and wavelength λ .
- Near field** = the first two zones. **Far field** = the third.

Zone	Boundary
Reactive near field	$R < 0.62\sqrt{D^3/\lambda}$
Radiating near field (Fresnel)	$0.62\sqrt{D^3/\lambda} < R < 2D^2/\lambda$
Far field (Fraunhofer)	$R > 2D^2/\lambda$



Reactive and radiating near field, then far field. Karkee, Ch-7 PDF p5 (white on black in the source, inverted for print).

What each zone is like (the “differentiate” answer, 79 Ch, 72 Ma)

	Reactive near field	Radiating near field	Far field
Energy	stored, sloshes back and forth to the antenna each cycle	radiated, but beam still forming	radiated, travelling away
E and H	90° out of phase, not in fixed ratio	in phase, ratio settling	in phase, $E \perp H \perp$ direction
E/H	not 377Ω ; measure E and H separately	approaching 377Ω	$= 377 \Omega$, so $S = E^2/377$
Fall with R	fast ($1/R^2$ to $1/R^3$)	irregular, local peaks	$S \propto 1/R^2$
Pattern	not defined	changes with distance	fixed shape
Hazard	highest : body close enough couples to stored energy	high: full beam power in a column about D wide	lowest, and predictable from $S = P_t G / 4\pi R^2$

- **Hazardous vs safe** (deck): reactive and radiating near field are treated as **hazardous**, the far field as the **safe** side, where exposure falls as $1/R^2$ and can be checked against a limit (§7.4).
- **Validity** verified/added : the two formulas assume an antenna **larger than a wavelength**. For a small antenna ($D < \lambda$, e.g. a quarter-wave whip) they give a “far field” that starts inside the reactive region. Use instead: reactive field out to $\lambda/2\pi$, far field beyond about 2λ . The 82 Ba quarter-wave question and the deck’s 150 MHz example fall here (ant.py).

Radiation zones of a microwave oven; radiation fields round a GSM BTS 3 4 4 4

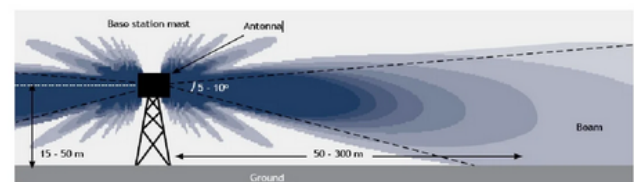
74 Ma, 77 Ch (oven); 81 Ba, 81 Bh, 82 Ba (BTS) · full working in the numerical companion

Microwave oven, $f = 2.45$ GHz, $D = 0.5$ m

- $\lambda = 0.1224$ m, $D > \lambda$ so the formulas hold.
- Reactive: $R < 0.63$ m. Fresnel: 0.63 to 4.08 m. Far field: $R > 4.08$ m.
- **In practice** the cavity is a closed metal box (Faraday cage) and the door mesh holes are far smaller than $\lambda = 12$ cm, so almost nothing leaks. Leakage limit (US FDA): **1 mW/cm^2 at 5 cm** when sold, 5 mW/cm^2 over its life. verified/added
- Zones matter only for leakage at a damaged door seal or hinge: stay out of the reactive zone (arm’s length) while it runs.

GSM / LTE base station (BTS)

- Panel antenna on a mast, main beam tilted a few degrees down, hitting the ground tens to hundreds of metres away (figure).
- **Reactive and Fresnel zones**: within a few metres **in front of** the panel, reached only by riggers on the tower or a rooftop → **occupational** exclusion zone.
- **Far field**: everywhere the public is. Power density there is far below the limit (81 Bh: zones 0.65 m and 4.33 m).
- **Radiation parameters** (81 Ba): frequency, transmit power, antenna gain, EIRP, beam tilt and beamwidth, polarization, power density (W/m^2), field strength (V/m), SAR.



Propagation of “main beam” from antenna mounted on a tower or roof top

Main beam of a mast-mounted antenna. KEC Ch-7 deck, slide 32.

7.2 How Microwave Radiation Is Hazardous: HERP, HERO, HERF

How are microwaves hazardous to humans (living body)? Microwave radiation hazards (short note)

3 4 5 6 8

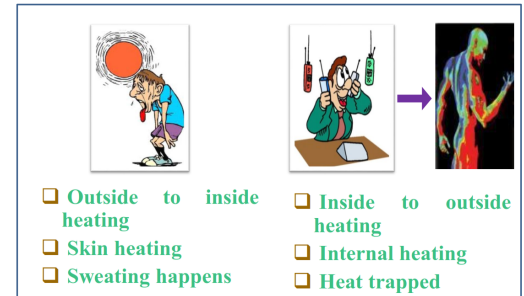
TOP 17/24 69 Bh, 70 Bh, 70 Ma, 71 Bh, 72 Ma, 73 Ma, 73 Bh, 74 Ma, 74 Bh, 75 Bh, 77 Ch, 78 Ch, 79 Bh, 80 Bh, 81 Bh, 82 Ba, 82 Bh · **the most repeated ask of the chapter**

1. Microwaves are non-ionizing

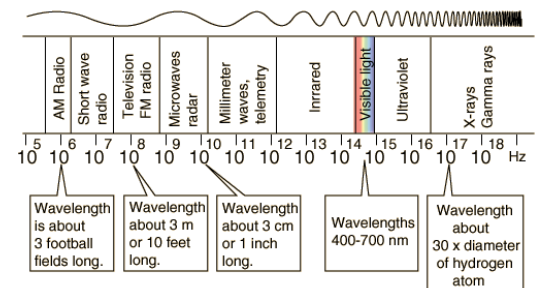
- Photon energy $E = hf$: at 2.45 GHz only 1.0×10^{-5} eV.
- Ionizing a molecule needs about 10 eV → a microwave photon is a **million times** too weak. It cannot break bonds or DNA directly (unlike X-rays, gamma rays).
- So the established hazard is **heat**, not ionization.

2. Thermal effect: dielectric heating

- The field **rotates polar water molecules** and drives **ionic currents** in tissue → friction and I^2R loss → heat.
- Absorbed power per kg = $\sigma E^2 / \rho = \text{SAR}$ (§7.3).
- Microwaves heat **inside out**. Sunlight heats **outside in**, where skin sensors warn and sweat cools (figure).



Sun heating (outside in, felt) vs microwave heating (inside out, trapped). Gangaju, Ch-7 deck, slide 4.



Microwaves sit far below visible light and the ionizing bands. KEC Ch-7 deck, slide 5.

3. Why the heating is dangerous

- No internal heat sensation:** the body senses heat only in the skin, so deep tissue can be damaged **before the person notices**.
- Poorly cooled organs** (little blood flow to carry heat away):
 - Eye lens** → cataract.
 - Testes** → temporary sterility.
 - Stomach, intestines, bladder, gall bladder → thermal damage at high power.
- Body resonance:** absorption peaks when body size is comparable to λ (whole adult body about 70 to 100 MHz). The deck adds: significant absorption even when body size is $\lambda/10$. That is why every limit chart dips lowest at 30 to 300 MHz (§7.4).
- Penetration falls with frequency:** deep at UHF (several cm), about 1 to 2 cm at 2.45 GHz, only the skin above about 10 GHz → lower microwave bands heat deep tissue unfelt, higher bands burn the surface. verified/added
- Long low-frequency magnetic exposure can also induce damaging currents (deck).

4. Other effects

- **RF burns and shocks:** touching a metal object (fence, mast, vehicle) in a strong field drives **contact current** through the hand.
- **Implants:** pacemakers and other active implants can malfunction; metal implants concentrate the field and heat.
- **Microwave hearing:** pulsed radar is heard as clicks (thermoelastic expansion in the head).
- **Behavioural complaints** reported by RF workers: headache, irritability, fatigue, sleeplessness (deck). Not reproduced in blind studies.
- **Cancer:** IARC (WHO) 2011 classes RF fields **Group 2B**, “possibly carcinogenic”, meaning limited evidence. **Correction:** one student deck calls microwaves “known” to cause cancer and another slide calls non-thermal effects “several times more harmful” than thermal; neither is established. verified/added

Link to the zones (69 Bh, 72 Ma, 81 Bh): “how is non-radiating radiation hazardous?” 4 5 8

- The **reactive near field** does not radiate, but its stored E and H fields are strongest there. A body inside it **loads** the antenna and **draws energy out** of the stored field → absorbed as heat, although nothing is “radiated”.
- Radiating near field: full beam power confined to a column about the antenna’s size → highest power density a person in front of the antenna meets.
- Far field: $S = P_t G / 4\pi R^2$ falls with distance → distance is the simplest protection (§7.5).

Types of electromagnetic radiation hazard (RADHAZ); HERP short note 4

74 Ma (types), 80 Bh (HERP)

HERP: to personnel

- Harmful **biological** effects in humans.
- Cause: thermal effect of absorbed energy (points 2 and 3 above).
- Eg: eye lens exposed → too little blood flow to cool it → cataract.
- Controlled by exposure limits (SAR, power density).
- “Hazards of Electromagnetic Radiation to Personnel”.

HERO: to ordnance

- RF induces current in the leads of **electro-explosive devices** (EEDs: detonators, squibs, airbag igniters, missile motors) → **premature firing**.
- Ordnance is **more sensitive** than people: no blood flow to shed heat.
- But easier to protect: metal enclosures reflect the field.
- A chain explosion is possible → HERO limits are **lower** than personnel limits.

HERF: to fuel

- RF-induced **arcs** ignite **fuel vapour** during refuelling near radar or radio transmitters.
- Ignition likely for arcs above about **50 VA**.
- Rules (deck):
 - no transmitting from a vehicle or aircraft being fuelled, or the one beside it
 - no connecting or breaking cables or ground wires while fuelling
 - radar lighting the area at 5 W/cm² peak: shut off
 - antennas of 250 W or less at least 50 ft from fuelling (500 W at sea)

7.3 Specific Absorption Rate (SAR)

Define SAR. Discuss microwave radiation hazards based on SAR exposures; what is non-ionizing SAR

4 6 10

HOT 3/24 74 Bh, 80 Ch, 81 Ba · 80 Ch sets the whole 10 marks on it

Definition

- **SAR** = rate at which RF energy is **absorbed per unit mass** of body tissue, in **W/kg**.

$$\text{SAR} = \frac{d}{dt} \left(\frac{dW}{dm} \right) = \frac{\sigma E^2}{\rho} = c \frac{dT}{dt}$$

- σ = tissue conductivity (S/m), E = **rms field inside the tissue** (V/m), ρ = mass density (**kg/m³**), c = specific heat (J/kg K), dT/dt = initial temperature rise rate.

- **Correction:** the deck prints the density unit as kg/m^2 ; a mass density is kg/m^3 , which is what makes $\sigma E^2 / \rho$ come out in W/kg . verified/added
- **Non-ionizing SAR** (81 Ba): SAR measures **heating** from non-ionizing RF. It is the dose quantity for RF the way absorbed dose (Gy) is for ionizing radiation.

Averaging and measurement

- **Whole-body SAR:** total absorbed power / body mass.
- **Localized (peak spatial) SAR:** averaged over any **1 g** (FCC) or **10 g** (ICNIRP, IEEE 2019) cube of tissue: the phone-against-head case.
- Averaged over time (6 min in ICNIRP 1998).
- **Measured** with a phantom head or body filled with tissue-simulating liquid: a robot moves an E -field probe through it ($\text{SAR} = \sigma E^2 / \rho$), or a temperature probe logs dT/dt ($\text{SAR} = c dT/dt$).
- Eg (ant.py): muscle at 900 MHz, $\sigma \approx 0.94 \text{ S/m}$, $\rho \approx 1040 \text{ kg/m}^3$; $E = 10 \text{ V/m}$ inside gives 0.09 W/kg .

Hazards based on SAR exposure (the 10-mark body)

Whole-body SAR	Effect and what the standards do with it
below 0.08 W/kg	public limit. No established health effect.
0.4 W/kg	occupational limit for trained workers.
about 1 to 4 W/kg	body temperature rises up to about 1°C , thermoregulation (blood flow, sweat) copes in healthy adults.
4 W/kg	threshold: about 1°C core rise, animals show disrupted behaviour. With no heat loss, 4 W/kg warms tissue 1°C in about 15 min ($c \approx 3500 \text{ J/kg K}$).
well above 4 W/kg	heat stress, heat stroke; local hot spots burn tissue, cataract in poorly cooled lens.

- **Safety factors:** threshold $4 \xrightarrow{\div 10} 0.4$ (occupational) $\xrightarrow{\div 5} 0.08 \text{ W/kg}$ (public). The student deck's "one-tenth of the observed threshold SAR (0.4 W/kg)" means the **limit** is 0.4; the threshold is 4. verified/added
- **Localized SAR limits:** public 2 W/kg over 10 g head and trunk (ICNIRP, IEEE 2019), **1.6 W/kg over 1 g** (FCC, US and India phones); workers 10 W/kg.
- **Mobile phones:** SAR is declared by the maker (typically 0.2 to 1.6 W/kg). Using a hands-free kit or speaker cuts it sharply because the head leaves the reactive near field of the handset antenna.

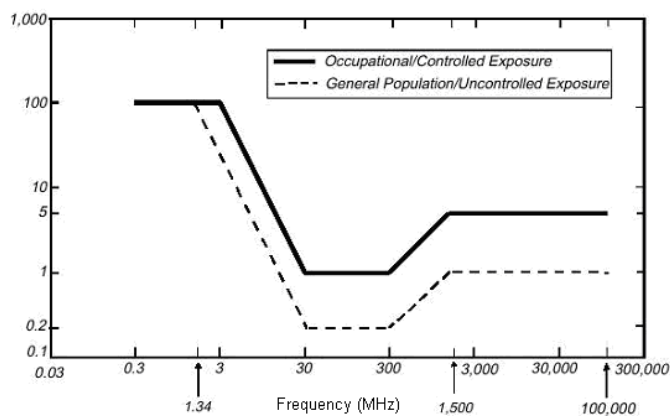
7.4 International EMR Safety Standards

International standards and recommended practices (SARPs) for RF/microwave exposure; IEEE safety standard 2 3 3+3 5+5

HOT 4/24 70 Ma, 76 Bh, 80 Ba, 82 Ba · practices half in §7.5

- **Basis of all of them:** adverse effects scale with absorbed power → **basic restriction** in SAR (W/kg), and because SAR cannot be measured in a person, a **reference level** (MPE) in measurable power density (mW/cm^2 , W/m^2) or field strength (V/m , A/m).
- **Two tiers:** **occupational / controlled** (trained workers who know they are exposed) and **general public / uncontrolled** (everyone else, 24 hours a day, including children, the ill and the elderly) → public limit 5 times lower.
- $1 \text{ mW/cm}^2 = 10 \text{ W/m}^2$.

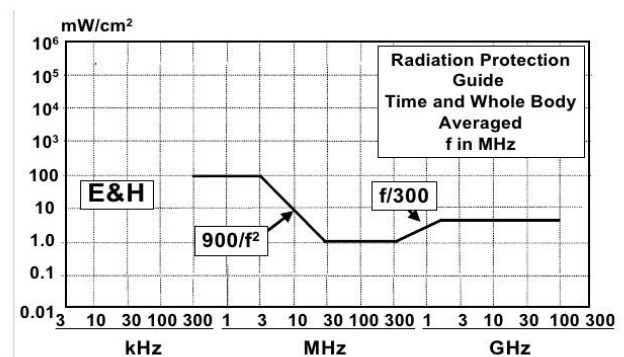
Body and standard	Range	Key numbers
IEEE / ANSI C95.1 (US; 1982, 2005, 2019)	3 kHz to 300 GHz (0 Hz to 300 GHz in 2019)	1982 single tier: 1 mW/cm ² at 30 to 300 MHz, $f/300$ to 1.5 GHz, 5 mW/cm ² above. 2005/2019: two tiers, whole body 0.08 / 0.4 W/kg, local 2 / 10 W/kg over 10 g (2019).
FCC (US, OET Bulletin 65)	300 kHz to 100 GHz	public 0.2 mW/cm ² at 30 to 300 MHz, $f/1500$ to 1.5 GHz, 1.0 mW/cm ² above; occupational 5 times higher. Phone SAR 1.6 W/kg over 1 g. Averaging 6 min (workers), 30 min (public).
ICNIRP (international, recognized by WHO; 1998, revised 2020)	100 kHz to 300 GHz	public $f/200$ W/m ² at 400 to 2000 MHz (4.5 at 900, 9 at 1800), 10 W/m ² above 2 GHz; workers 5 times. SAR 0.08 / 0.4 W/kg whole body, 2 / 10 W/kg local. Adopted by most countries.
IRPA / INIRC (1988)		ICNIRP's predecessor: first international occupational and public limits.
OSHA (US workplaces)	10 MHz to 100 GHz	10 mW/cm ² averaged over 6 min (older guide).
ITU-T K.52, K.70	telecom sites	how to check a BTS site against ICNIRP limits, and how to mitigate (compliance distance, exclusion zones).
WHO EMF Project, IARC		reviews health evidence; IARC Group 2B (2011).



FCC MPE (power density, mW/cm², vs MHz): solid occupational, dashed public.

Karkee PDF p18.

- **Read the charts:** both dip at **30 to 300 MHz**, the body-resonance band (§7.2), and rise again above 300 MHz as absorption becomes shallower.
- **Correction:** the deck's "IRPA permissible exposure level" table (PDF p17: 200 μ T, 5 kV/m, 20 mV/m in the head) is ICNIRP's **50 Hz power-line** limit, not a microwave one. It is left out here. verified/added
- **SARPs (80 Ba) = Standards And Recommended Practices:** answer with the ICNIRP / IEEE limits above as the standards and §7.5 as the practices.



ANSI C95.1-1982 Radiation Protection Guide. Karkee PDF p19.

7.5 RF/Microwave Safety Practices

RF/MW radiation hazards and safety practices; list and describe five major RF safety practices 3 5

6 5+5

TOP 7/24 70 Bh, 71 Bh, 73 Bh, 74 Bh, 76 Bh, 79 Ch, 79 Bh · short-note form mostly; hazard half is §7.2

Five major practices (79 Ch)

1. **Time, distance, shielding:**
 - cut exposure time
 - keep out of the main beam and out of the near field (power density $\propto 1/R^2$)
 - metal panels or mesh, RF absorbers, Faraday enclosures
2. **Engineering controls:**
 - safety **interlocks** that cut RF when a door or panel opens
 - RF gaskets, choke joints, oven door mesh
 - test into a **dummy load**, never an open waveguide
 - grounding straps not bent at sharp angles (they radiate)
3. **Administrative controls:**
 - RF survey of the site, marked **exclusion zones**, warning signs (figure)
 - **lockout / tagout** before work on a tower or transmitter
 - never look into a live waveguide, horn or dish
 - training and exposure records
4. **Personal protective equipment:** RF protective suit (stainless steel woven into fire-retardant fibre) for high-field areas; personal RF monitor that alarms.
5. **Regular inspection:**
 - **visual:** housekeeping, interlocks present, high-voltage grounding rods, condition of cables and coax, RF shielding intact
 - **operational:** RF level survey in occupied spaces, interlock tests, lockout used properly



RF warning sign at an exclusion-zone boundary.

KEC Ch-7 deck, slide 24.

More practices (deck)

- Operate stray-RF sources (induction and dielectric heaters) **remotely**.
- Keep workers below the recommended limits (§7.4).
- **HERO:** keep EEDs away from transmitters.
- **HERF:** no RF devices in flammable atmospheres; fuelling rules of §7.2.
- RF-sensitive equipment and people with **implants** away from sources.
- Switch equipment off when not in use.
- **Public:** antennas high and beams tilted clear of rooftops; hands-free phones; awareness and school teaching.

7.6 Microwave Antennas

Microwave antenna types syllabus, no PYQ yet

syllabus 7(a) · the deck leaves it as an assignment and refers to Pozar

- Conventional wire antennas work at microwaves too, but the short wavelength allows **aperture antennas** a few λ across → high gain, narrow beams in a small size.
- **Parameters:** gain G and directivity, beamwidth, efficiency, polarization, input impedance / VSWR, bandwidth. Aperture gain $G = 4\pi A_e / \lambda^2$ with $A_e = \eta \pi D^2 / 4$.

Type	How it works	Typical use
Horn	flared end of a waveguide; flare matches guide to free space. E-plane, H-plane, pyramidal, conical. 10 to 25 dBi	dish feed, gain standard in measurements
Parabolic reflector	feed (horn) at focus; reflector turns spherical wave into plane wave. $G = \eta(\pi D/\lambda)^2$, HPBW $\approx 70\lambda/D$ degrees. Front-fed or Cassegrain	satellite earth stations, radar, point-to-point links
Lens	dielectric (slows wave) or metal-plate (speeds it) lens collimates like an optical lens	mm-wave radar, automotive
Slot	slot cut in a waveguide wall or ground plane; complement of a dipole (Babinet); flush	aircraft, slotted-guide arrays
Microstrip patch	metal patch about $\lambda_g/2$ long over a ground plane on a substrate; low profile, narrow band (1 to 5%), low power; arrays	phones, GPS, WLAN, 5G arrays
Helical	wire helix; axial mode when circumference $\approx \lambda \rightarrow$ circular polarization	satellite, space telemetry

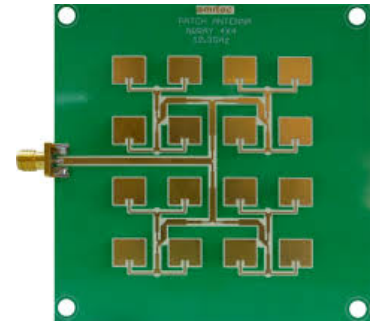
Eg (ant.py): 1.2 m dish at 12 GHz, $\eta = 0.55$: $G = 41.0$ dBi, HPBW = 1.46°.



Pyramidal horn.



Parabolic reflector.



16-element patch array. All three: 2078 student presentation.

- **Corrections to the student deck** verified/added : a microstrip antenna is a **microwave** antenna (at 100 MHz a half-wave patch would be over a metre long), not one “used at low frequency”; and its “MIMO” slide shows a Yagi array. MIMO is a way of using several antennas (spatial multiplexing, diversity), not an antenna type.

Choose a proper microwave measurement tool to test an antenna as a DUT; explain its working principle 8

PIN 1/24 74 Bh · instrument principles in Ch 8

Tool 1: vector network analyzer (VNA), port 1 to the antenna

- Measures S_{11} vs frequency $\rightarrow$ return loss, VSWR = $(1 + |S_{11}|)/(1 - |S_{11}|)$, input impedance on a Smith chart, and the **bandwidth** where VSWR < 2.
- Working: internal source sweeps; directional couplers split incident and reflected waves; receivers compare amplitude and phase; calibration (short, open, load) removes cable and coupler errors. Detail in Ch 8.

Tool 2: anechoic chamber (pattern and gain)

- Walls lined with RF absorber cones (no reflections). Known source horn at one end, antenna under test on a rotating **positioner** at the other.
- Separation $R > 2D^2/\lambda$ so the test is in the **far field** (§7.1).
- Rotate and record received power (VNA S_{21} or spectrum analyzer) $\rightarrow$ **radiation pattern**, beamwidth, side lobes.
- **Gain**: compare with a standard gain horn, or three-antenna method with Friis.

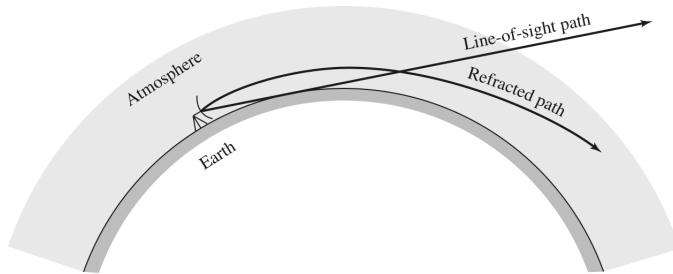
7.7 Propagation Characteristics of Microwaves

Propagation characteristics: line of sight, atmosphere, ground and plasma effects syllabus, no PYQ yet

syllabus 7(b) · deck refers to Pozar; numbers from `ant.py`

Line-of-sight (space wave)

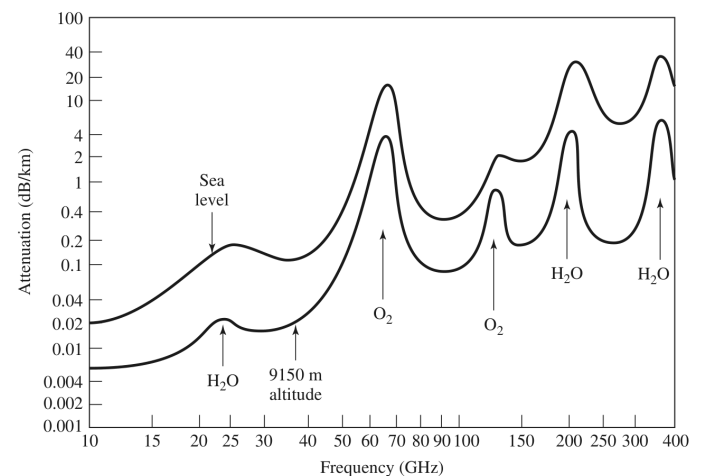
- Ground wave dies within a short range and the ionosphere does not reflect microwaves (they pass through) → terrestrial links are **LOS**, satellites use the fact that they pass.
- Radio horizon** with $k = 4/3$ earth (refraction bends the ray down): $d \approx 4.12\sqrt{h}$ km, h in m. Two masts: $d = 4.12(\sqrt{h_1} + \sqrt{h_2})$; 30 m and 50 m give 51.7 km.
- Fresnel clearance**: keep 60% of the first Fresnel zone $r_1 = \sqrt{\lambda d_1 d_2 / (d_1 + d_2)}$ clear of obstacles; 50 km hop at 6 GHz needs $r_1 = 25$ m at mid-path.
- Free-space path loss**:
 $L = 32.44 + 20 \log d_{km} + 20 \log f_{MHz}$ dB; Friis
 $P_r = P_t G_t G_r (\lambda / 4\pi d)^2$.



The refracted path bends down toward the earth, beyond the straight line-of-sight path: why the 4/3-earth horizon is longer than the geometric one. Pozar Fig 14.28.

Atmosphere, ground and plasma effects

- Gas absorption**: water vapour peak at 22 GHz, **oxygen at 60 GHz**; little below 10 GHz. The 60 GHz peak suits short private links and satellite crosslinks.
- Rain and fog**: drops comparable to λ scatter and absorb → rain fade, serious above about 10 GHz (Ku, Ka bands).
- Refraction**: ducting and super-refraction extend or bend paths → fading and interference.
- Ground effect**: reflection off ground or water adds a second ray → **multipath fading**; earth bulge blocks long hops.
- Plasma effect**: ionosphere rotates polarization (Faraday rotation) and causes scintillation at low GHz; the plasma sheath round a re-entering spacecraft blocks signals (blackout).
- Diffraction** round obstacles is weak at short λ → shadowing behind buildings and hills.



Gaseous attenuation (dB/km) against frequency, at sea level and at 9150 m: the H₂O peak at 22 GHz, the O₂ peak at 60 GHz, windows between them. Pozar Fig 14.29.

7.8 PYQ Mapping

Theory questions, covered in this document

- Microwave radiation fields; differentiate reactive, near and far field** → §7.1. 69 Bh 8, 71 Ma 5, 72 Ma 5, 76 Bh 5, 79 Ch 5
- Radiation zones of a microwave oven** → §7.1 + companion Problem 1. 74 Ma 4, 77 Ch 3
- BTS fields: classify, radiation parameters; 900 vs 1800 MHz; hazardous and safe zone at 2600 MHz** → §7.1 + companion Problems 1 and 2. 81 Ba 4, 81 Bh 4, 82 Ba 4
- RF radiation fields to public exposure** → §7.1 and §7.4. 80 Ba 3
- How microwaves are hazardous; microwave radiation hazards (short note)** → §7.2. 69 Bh 8, 70 Ma 5, 72 Ma 5, 73 Ma 5, 75 Bh 5, 77 Ch 3, 78 Ch 5, 81 Bh 4, 82 Ba 2, 82 Bh 4
- Hazards + safety practices (short note)** → §7.2 and §7.5. 70 Bh 5, 71 Bh 5, 73 Bh 6, 74 Bh 6, 79 Bh 3+3
- Types of EM radiation hazard; HERP** → §7.2. 74 Ma 4, 80 Bh 4
- SAR: define, effect as hazard, non-ionizing SAR** → §7.3. 74 Bh 6, 80 Ch 10, 81 Ba 4
- International standards, SARPs, IEEE standard** → §7.4. 70 Ma, 76 Bh 5, 80 Ba 3, 82 Ba 2

- **Five major RF safety practices** → §7.5. 79 Ch 5, 76 Bh
- **Antenna as DUT: measurement tool and principle** → §7.6. 74 Bh 8

Numerical content

- **Radiation-zone boundaries** (BTS at 2600 MHz, BTS at 900 / 1800 MHz with quarter-wave antennas, microwave oven) → numerical companion, 2 problems. Every number is asserted in `src/ant.py`.

Corrections, and fixes made to the PYQ documents while building this chapter

- Karkee PDF p6: SAR mass density is kg/m^3 , not kg/m^2 .
- Karkee PDF p17: the “IRPA” table is a 50 Hz power-line limit.
- Deck example (150 MHz, $D = 0.5$ m): $D < \lambda$, so its zone boundaries are not valid; small-antenna rule applies.
- Student decks: cancer “known” (IARC says possibly, 2B); non-thermal effects “several times more harmful” (not established); “threshold SAR 0.4” (0.4 is the limit, 4 the threshold); microstrip “used at low frequency”; Yagi pictured as “MIMO”.
- Sorted PYQ documents: the hazard short notes were tagged 71 Ma and 72 Ash, which ask none. They are 70 Bh, 71 Bh and 73 Ma (73 Ma was missing). 70 Ma’s note is [5], not [8]. Bold tags for 69 Bh, 71 Bh and 73 Bh made consistent (Regular papers).

8. RF/Microwave Measurements

6 hrs · asked in **22 of 24 papers** · 9 syllabus sub-topics · **181 marks, 9.5% of the archive** → 149 of those 181 marks are **power measurement**, and the tool is picked by one number: how many milliwatts.

READ THIS FIRST

Chapter 8 is descriptive and it repeats harder than any other chapter: five short-note titles (bolometry, double-channel bolometer, circulating calorimetry, spectrum analyzer, VNA) and three long-answer shapes cover everything asked in 24 papers.

1. **“Choose a proper power measuring tool and explain its working principle”** (72 Ma 7.5 mW, 74 Ma airport radar, 73 Bh, 82 Bh BTS) → one table, §8.2. Say which band the number falls in, name the sensor, then give that sensor’s working principle from §8.3 or §8.5. **The table is the mark.**
2. **Calorimetry** (9 papers) → §8.5. Static vs circulating, one formula each, and they are the same formula with m/t replaced by $v d$.
3. **Bolometer bridge** (9 papers) → §8.3. Barretter vs thermistor, single bridge, its two faults, double bridge that cures them.

Three things worth knowing before the paper starts:

- **Every microwave power meter is a substitution instrument.** It never measures RF. It measures how much **dc or low-frequency** power had to be taken away to keep a bridge balanced, or how much a **temperature** rose. Open any power answer with that sentence.
 - **The three bands are the whole decision:** low < 10 mW, medium 10 mW to 10 W, high > 10 W. A radar is quoted in **peak** power, so convert with the duty cycle before choosing (§8.2).
 - Measurement questions are worded as “describe how X and Y are measured”. X and Y are almost always **VSWR meter** and **some power sensor**, which is two separate answers glued together (72 Ash, 70 Bh).
-

8.1 Microwave Measurements Vs Low-Frequency Measurements

How are microwave measurements different from low-frequency measurements? List the major RF/MW measurement parameters **2 3 3**

HOT 4/24 71 Bh, 73 Bh, 79 Bh, 81 Bh · always the opener to a longer measurement question

Why voltage and current stop being useful

- At low frequency the circuit is **lumped**: V and I are the same everywhere in a wire, so measure them and compute Z , power factor and phase angle.
- At microwaves the line is **distributed**: V and I are functions of **distance** along the line, so a “the voltage” does not exist. In a waveguide there is no unique voltage at all.
- **Power is the invariant**: on a lossless line the transmitted power is the same at every plane. So measure **power**, and get everything else from ratios of powers and from the standing-wave pattern.
- Probes and leads are a **significant fraction of a wavelength**, so any contact instrument loads and detunes what it measures. Measurement is made by **directional coupling** and by **sampling** the field, not by touching nodes.
- Circuit behaviour is described by **S-parameters**, which are themselves defined from travelling waves rather than terminal V and I (§3.1).
- **Cost dodge used in every teaching laboratory**: rather than process microwave signals directly, the source is **1 kHz square-wave modulated**, the microwave signal is detected by a square-law crystal, and all the amplitude information is then read with an ordinary 1 kHz low-frequency amplifier, the **VSWR meter**.

The measurables and their tools (81 Bh asks exactly this list)

Measurable	Instrument
Power (low)	bolometer (thermistor / barretter) bridge, Schottky diode sensor, thermocouple sensor
Power (medium, high)	calorimeter (static, circulating), calorimetric wattmeter, thermistor behind a calibrated attenuator or directional coupler
VSWR, $ \Gamma $	slotted line + VSWR meter, reflectometer
Impedance, unknown load	slotted line (minima-shift method), VNA, magic tee
Frequency	cavity (absorption) wavemeter, frequency counter, slotted line via λ_g
Spectrum, harmonics	spectrum analyzer
S-parameters, insertion and return loss	network analyzer (VNA)
Attenuation, insertion loss	two directional couplers + VSWR meter
Noise figure	noise-source + Y-factor measurement (§8.9)
Field strength, radiation zones	EMF survey meter with isotropic probe (§8.10)

8.2 Microwave Power: Bands, Average Vs Peak, Choosing the Sensor

Choose a proper power measurement tool for the stated source and explain its working principle **8**

10

HOT 4/24 72 Ash, 72 Ma, 73 Bh, 74 Ma · 72 Ma 7.5 mW, 74 Ma airport surveillance radar, 73 Bh “low microwave power”

- **Power** = energy dissipated or stored per unit time. At microwaves it is the **average** power that a sensor responds to:

$$P_{av} = \frac{1}{T} \int_0^T v(t) i(t) dt, \quad P_{av} = P_{peak} \times \text{duty cycle}, \quad \text{duty} = \tau f_r = \tau/T$$

- Unit: $P(\text{dBm}) = 10 \log_{10}[P(\text{mW})/1 \text{ mW}]$, so 0 dBm = 1 mW, 30 dBm = 1 W, –30 dBm = 1 μW .

The decision table: this is the answer to every “choose a tool” question

Band	Range	Sensor, and why	Section
Low	< 10 mW	resistance changes with power: thermistor or barretter bolometer bridge (floor about -20 dBm), Schottky diode detector (down to -70 dBm = 100 pW), thermocouple sensor	§8.3, §8.4
Medium	10 mW to 10 W	thermistor bolometer behind a calibrated attenuator or 20 dB directional coupler ; static or circulating calorimeter	§8.3, §8.5
High	> 10 W	temperature changes with power: calorimeter (static for moderate, circulating / flow for kW), calorimetric wattmeter	§8.5

Measurement procedure, common to every power meter (the “zeroing” marks)

1. Attach a **calibrated sensor** to the line port where the unknown power appears; connect it to the power meter.
2. **Turn the RF off and zero the meter** (“zero setting”): this removes sensor self-heating and ambient drift from the reading.
3. Turn the RF on. The sensor reacts, the meter reads, and the reading is by **substitution**: the dc or low-frequency power displaced, or the temperature rise.
4. Correct for the coupler or attenuator in front of the sensor, and for the **calibration factor** (mismatch plus sensor efficiency).

The three PYQ cases worked (numbers checked in meas.py)

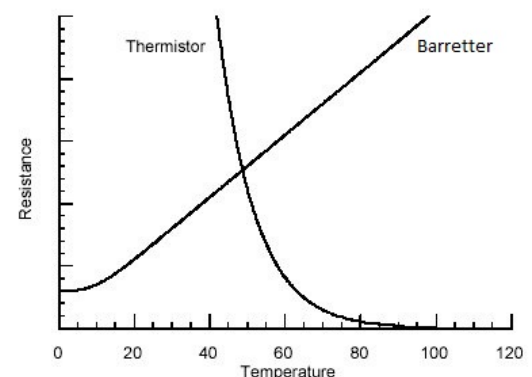
- **72 Ma, “about 7.5 mW”** 10 : $7.5 \text{ mW} = 8.75 \text{ dBm}$, below 10 mW, so **low power**. Answer: **thermistor bolometer bridge**, sensitivity floor about $-20 \text{ dBm} = 0.01 \text{ mW}$, so 7.5 mW sits comfortably inside its range. Then give the bridge working principle (§8.3). A Schottky diode sensor is the alternative.
- **74 Ma, airport surveillance radar** 8 : a radar is quoted in **peak** power and is pulsed. Take a representative set, 1 MW peak, $\tau = 1 \mu\text{s}$, PRF 1 kHz: duty = 10^{-3} , so $P_{av} = 1 \text{ kW}$, which is **high power**. Answer: a **circulating (flow) calorimeter** on a dummy load, or a thermistor behind a high-directivity coupler plus a calibrated attenuator; report $P_{peak} = P_{av}/\text{duty}$. Never put a bolometer directly in the line: it would burn out.
- **73 Bh, “low microwave power measurement device”** 8 : name the parameters first (§8.1), then the **bolometer bridge** in full.

8.3 Bolometer Bridge Method

Explain the working principle of bolometry used in microwave measurements 5 6 8 10

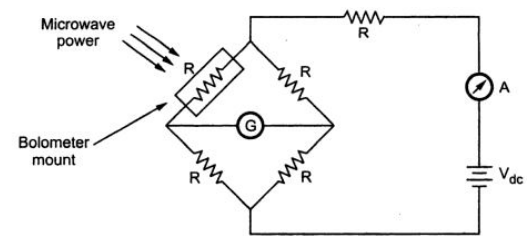
HOT 6/24 70 Bh, 72 Ash, 72 Ma, 73 Bh, 79 Ch, 80 Ch · 79 Ch for 10 marks, 80 Ch as a 5 mark short note

- A **bolometer** is a power sensor whose **resistance changes with temperature** as it absorbs microwave power. RF energy is turned to heat inside the element, the element warms, its resistance moves, and that resistance change is read.
- Bolometer impedances are **100 to 200 Ω** . The **mount** provides a good impedance match, low loss, thermal and mechanical isolation and shielding against leakage.
- **Two types:**
 - **Barretter**: a short thin metal wire (platinum or tungsten, like a fuse), **positive** temperature coefficient. More delicate, so used only for very low power, a few mW at most.
 - **Thermistor**: a semiconductor bead, **negative** temperature coefficient. Small, rugged, easily mounted in a waveguide or coaxial mount; the usual choice. Sensitivity floor about -20 dBm .



Resistance vs temperature: thermistor falls (negative coefficient), barretter rises (positive). Karkee, Ch-8 deck slide 8.

- **Principle:** the element sits in one arm of a **Wheatstone bridge**. With no RF, R_5 is adjusted to balance the bridge. Incident microwave power heats the element, its resistance changes, the bridge unbalances, and the galvanometer or voltmeter current is **calibrated directly in power**.
- **Substitution:** because the element responds only to total heating, the RF power is exactly the **dc power that has to be withdrawn** to restore balance. This is what makes it an absolute measurement rather than a comparison.



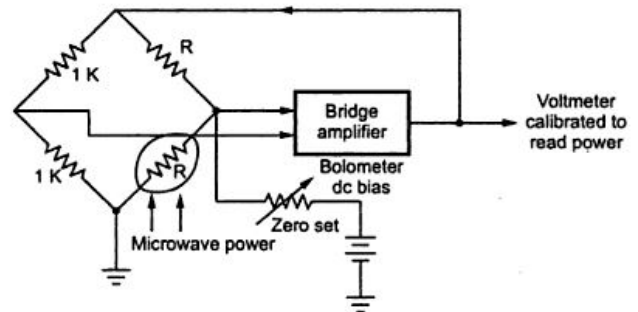
Bolometer mount in one arm of a Wheatstone bridge; the galvanometer current measures the incident power. Deck slide 9.

Single bridge, its two faults, and the double-channel (double bridge) cure 2 4 8

HOT 3/24 80 Ba, 80 Bh, 81 Ba · 80 Ba asks only the limitations 2; 81 Ba is a 4 mark short note

Single bridge

- Balanced under **zero incident power** by adjusting R_5 ; the bolometer arm is R_4 .
- Microwave power into R_4 changes its resistance, unbalances the bridge, and the non-zero output is read on a voltmeter calibrated in power.
- With dc bias voltage E_1 at balance and E_2 after RF is applied, the change $(E_1 - E_2)$ is directly proportional to the microwave power.
- **Two faults** (this is the whole of 80 Ba's 2 marks):
 1. A **mismatch** at the microwave input reflects power, so the element absorbs less than is incident: the reading is wrong and there is no way to tell.
 2. The thermistor is **sensitive to ambient temperature**: room drift is read as RF power. A false reading with no warning.



Self-balancing single bolometer bridge. Deck slide 10.

Double-channel bolometer bridge

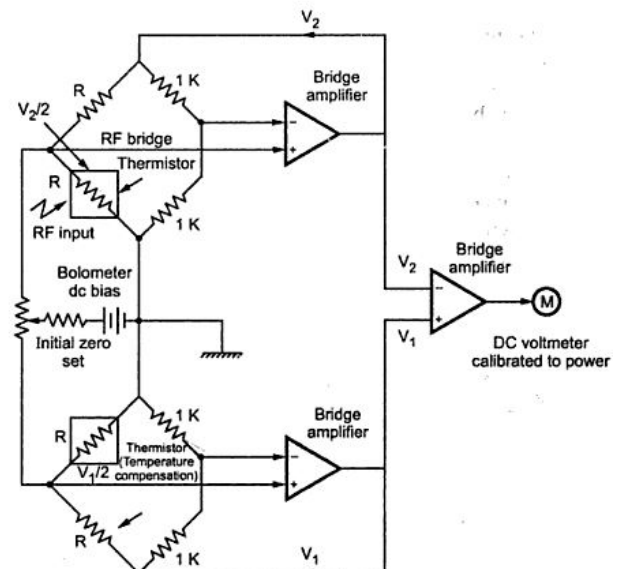
- **Two identical bridges.** The **upper** one carries the RF sensing thermistor and measures the power. The **lower** one carries a matched thermistor with **no RF**, exposed to the same ambient, and compensates temperature drift by forcing $V_1 = V_2$.
- The extra heating caused by mismatch is taken out automatically by **negative dc feedback**: the self-balancing loop lowers the dc power in the sensing thermistor until balance is restored, that is until the **net resistance change is zero**.
- **Zero setting:** with no input, adjust $E_1 = E_2 = E_0$. Without RF the dc voltage across the sensor at balance is $E_1/2$; with RF it is $E_2/2$.
- Average input power is the dc power the RF displaced:

$$P_{av} = \frac{E_1^2}{4R} - \frac{E_2^2}{4R} = \frac{(E_1 - E_2)(E_1 + E_2)}{4R}$$

- If ambient drift moves both rails by ΔE ,

$$P_{av} + \Delta P = \frac{(E_1 - E_2)(E_1 - E_2 + 2\Delta E)}{4R}$$

and since $E_1 + E_2 \gg \Delta E$ in practice, $\Delta P \approx 0$: the meter reads the line above. **That inequality is why the second bridge works.**



Double-channel bolometer bridge: upper bridge measures, lower bridge compensates ambient temperature. Deck slide 12.

- **Worked, to fix the formula** (`meas.py`): $R = 200\ \Omega$, $E_1 = 6\ \text{V}$, $E_2 = 4\ \text{V}$. Sensor dc power falls from 45 mW to 20 mW, so $P_{av} = (36 - 16)/800 = \mathbf{25\ mW}$. A +50 mV ambient drift on both rails would move a **single** bridge to 25.25 mW, an error of 1%, which the second bridge removes.

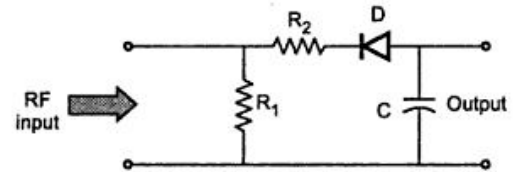
8.4 Schottky Diode Detector and Thermocouple Sensor

Other low-power sensors: the square-law diode detector and the thermocouple syllabus 8.4

no question names them, but both are legitimate answers to “a low power measurement device” (73 Bh, 72 Ash, 72 Ma)

Schottky barrier diode detector

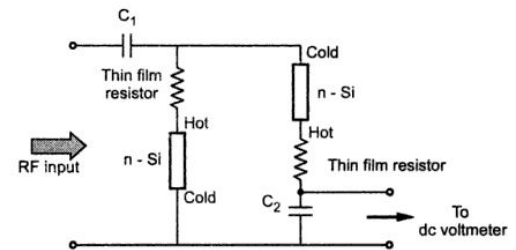
- A zero-biased **metal-semiconductor (point contact) diode** used as a **square-law detector**: output is proportional to the **square** of the input voltage, therefore to input **power**.
- RF input is applied to R_1 and passes through R_2 ; the diode rectifies, and the resulting low-frequency or dc current is read.
- Sensitivity down to **−70 dBm (100 pW)**, up to about 18 GHz. It is the fastest and most sensitive of the low-power sensors.
- The circuit is arranged so the input match does not depend on the diode resistance, which is a strong function of temperature.
- **Caution**: it is square-law only over a limited range. Above roughly −20 dBm it enters a transition region and eventually becomes **linear**, which is exactly why the VSWR meter misreads a high SWR (§8.6).



Schottky barrier diode detector. Deck slide 6.

Thermocouple sensor

- A **thermocouple** is a junction of two dissimilar metals or semiconductors (here n-type Si). A thin film of **tantalum-nitride** resistive load is deposited on a Si substrate and forms one electrode.
- Microwaves are absorbed in the resistive load, heat one end of the junction more than the other, and the junction generates an **emf proportional to incident power**.
- C_2 is the RF bypass capacitor, C_1 the input coupling capacitor (dc block). Emfs of the parallel thermocouples add across C_2 ; the output leads are at RF ground, so the dc voltmeter reads pure dc.
- For a square-wave modulated signal the peak power follows from the average: $P_{peak} = P_{av} T/\tau$, T the period and τ the pulse width. **Same relation as the radar case in §8.2.**
- Wider dynamic range and better match stability than a bolometer, and no dc bias, but no substitution: it must be calibrated.



Thermocouple power sensor. Deck slide 14.

8.5 Calorimetric Methods

What is calorimetry in microwave? Explain how microwave power is measured with a static calorimeter 2 5 6 7 10

HOT 6/24 70 Ma, 71 Ma, 71 Bh, 76 Bh, 77 Ch, 81 Bh · 76 Bh says “static and dry”, which is the same instrument: a dry dielectric load, no flowing water

Calorimetry: the 2 mark definition

- **Calorimetry** measures microwave power by **converting it entirely into heat**, absorbing that heat in a known medium, and measuring the medium’s **temperature rise**. It is the standard method for **medium and high power**, and it is an **absolute** measurement: nothing is compared against a reference source, only against the specific heat of a fluid.
- **Two heating arrangements** (both used with either calorimeter below):
 - **Direct heating**: the rate of heat production is found from the temperature rise of the **dissipating medium itself**.
 - **Indirect heating**: heat is **transferred to another medium** before it is measured.

Static (dry) calorimeter

- A $50\ \Omega$ coaxial line terminated in a lossy dielectric load with a high hysteresis loss, thermally isolated from its surroundings. No fluid circulates, hence “dry”.
- The load dissipates all the microwave power; a thermometer or thermocouple reads the rise T of the thermometric medium over a time t :

$$P = \frac{4.187 m C_p T}{t} \text{ watts}$$

m mass in grams, C_p specific heat in cal/g, T rise in $^{\circ}\text{C}$, t in seconds, and **4.187 is joules per calorie**.

- **Where the formula comes from:** mC_pT is the heat in **calories**, dividing by t makes it calories per second, and 4.187 converts to watts. Quote this and the formula cannot be mis-remembered.
- **Worked** (meas.py): 100 g of water raised 5°C in 60 s is $4.187 \times 100 \times 1 \times 5/60 = \mathbf{34.89\ W}$.
- **Limits:** slow (thermal inertia), and the load must stay isolated, so it suits steady medium power rather than a kilowatt transmitter.

Circulating (flow) calorimeter; circulating Vs flow calorimetry 4 5 8

HOT 4/24 70 Ma, 75 Bh, 81 Bh, 82 Ba · 82 Ba is a 4 mark short note; 81 Bh pairs it with the measurables list of §8.1

- The calorimeter fluid (usually **water**) is **pumped continuously** through a water load in the waveguide. Microwave power heats it in transit, so the **outlet** temperature T_2 exceeds the **inlet** T_1 :

$$P = 4.187 v d C_p (T_2 - T_1) \text{ watts}$$

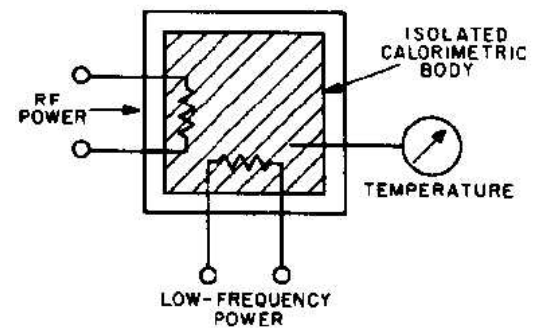
v flow rate in cc/s, d specific gravity in g/cc, C_p specific heat in cal/g.

- **Same law as the static one:** vd is grams per second, which is the static formula's m/t . Confirmed in meas.py.
- **Worked:** 10 cc/s of water with a 2°C rise is **83.7 W**. To take 1 kW with a 10°C rise needs **23.9 cc/s**.
- **Instrumentation:** pump, flow meter, glass tube through the guide, inlet and outlet thermocouples, heat exchanger to return the water cool.

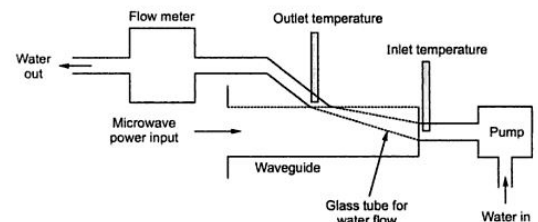
Static Vs circulating (the “differentiate” answer)

	Static (dry)	Circulating (flow)
Load	fixed dry dielectric, thermally isolated	water pumped through continuously
Reading	temperature rise per unit time	steady inlet to outlet difference
Formula	$4.187 m C_p T / t$	$4.187 v d C_p (T_2 - T_1)$
Power handled	medium; the load stores heat and saturates	high , kW upward: the flow carries heat away
Speed	slow, must wait for a measurable rise	continuous once flow is steady
Errors	heat leakage, isolation, thermal inertia	flow-rate stability, thermometer placement, thermal inertia

- **On the wording “circulating Vs flow”** (70 Ma) **verified/added**: the sources use the two words for the **same** instrument. Answer the question that is meant, which is **static vs circulating**, and say so in one line: if a distinction is forced, **circulating** returns the same fluid round a closed loop through a heat exchanger, while a plain **flow** calorimeter passes fluid once and discards it.



Static calorimeter: RF into an isolated calorimetric body, compared against low-frequency power. Deck slide 18.

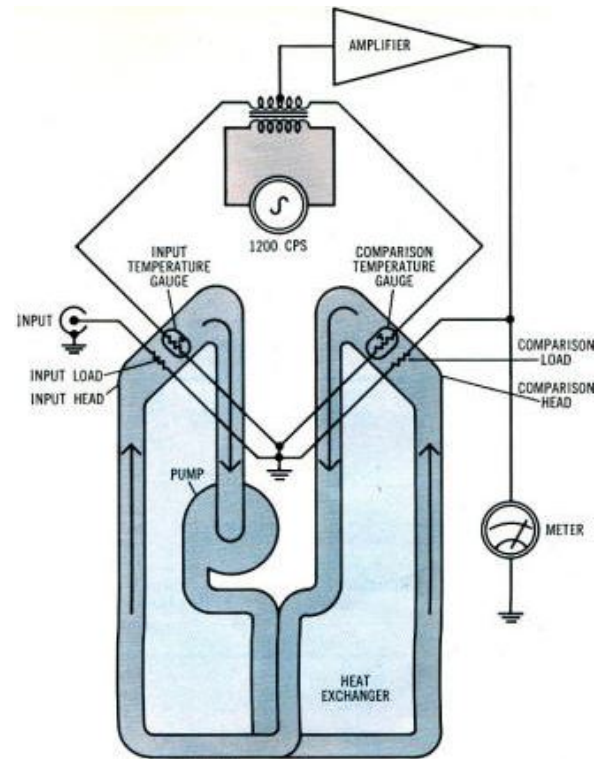


Circulating (flow) calorimeter: water load, pump, flow meter, inlet and outlet thermometers. Deck slide 19.

Power measurement using a calorimeter wattmeter 6

PIN 1/24 80 Ba · paired with the single-bridge limitations of §8.3

- A calorimeter run as a **self-balancing comparison bridge**, so the answer comes out as a meter reading instead of a temperature calculation.
- The unknown RF power is checked against a **1200 cps (Hz) comparison power** in a bridge circuit. Two **temperature-sensitive resistors** serve as gauges.
- **Operation:**
 1. The unknown RF heats an **input load resistor**.
 2. That resistor and one gauge sit in close thermal contact, so the gauge warms and **unbalances the bridge**.
 3. The unbalance signal is **amplified** and applied to the **comparison load resistor**, which is next to the second gauge, and **rebalances** the bridge.
 4. The meter reads the **power supplied to the comparison load**, which at balance equals the unknown RF power.
- Both loads and gauges are **immersed in an oil stream** for efficient, equal heat transfer.
- **Why it is better than a bare calorimeter:** it is a null method, so gauge calibration and ambient drift cancel, and it reads continuously.



Calorimetric wattmeter: input load and comparison load, two temperature gauges, 1200 cps comparison power, oil stream. Deck slide 20.

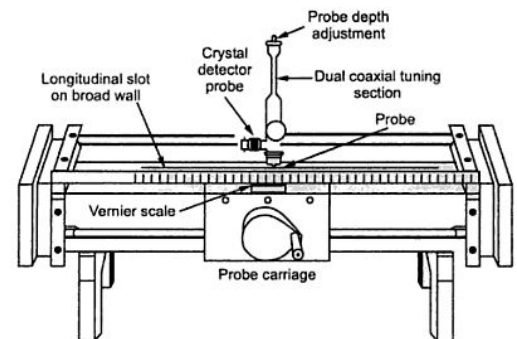
8.6 VSWR Measurement: Slotted Line and VSWR Meter

Describe how standing waves and microwave power are measured with a VSWR meter 2 4

HOT 3/24 70 Bh, 72 Ash, 79 Bh · always part A, with a power sensor as part B: 72 Ash "low power measurement" 8, 70 Bh "bolometry" 6

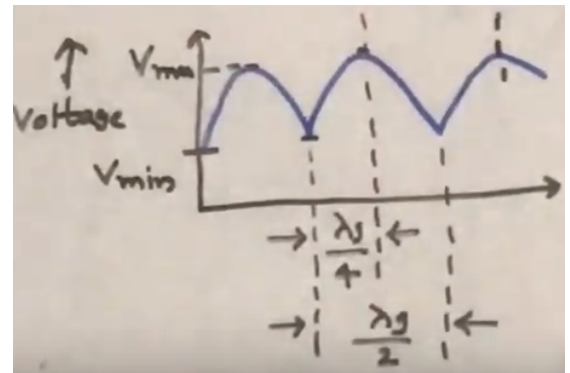
Slotted line carriage: where the standing wave is seen

- A **coaxial E-field probe** penetrates a rectangular waveguide (or coaxial line) through a **longitudinal slot** cut in the centre of the broad wall, over 2 to 3 wavelengths.
- The slot lies where the **wall current has no transverse component**, so cutting it does not disturb the field. Both ends are **tapered** to zero width to reduce discontinuity, and the probe is thin and shallow for the same reason.



Slotted-line carriage: longitudinal slot on the broad wall, probe, crystal detector, vernier scale. Deck slide 22.

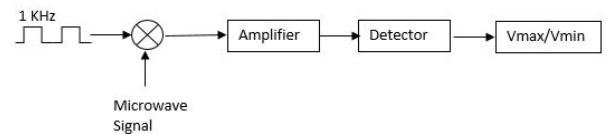
- The carriage moves the probe along the slot at constant depth, so coupling is uniform, and a **vernier scale** reads its position. It carries a tunable stub and a crystal detector, whose 1 kHz output goes to the VSWR meter.
- **It measures:** standing-wave pattern and **VSWR**, **wavelength** ($\lambda_g = 2 \times$ distance between successive minima), and by the minima-shift method **impedance**, **reflection coefficient** and **return loss** (§8.7).



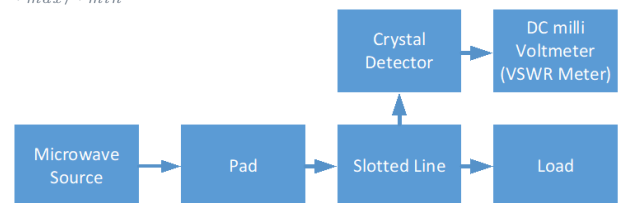
The pattern the probe traces: maxima and minima $\lambda_g/2$ apart, each minimum $\lambda_g/4$ from its neighbouring maximum. Deck slide 27.

VSWR meter

- A **high-gain, high- Q , low-noise voltage amplifier** tuned to a **fixed 1 kHz**, the frequency at which the microwave source is square-wave modulated. Overall gain about 125 dB, switchable in 10 dB steps.
- Input is the **detected** output of the crystal detector; output goes to a **square-law calibrated** voltmeter, which therefore reads V_{max}/V_{min} directly once it has been set to 1.0 at a maximum.
- **Procedure** (low VSWR):
 1. Set the attenuator to about 10 dB, tune the probe stub for maximum reading.
 2. Slide the carriage to a **maximum** and adjust the **gain** until the meter reads **1.0** (0 dB).
 3. Move to the adjacent **minimum**. The meter now reads S **directly**.
- **Three scales:** SWR NORMAL top (1 to 4), SWR NORMAL bottom (3.2 to 10), EXPANDED (1 to 1.3) for a nearly matched load, plus a dB scale. For S between 10 and 40 increase sensitivity by 20 dB, read on the 1 to 4 scale and **multiply by 10**.



1 kHz modulation, mixer, amplifier, detector: the meter reads V_{max}/V_{min} . Deck slide 31.

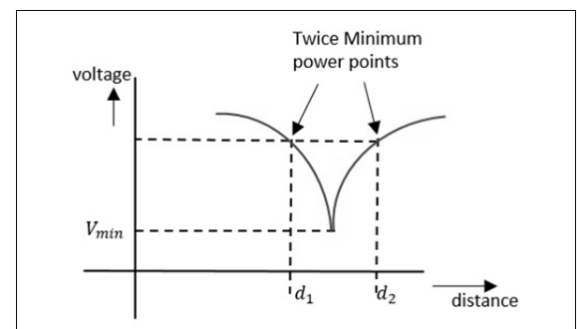


Low-VSWR bench: source, attenuator, slotted line, crystal detector, power meter. Deck slide 28.

How is the VSWR of a microwave transmitter measured when $VSWR > 10$? 6

PIN 1/24 79 Bh · **the double-minimum method:** the one question in the chapter with a derivation in it

- **Why the direct reading fails.** With $S > 10$ the gap between V_{max} and V_{min} is large. To read V_{max} the detector is driven out of its **square-law** region (diode current past about 20 mA, response turning linear), while at V_{min} it is in the noise. A meter calibrated on $I = kV^2$ is then simply wrong, and the error grows with S .
- **The cure:** never look at the maximum. Take **two readings either side of a single minimum**, where the level is low and the detector is safely square-law. This is the **double-minimum method**.
- **Procedure:**
 1. Terminate the line with the load and find a **voltage minimum** on the slotted line; note its position and meter reading.
 2. Move the probe either side until the detected **power is twice the minimum** (voltage $\sqrt{2} V_{min}$). Record the two positions d_1 and d_2 ; $\Delta x = d_2 - d_1$.
 3. Short the line and measure λ_g as **twice** the distance between two successive minima.
 4. Compute S .



The two “twice minimum power” points either side of a minimum, $\Delta x = d_2 - d_1$. Deck slide 29.

- **The formula, and where it comes from.** Near a minimum the detected quantity is $|V|^2 = V_{min}^2 + (V_{max}^2 - V_{min}^2) \sin^2 \beta \delta$ with δ measured from the minimum. Setting $|V|^2 = 2V_{min}^2$ gives $(S^2 - 1) \sin^2 \beta \delta = 1$, so with $\Delta x = 2\delta$ and $\beta = 2\pi/\lambda_g$:

$$S = \sqrt{1 + \frac{1}{\sin^2(\pi \Delta x / \lambda_g)}} \quad \rightarrow \quad S \approx \frac{\lambda_g}{\pi \Delta x} \text{ for large } S$$

- **Both forms verified** in `meas.py` against a simulated standing wave: at $S = 25$ the exact form returns 25.00 and the short form 24.97 (0.12% error); at $S = 60$, 0.04%. **The short form is only for large S :** at $S = 2$ it returns 1.63, 19% out, which is precisely why the direct method is used there instead.
- **Threshold:** the deck sets the changeover at $S > 10$ (which is the wording of 79 Bh), Das at $S > 20$. Say “above about 10, where the detector leaves its square-law region” and the distinction never matters.

8.7 Impedance, Reflection Coefficient and Frequency

Measurement of an unknown load, of reflection coefficient, and of frequency syllabus 8.5–8.8

no direct PYQ, but it is four of the nine syllabus sub-topics, and it is the reasoning behind 74 Bh’s “antenna as DUT” 8 in §7.8

Unknown impedance from the slotted line (minima-shift method)

- Impedance is complex, so **two** numbers are needed: the slotted line supplies S (giving $|\Gamma|$) and d_{min} (giving the phase).
- **Set-up 1:** load connected, measure S and the position of the first minimum. **Set-up 2:** replace the load with a **short**, which puts a minimum at the load plane, and measure the **shift**.
- Shift **towards the generator** means the load is **capacitive**; shift **towards the load** means **inductive**.
- The relations:

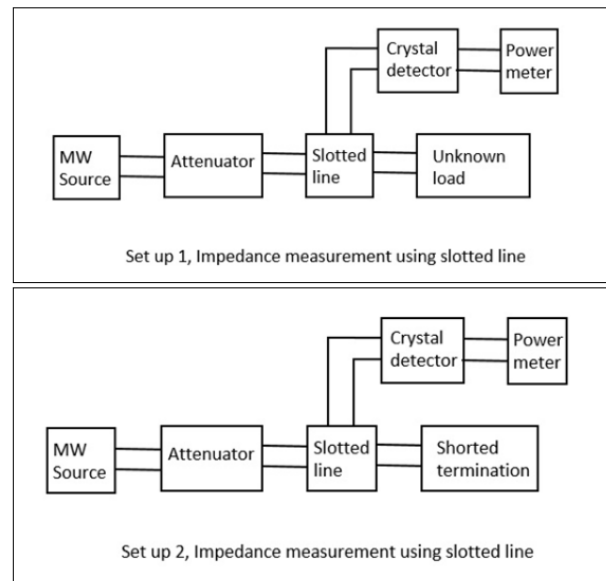
$$\rho = \frac{S-1}{S+1}, \quad \phi_L = 2\beta d_{min} - \pi, \quad \beta = \frac{2\pi}{\lambda_g}, \quad Z_L = Z_0 \frac{1+\Gamma}{1-\Gamma}$$

with $\lambda_g = 2 \times (\text{distance between successive minima})$,
 $\Gamma = \rho e^{j\phi_L}$.

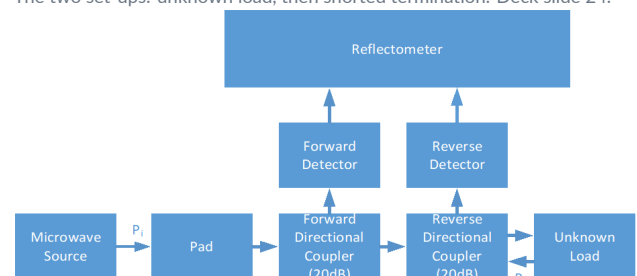
- Equivalent closed form, useful when no chart is at hand,
 $t = \tan \beta d_{min}$:

$$Z_L = Z_0 \frac{1 - jSt}{S - jt}$$

- **Both routes checked** in `meas.py`: a simulated pattern from $Z_L = 60 - j80$ on a 50Ω line is inverted back to $60 - j80$ by each.
- On the Smith chart: draw the S circle, start at the short-circuit point and move **towards the load** by d_{min}/λ_g ; the intersection is z_L .



The two set-ups: unknown load, then shorted termination. Deck slide 24.



Reflectometer: two 20 dB directional couplers sample forward and reverse waves. Deck slide 26.

Reflectometer: $|\Gamma|$ without a slotted line

- **Two directional couplers** face opposite ways: the first samples the **forward** wave at port 4, the second the **reverse** wave at port 3.
- With coupling $C \ll 1$ the ratio of the two detected outputs is the reflection coefficient itself:

$$\left| \frac{b_3}{b_4} \right| = \sqrt{1 - C^2} |\Gamma_L| \approx |\Gamma_L|$$

- Calibrate by shorting port 2 ($|\Gamma| = 1$) and setting the meter to unity, then connect the load. Everything else follows:

$$\text{VSWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}, \quad \text{RL(dB)} = -20 \log_{10} |\Gamma_L|$$

- **Broadband and fast**, but accuracy is set by coupler **directivity**, so it suits **low VSWR**; a high SWR still wants the slotted line.
- Related loss family, all measured the same way: **insertion loss** = reflection loss + attenuation loss, with reflection loss = $-10 \log(1 - |\Gamma|^2) = -10 \log[4S/(1 + S)^2]$. **Return loss compares P_r with P_i ; insertion loss compares P_0 with P_i (§3.4).**

Frequency measurement

- **Cavity (absorption) wavemeter:** a cylindrical cavity with a **variable short**, coupled to the guide through a hole in the narrow wall. Tuning the plunger changes the resonant frequency; at resonance the cavity **absorbs**, and the power meter downstream shows a **dip**. The plunger scale is calibrated in frequency.
- TE_{011} has the highest Q and no axial current, but it is a higher-order mode, so practical wavemeters use the dominant TM_{010} , excited by magnetic coupling. Polytron absorber behind the plunger stops oscillation above it.
- Accuracy follows Q : **1% at $Q = 1000$, 0.005% at $Q = 50\,000$.**
- **Slotted line:** short the guide, measure λ_g as twice the minima spacing, then

$$\frac{1}{\lambda^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}, \quad f = \frac{c}{\lambda}, \quad \lambda_c = 2a (TE_{10})$$

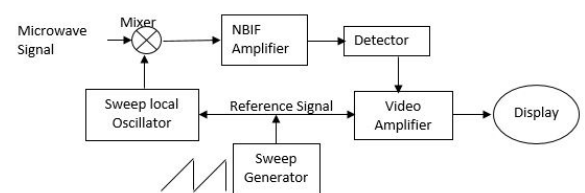
Worked (meas.py): WR-90 ($a = 2.286$ cm) with $\lambda_g = 4.00$ cm gives **9.96 GHz**.

- **Frequency counter with a down-conversion front end:** the unknown is mixed with a selected harmonic nf_1 of a known oscillator, and the counter reads $f_x = \text{counter} + nf_1$. Direct and most accurate.

8.8 Spectrum Analyzer**Spectrum analyzer (short note): working principle and block diagram** 4 5

PIN 2/24 80 Bh, 80 Ch · short note both times, so a block diagram plus six lines is the full answer

- A **swept-tuned superheterodyne receiver** that plots **amplitude against frequency**: frequency on the horizontal axis, amplitude on the vertical. It shows what an oscilloscope cannot, namely the **signal spectrum**.
- **Working:**
 1. The input microwave signal is mixed with a **local oscillator swept electronically** between two limits at a linear rate by a **sawtooth sweep voltage**.
 2. The difference product is amplified by a **narrow-bandwidth IF amplifier**, then detected and **video amplified**.
 3. The same sawtooth drives the horizontal deflection, with **zero flyback time**, so horizontal position tracks the LO frequency while the detected amplitude gives vertical deflection.
- **Measures:** carrier power, harmonics, spurious products, sidebands, phase noise, occupied bandwidth; on a modern analyzer, demodulated WLAN, GSM or LTE signals.
- **Four design parameters:** sweep rate, sweep range, IF bandwidth, IF centre frequency.
- **Resolution is the IF bandwidth (RBW).** Narrow RBW resolves closer tones but needs a **slower sweep**, because the IF filter must have time to settle: sweep time rises as $1/\text{RBW}^2$. Keep the span as small as the measurement allows.
- **Image response:** any signal $f_i = f_0 \pm f_{IF} = f_s \pm 2f_{IF}$ also beats to the IF. Choose f_{IF} **high** to push the image out of band. Worked: $f_{IF} = 450$ kHz puts the image **900 kHz** from the signal; $f_{IF} = 2$ MHz moves it to **4 MHz** away, easily filtered (meas.py).



Block diagram: mixer, swept local oscillator, narrow-band IF amplifier, detector, video amplifier, display. Deck slide 33.

- **Types:** swept-tuned (superheterodyne), FFT, real-time, audio.

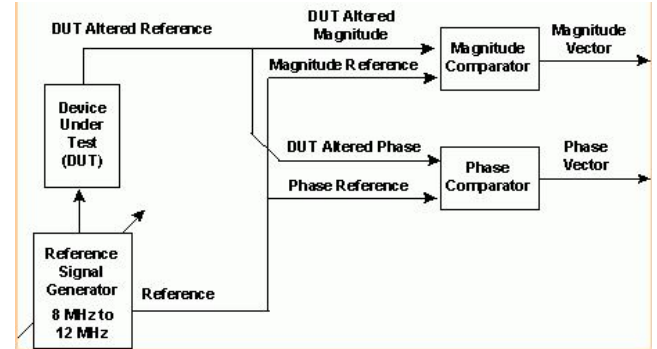
8.9 Network Analyzer, VNA and the Two-Port Model

Describe the working principle of a network analyzer; VNA (short note) 5 8

HOT 3/24 71 Bh, 73 Ma, 78 Ch · 78 Ch as a 5 mark short note on “VNA type network analyzer”; 71 Bh asks the two-port model below

Why it exists, and how it works

- The slotted line measures amplitude and phase **one frequency at a time**, so broadband testing with it is slow and expensive. A **network analyzer** measures both **over a wide band** in a reasonable time.
- **Principle: everything is measured against a reference.**
 1. A **sweep oscillator** feeds a **power divider**, splitting the signal into a **test channel** and a **reference channel**.
 2. The test signal passes through the **DUT**; the reference passes through a **phase-equalising length** of line.
 3. Microwave frequencies cannot be processed directly, so a **harmonic frequency converter** brings both to a fixed IF. A **phase-locked loop** makes the local oscillator track the reference channel, which is what allows a **swept** measurement.
 4. Conversion is in two steps, RF to an MHz IF, then to a kHz IF, where the **magnitude and phase comparators** compare test against reference.
- **Reflection and transmission** are separated by a test unit built from **dual directional couplers**: S_{11} and S_{21} from one direction, then source and load are interchanged for S_{22} and S_{12} . Each S_{ij} is a **ratio of a coupler output to the reference**, with the phase as the difference $\phi_j - \phi_i$.
- **Two types: SNA** (scalar) measures amplitude only; **VNA** (vector) measures **amplitude and phase**, so it can apply **vector error correction** and is far more accurate.



VNA: DUT alters the reference in magnitude (resistive) and phase (reactive); two comparators output the magnitude and phase vectors. Deck slide 36.

	Spectrum an.	Network an.
Measures	unknown signals: power, harmonics, sidebands, noise	known stimulus: Γ , S -parameters, insertion and return loss
Accuracy	lower	higher, vector error correction
IF BW	wider	narrower
Quantity	scalar only	amplitude and phase
Sweep	frequency	frequency and power

Design a two-port network model and derive the required parameters 10

PIN 1/24 71 Bh · the only 10 mark question in §8.9; it wants the set of two-port descriptions and why S is the microwave one

- **The model:** a black box with two ports, port 1 (V_1, I_1) and port 2 (V_2, I_2), both currents taken as flowing **into** the box. Any two of the four are independent, which is what gives the six standard parameter sets.

Set	Defining equations	Element found by	Needs
Z (open-circuit)	$V_1 = Z_{11}I_1 + Z_{12}I_2$ $V_2 = Z_{21}I_1 + Z_{22}I_2$	$Z_{11} = V_1/I_1 _{I_2=0}$	open circuit at a port
Y (short-circuit)	$I_1 = Y_{11}V_1 + Y_{12}V_2$ $I_2 = Y_{21}V_1 + Y_{22}V_2$	$Y_{11} = I_1/V_1 _{V_2=0}$	short circuit at a port
h (hybrid)	$V_1 = h_{11}I_1 + h_{12}V_2$ $I_2 = h_{21}I_1 + h_{22}V_2$	$h_{21} = I_2/I_1 _{V_2=0}$	one short, one open
$ABCD$ (chain)	$V_1 = AV_2 - BI_2$ $I_1 = CV_2 - DI_2$	cascades by matrix product	short and open
S (scattering)	$b_1 = S_{11}a_1 + S_{12}a_2$ $b_2 = S_{21}a_1 + S_{22}a_2$	$S_{11} = b_1/a_1 _{a_2=0}$	matched Z_0 termination

- **Why S at microwaves** (the same argument as §3.1, and the reason this question sits in the measurements chapter): Z , Y , h and $ABCD$ all need a **short or open circuit** at a port. At microwaves a short is not a short (lead inductance), an open radiates, and both are **frequency dependent** and may destabilise an active device. A **matched load** is realisable at every frequency, and a , b are travelling-wave quantities that a directional coupler can sample: that is exactly what the network analyzer above does, which is why S -parameters are the set that can actually be **measured**.

- **Reciprocity** $S_{12} = S_{21}$, **losslessness** $[S]^\dagger [S] = [I]$, **matched** $S_{ii} = 0$; conversions to and from Z exist, $[S] = ([Z] - [I])([Z] + [I])^{-1}$ for Z normalised to Z_0 (§3.2).

Noise figure measurement syllabus 8.9, never asked

- **Y-factor method**: drive the DUT with a calibrated **noise source** that switches between hot and cold. Y is the ratio of the two output powers, and with the source's **excess noise ratio**:

$$F = \frac{\text{ENR}}{Y - 1}, \quad T_e = \frac{T_h - YT_c}{Y - 1}, \quad F = 1 + \frac{T_e}{T_0}$$

- Worked (meas.py): ENR 15 dB with $Y = 10$ dB gives $F = 3.51$, i.e. **5.46 dB**, equivalently $T_e = 729$ K. The floor any such measurement sits on is $kT_0 = -174$ dBm/Hz.

8.10 Surveying the Radiation Zones Round a BTS

Working principle of a microwave measuring device to map the radiation zones around a GSM BTS 8

PIN 1/24 82 Bh · paired with “how microwaves are hazardous” 4 from §7.2; the zone formulas are §7.1

- **The instrument**: a **broadband EMF survey meter** (field-strength meter), which is a calibrated probe plus a display, usually backed by a **spectrum analyzer** when the individual operators' carriers must be separated.
- **Probe**: an **isotropic (three-axis) probe**, three mutually perpendicular dipoles (E field) or loops (H field) whose outputs are squared and summed, so the reading does not depend on how the probe is held. Diode or thermocouple detection, high-resistance leads so the cable does not perturb the field.
- **Why both E and H must be measured in the near field**: only in the far field is $E/H = 377 \Omega$ and $S = E^2/377$. Closer in, the ratio is not fixed, so the two are measured **separately** and compared against separate limits (§7.1).
- **Working principle**: the probe dipoles develop a voltage proportional to the field; a square-law diode detector gives a dc output proportional to **power density**; high-resistance lines carry it to a meter calibrated in V/m, A/m or W/m².
- **Procedure**: zero the meter with the transmitter off, then walk a **radial grid** out from the tower at a fixed height, log position with GPS, and plot contours. Compare each contour with the ICNIRP or FCC limit for that frequency, and mark the compliance boundary.
- **Worked** (meas.py, same formulas as §7.1): a 1.3 m panel at 1800 MHz has $\lambda = 16.7$ cm, so the reactive near field ends at $0.62\sqrt{D^3/\lambda} = \mathbf{2.25\ m}$ and the far field starts at $2D^2/\lambda = \mathbf{20.3\ m}$. With 20 W into a 17 dBi sector, S at 50 m is 0.032 W/m², against the ICNIRP public limit of $f/200 = 9$ W/m²: a margin of **about 280 times**.
- **Related ask**: 74 Bh's “antenna as a DUT, choose a measurement tool” 8 wants a **VNA** instead (§8.9): return loss and S_{11} give the match and bandwidth, and an anechoic range gives the pattern. It is filed in §7.8.

8.11 PYQ Mapping

Theory questions, covered in this document

- **Microwave vs low-frequency measurements; major measurement parameters** → §8.1. 71 Bh 3, 79 Bh 2, 81 Bh 3
- **Choose a proper power measuring tool and explain it** → §8.2. 72 Ma 10 (7.5 mW), 74 Ma 8 (airport radar), 73 Bh 8 (low power)
- **Bolometry; low power measurement** → §8.3. 70 Bh 6, 72 Ash 8, 79 Ch 10, 80 Ch 5
- **Single-bridge bolometer limitations; double-channel bolometer bridge** → §8.3. 80 Ba 2, 80 Bh 8, 81 Ba 4
- **Calorimetry; static (dry) calorimeter** → §8.5. 70 Ma 2, 71 Ma 5, 71 Bh 7, 76 Bh 10, 77 Ch 6, 81 Bh 6
- **Circulating (flow) calorimetry** → §8.5. 70 Ma 8, 75 Bh 5, 82 Ba 4
- **Calorimeter wattmeter** → §8.5. 80 Ba 6
- **Standing waves and VSWR meter** → §8.6. 70 Bh 4, 72 Ash 2
- **VSWR of a transmitter with VSWR > 10 (double-minimum method)** → §8.6. 79 Bh 6
- **Spectrum analyzer** → §8.8. 80 Bh 4, 80 Ch 5
- **Network analyzer; VNA** → §8.9. 73 Ma 8, 78 Ch 5
- **Two-port network model and its parameters** → §8.9. 71 Bh 10
- **Measuring device to map radiation zones round a GSM BTS** → §8.10. 82 Bh 8

Syllabus topics with no PYQ in 24 papers

- Thermocouples (§8.4), impedance measurement, measurement of unknown loads, measurement of reflection coefficient and frequency measurement (§8.7), and noise (§8.9). All six are written up, briefly, because the “choose a tool” questions can reach any of them.

Numerical content

- **This chapter has no numerical component:** no paper in 24 sets a measurement calculation. The arithmetic above (duty cycle, the two calorimeter laws, the bridge substitution, the double-minimum formula, λ_g to frequency) exists to make the descriptive answers specific, and every number is re-derived in `src/meas.py`.

Corrections and source defects found while building this chapter

- **Karkee's Chapter-8 PDF export is missing seven slides** that the pptx has: impedance measurement, the reflectometer and the low and high VSWR methods (deck slides 23 to 30). The **double-minimum method**, which is the whole of 79 Bh's 6 marks, appears **only** in the pptx. Figures are pulled from the pptx for this reason.
- Deck slide 30, "Measurement of frequency", is an **empty slide**. §8.7 is written from Das ch. 13 instead verified/added.
- Das Fig. 13.21's worked chart example ($S = 2$, $d_{min} = 0.2\lambda_g$) prints $Z_L = Z_0(1.0 + j0.7)$. Its own equations 13.46 to 13.50 give $Z_0(1.55 - j0.69)$, and the closed form $Z_0(1 - jSt)/(S - jt)$ agrees with that. The printed point does sit on the $S = 2$ circle, but at 70.7° , which needs $d_{min} = 0.348\lambda_g$, not 0.200. **The answer does not follow from its own data**; both routes are asserted in `meas.py`.
- Das §13.8 writes insertion loss as $10 \log(P_0/P_i)$, which is negative for a lossy network. Losses are quoted positive: $10 \log(P_i/P_0)$. Same slip as the Chapter-3 deck's (§3.4).
- Deck slide 19 titles the circulating calorimeter "**Circular** Calorimeter"; no such instrument. It is circulating, or flow.
- The deck sets the high-VSWR threshold at $S > 10$, Das at $S > 20$. Both are quoted in §8.6 rather than one being picked.

9. Last-Minute Revision

Built from the frequency analysis of all 24 papers (2069–2082 BS), EX 716 (BEI) and EX 752 (BEX), plus EG 785 for 69 Bh. Nothing here is new: every line points back into a chapter.

R.1 Must Know (every TOP-tier topic: these carry the paper)

A topic is TOP tier when 7 or more of the 24 papers ask it. There are 22 of them and every chapter has at least one, except Chapter 8, whose marks are spread thin over many small instruments instead.

Ch	Topic	Papers	The one thing to be able to do
6	Stability and maximum gain of a transistor	22/24	Δ , K , then $G_{T,max}$; if $K < 1$ say MSG, do not quote MAG
6	Insertion loss method; Butterworth Vs Chebyshev prototype	20/24	P_{LR} , order from the attenuation curve, g -values, scale and transform
1	Microwave Vs conventional low-frequency circuits	18/24	The d/λ criterion first, then the comparison table
7	How microwaves are hazardous; HERP, HERO, HERF	17/24	Non-ionizing $\rightarrow$ dielectric heating $\rightarrow$ no internal heat sensors $\rightarrow$ poorly cooled organs
2	Double-stub matching, all steps	17/24	Chart construction with the spacing circle; two lengths and the forbidden region
4	Rectangular waveguide TE and TM field equations	14/24	Six steps, then the characteristic-equation table
6	Microstrip implementation of the filter	13/24	High- Z line = series L , low- Z line = shunt C ; draw the named section
3	Why S-parameters, and S Vs h-parameters	13/24	A matched load exists at every frequency; a short and an open do not
7	Radiation fields (zones) of an antenna	11/24	$0.62\sqrt{D^3/\lambda}$ and $2D^2/\lambda$, and the $D > \lambda$ validity check
1	Real-world microwave systems and protocols	11/24	Three or five systems with band, modulation, data rate and standard
6	Amplifier design flowchart; stability analysis	10/24	Boxes joined by arrows: Δ , K , circles, Γ_S , Γ_L , networks
2	S-matrix of the network before and after matching	10/24	After matching $S_{11} = S_{22} = 0$; before, $S_{11} = \Gamma_{in}$
1	Why S-parameters instead of Z, Y, h	10/24	Same answer as Ch 3's; the papers file it under both
5	Cavity magnetron and cross-field effect	9/24	$\phi = 2\pi n/N$, Hull cut-off, the π -mode
5	Bunching, velocity and density modulation	9/24	Two lines for 2 marks; $X = 1.841$ and $2I_0J_1(X)$ for 10
4	Magic tee and its S-matrix	9/24	Sign pattern down a column, never the port number
4	Dominant and degenerate modes; which cannot exist	9/24	f_c ranking; TM_{m0} and TM_{0n} do not exist
3	Properties of the S-matrix	8/24	Matched, reciprocal, unitary, reference-plane phase
7	Safety practices and standards	7/24	Name the body, the range and one number each
5	Two-cavity and multicavity klystron	7/24	Velocity modulate, drift, bunch, decelerate at the catcher
3	Derive S-parameters for a two-port	7/24	Five steps ending at $S_{11} = (Z - Z_0)/(Z + Z_0)$
1	IEEE band classification and microwave features	7/24	The band table, then advantages and disadvantages

Two of these rows are one question. “Why S-parameters” is tagged into Chapter 1 by ten papers and into Chapter 3 by thirteen, because the papers ask it as an opener to both a behaviour question and a network question. Learn it once.

R.2 Must Memorize (what cannot be derived in the hall)

Ch 1: introduction

- **Microwaves** 300 MHz to 300 GHz, λ 1 m down to 1 mm. IEEE letters: L 1-2, S 2-4, C 4-8, X 8-12, Ku 12-18, K 18-27, Ka 27-40 GHz.
- **The criterion:** lumped analysis holds while $d/\lambda \ll 1$ (below about 0.01). Above it the line is distributed and V, I depend on position.
- Parasitics that kill a lumped part: a resistor becomes R with lead L and shunt C ; an inductor self-resonates; a capacitor turns inductive above its SRF.

Ch 2: transmission lines and the Smith chart

- $Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan \beta d}{Z_0 + jZ_L \tan \beta d}$, $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$, $S = \frac{1 + |\Gamma|}{1 - |\Gamma|}$.
- **Special lengths:** short $\rightarrow jZ_0 \tan \beta d$; open $\rightarrow -jZ_0 \cot \beta d$; $\lambda/2 \rightarrow Z_L$; $\lambda/4 \rightarrow Z_0^2/Z_L$.
- **Chart conventions**, both easy to reverse: the WTG scale reads 0.000 at $\Gamma = -1$ and increases **clockwise**; on an **admittance** chart the terminations swap ends, open at $\Gamma = -1$, short at $\Gamma = +1$.
- **Double stub:** $B = \frac{1 \pm \sqrt{g(1+t^2) - g^2 t^2}}{t}$, $t = \tan \beta d$, a leading **1**, not g . Forbidden region $g > 1/\sin^2 \beta d$.

Ch 3: S-parameters

- $b_i = \sum_j S_{ij} a_j$; $S_{ij} = b_i/a_j$ with every other port **matched**, not shorted or open.
- **Four tests:** $S_{ii} = 0$ matched; $[S] = [S]^T$ reciprocal; $[S]^\dagger [S] = [I]$ lossless (columns unit, distinct columns orthogonal); reference plane shift multiplies S_{ij} by $e^{-j(\theta_i + \theta_j)}$.
- **No 3-port** can be lossless, reciprocal and matched at once. Give up one: lossy gives the resistive divider, non-reciprocal the circulator, unmatched the tee.
- **Magic tee** $\frac{1}{\sqrt{2}}$ with the E-arm column carrying $+1, -1$ and the H-arm column $+1, +1$. Read the **sign pattern**, not the port numbers.

Ch 4: waveguides and components

- $f_c = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{(m/a)^2 + (n/b)^2}$, $\lambda_c = 2a$ for TE_{10} ; $\lambda_g = \frac{\lambda}{\sqrt{1 - (f_c/f)^2}}$; $\frac{1}{\lambda^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}$.
- $v_p v_g = v^2$, $v_p > v > v_g$. Below f_c the mode is **evanescent**: a guide is a high-pass filter.
- $Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$, $Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$, $\eta_0 = 120\pi \approx 377 \Omega$.
- **TM_{m0} and TM_{0n} do not exist**; TE_{10} is dominant when $a > b$. Circular guide: TE_{11} dominant ($p'_{11} = 1.841$), TM_{01} next ($p_{01} = 2.405$).
- Directional coupler (input at 1, coupled 4, isolated 3): coupling $C = 10 \log P_1/P_4$, directivity $D = 10 \log P_4/P_3$, isolation $I = 10 \log P_1/P_3$, and $I = C + D$.

Ch 5: microwave generators

- $v_0 = \sqrt{2eV_0/m} = 0.593 \times 10^6 \sqrt{V_0}$ m/s.
- Bunching parameter $X = \frac{\beta_i V_1}{2V_0} \theta_0$; catcher fundamental $2I_0 J_1(X)$, maximum at $X = \mathbf{1.841}$ where it is $1.164 I_0$.
- **Reflex klystron** oscillates at $n + 3/4$ cycles in the repeller space; best mode efficiency about **22.7 %**.
- **Magnetron:** adjacent-cavity phase $\phi = 2\pi n/N$, so π -mode is $n = N/2$; Hull cut-off $B_c = \frac{\sqrt{8mV_0/e}}{b(1 - a^2/b^2)}$.

Ch 6: RF design practices

- $\Delta = S_{11}S_{22} - S_{12}S_{21}$; $K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|}$. Unconditionally stable needs $K > 1$ and $|\Delta| < 1$, both.
- $G_{T,max} = \frac{|S_{21}|}{|S_{12}|} (K - \sqrt{K^2 - 1})$, valid **only** when $K > 1$. If $K < 1$ quote $MSG = |S_{21}|/|S_{12}|$.
- Unilateral: $G_{TU,max} = \frac{|S_{21}|^2}{(1 - |S_{11}|^2)(1 - |S_{22}|^2)}$.
- $P_{LR} = 1/(1 - |\Gamma|^2)$. Butterworth $g_k = 2 \sin \frac{(2k-1)\pi}{2N}$; Chebyshev from the ripple. Order from the attenuation curve, never guessed.
- Microstrip: a **short high-impedance** line is a series L , a **short low-impedance** line a shunt C . That one sentence builds every filter layout.

Ch 7: antennas, propagation and hazards

- Zones: $R_1 = 0.62\sqrt{D^3/\lambda}$, $R_2 = 2D^2/\lambda$, **both needing** $D > \lambda$. For a small antenna use $\lambda/2\pi$ and 2λ instead.
- $S = \frac{PG}{4\pi R^2}$, $S = E^2/377$ **in the far field only**.
- SAR = $\frac{\sigma E^2}{\rho}$ W/kg. Whole-body **4 W/kg** warms the body about 1°C: divide by 10 for workers (0.4), by 50 for the public (0.08).
- ICNIRP 1998 public limit, 400 MHz to 2 GHz: $f/200$ W/m² with f in MHz.

Ch 8: measurements

- **Bands:** low < 10 mW, medium 10 mW to 10 W, high > 10 W. $P_{av} = P_{peak} \times \text{duty}$, $\text{duty} = \tau f_r$.
- Static calorimeter $P = \frac{4.187 m C_p T}{t}$; circulating $P = 4.187 v d C_p (T_2 - T_1)$. **Same law**, with m/t replaced by $v d$.
- Double bridge $P_{av} = \frac{E_1^2 - E_2^2}{4R}$.
- Double minimum, for S above about 10: $S = \sqrt{1 + \frac{1}{\sin^2(\pi \Delta x / \lambda_g)}} \approx \frac{\lambda_g}{\pi \Delta x}$.

R.3 Must Practice (the 56 solved problems, ranked by how many papers set them)

All of these are worked in the companion volume, RF and Microwave Engineering: Complete Numerical Problems and Solutions. Chapters 1, 5 and 8 set no numerical in 24 papers.

Ch	Family	Papers	Problems	Where it goes wrong
6	Transistor stability and maximum gain	22/24	1 to 4	Quoting MAG when $K < 1$. Three of the 13 sets are not unconditionally stable
2	Double-stub shunt tuner	17/24	8 to 24	The admittance chart's terminations are mirrored; a wrong end costs $\lambda/4$
6	Filter design and microstrip layout	13/24	10	Choosing N by habit rather than from the attenuation curve
4	Rectangular guide parameter sets	14/24	1 to 3	Using λ where λ_g belongs; forgetting the cut-off check
3	Magic tee and radar power splits	9/24	3 to 5, 9 to 11	Reading the port number instead of the sign pattern
4	Two magic tees joined; terminated arms	5/24	6 to 8	Not re-deriving after termination; use <code>sparam.terminate()</code>
2	Single-stub shunt tuner	6/24	2 to 7	Two valid answers exist; give both d and ℓ pairs
6	Amplifier matching networks and gain	10/24	5 to 8	Conjugate match at one port only; check $\Gamma_S = \Gamma_{in}^*$
3	Identify the device from its S-matrix	5/24	1, 6 to 8, 12	Every one of them is a magic tee except the modulator
7	Radiation zones with numbers	5/24	1, 2	$D < \lambda$ makes the two formulas invalid; give the small-antenna boundaries
4	Circular guide modes with numbers	3/24	4, 5	Bessel roots: $p'_{11} = 1.841$ for TE, $p_{01} = 2.405$ for TM
6	Negative-resistance oscillator	2/24	9	$ R_{out} \geq 1.2R_L$ is a rule, not a suggestion

Every published answer in that volume is machine checked. `rf.py` solves the lines exactly, `verify.py` walks each published stub design forward and asserts $y_{in} = 1 + j0$ **re-deriving the susceptance from the published length**, `sparam.py` re-derives each device matrix from its stated properties, `wg.py` tests every printed field component against both curl equations and the walls, `amp.py` and `filt.py` reproduce the deck examples, `tubes.py` simulates the electrons, `ant.py` and `meas.py` assert every number in Chapters 7 and 8.

R.4 Most Repeated PYQs (if you revise only fifteen questions, revise these)

1. Check the stability of a transistor with these S-parameters and compare maximum power gain for bilateral and unilateral modes. 22 papers → §6.4 and Problems 1 to 4.
2. Why is the insertion loss method used for microwave filters? Prototype a Butterworth or Chebyshev LPF and implement it in microstrip. 20 papers → §6.1 to §6.3.
3. Differentiate the behaviour of microwave circuits from conventional low-frequency circuits. 18 papers → §1.2.
4. How are microwaves hazardous to humans? 17 papers → §7.2.
5. Design a double-stub matching network for this load and prepare the S-matrix of the matched network. 17 papers → §2.7, §2.8, Problems 8 to 24.
6. Derive the field equations and characteristic parameters of a rectangular waveguide in TE (or TM) mode. 14 papers → §4.1.
7. Why are S-parameters used in microwave network analysis? Compare with h-parameters. 13 papers in Ch 3, 10 in Ch 1 → §3.1, §1.3.
8. Classify the microwave radiation fields around an antenna, oven or BTS. 11 papers → §7.1 and its two problems.
9. Give detailed protocols and specifications of three (or five) microwave communication systems. 11 papers → §1.5.
10. Draw a neat diagram of a magic tee and derive its S-matrix. 9 papers → §4.5, §3.6.
11. What is bunching effect? Explain velocity modulation. 9 papers → §5.2. Usually the 2-mark rider on a tube question.
12. Sketch a cross-sectional magnetron with 45° (or 60° , 90°) phase shift between adjacent cavities and explain its working. 9 papers → §5.5.
13. State and justify the properties of the scattering matrix; analyse a three-port network. 8 papers → §3.3, §3.5.
14. Describe the construction and operation of a multicavity klystron. 7 papers → §5.3.
15. Describe how standing waves and microwave power are measured (VSWR meter plus bolometry or calorimetry). the pairing itself in 72 Ash, 70 Bh and 79 Bh; some Chapter 8 question in 22 of 24 papers → §8.3, §8.5, §8.6.

R.5 One-Day Revision Checklist

Morning: the two heavy chapters (about 4 hrs)

- Chapter 6 (22.4 % of every mark ever set). Work Problem 1 end to end without looking: Δ , K , $|\Delta|$, verdict, $G_{T,max}$. Then Problem 10's ladder, and draw the six named microstrip layouts from memory.
- Chapter 2 (15.6 %). Draw one double-stub construction from a blank chart: SWR circle, spacing circle rotated d , constant- g arc, both stub lengths off the rim. Say the two chart conventions out loud before you start.
- If either feels shaky, stop here. These two are a third of the paper.

Afternoon: the reliable marks (about 3 hrs)

- Chapter 4: write the six derivation steps and the characteristic-equation table. Then the magic tee matrix, and one two-tee junction.
- Chapter 7: the zone figure, the hazard chain, HERP / HERO / HERF, SAR and the three limits. This is a guaranteed question and it is fast.
- Chapter 3: the four tests, the 3-port impossibility theorem, and the five-step S-parameter derivation.
- Chapter 8: the power decision table, the two calorimeter formulas, and the bolometer double bridge. Three answers cover 22 papers.

Evening: consolidation (about 1 hr)

- Chapter 5: velocity modulation, bunching, klystron, magnetron, TWT. All descriptive, all the same three-line skeleton: velocity-modulate, bunch, decelerate.
- Chapter 1: the band table, the d/λ criterion, and the five systems with their specifications.
- Re-read R.2 straight through. Those are the lines that cannot be rebuilt in the hall.

In the hall

- Read the marks split first. A $[2 + 8]$ is two answers; give the 2-mark definition its own line and heading, then the long part.
- Draw first, write second. Nearly every question here has one figure that earns marks on its own: the chart construction, the guide with a and b , the magic tee, the zone diagram, the bridge circuit.
- Say which case you are in before computing: $K > 1$ or not, $a > b$ or not, $D > \lambda$ or not, which power band. Most lost marks in this subject are a correct method applied to the wrong case.

- If a number will not come out, **write the formula and the substitution anyway**. In this subject the method carries most of the mark.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave
Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector
Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.